

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277019

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Lerner; Lorena et al.

ANTI-GDF-15 ANTIBODIES

Abstract

Monoclonal antibodies that bind and inhibit the activity of human GDF15 are disclosed. The antibodies can be used to treat body weight loss, including cachexia, associated with the over-expression of human GDF15.

Inventors: Lerner; Lorena (Newton Centre, MA), Abbott; Sandra (Boston, MA), Bai; Ailin (Newton, MA), Chen; Ting (Acton, MA), Chiu; Maria Isabel (Newton Centre, MA), Liu; Qing (Acton, MA), Poling; Laura (Boston, MA), Tao; Nianjun (Brighton, MA), Weiler; Solly (Newton, MA), Weng; Zhigang (Brookline, MA), Winston, JR.; William M. (Marlborough, MA), Gyuris; Jeno (Lincoln, MA)

Applicant: AVEO Pharmaceuticals, Inc. (Boston, MA)

Family ID: 50069279

Appl. No.: 19/048087

Filed: February 07, 2025

Related U.S. Application Data

parent US continuation 18336713 20230616 parent-grant-document US 12258391 child US 19048087

parent US continuation 16824034 20200319 parent-grant-document US 11725047 child US 18336713

parent US continuation 15655263 20170720 parent-grant-document US 10597444 child US 16824034

parent US continuation 14863870 20150924 parent-grant-document US 9725505 child US 15655263

parent US division 14137415 20131220 parent-grant-document US 9175076 child US 14863870
us-provisional-application US 61827325 20130524

us-provisional-application US 61745508 20121221

Publication Classification

Int. Cl.: C07K16/22 (20060101); A61K39/00 (20060101); A61K39/395 (20060101); A61K45/06 (20060101); A61P3/02 (20060101); C12N5/10 (20060101); C12N15/10 (20060101); C12N15/63 (20060101)

U.S. Cl.:

CPC C07K16/22 (20130101); A61K39/3955 (20130101); A61K45/06 (20130101); A61P3/02 (20180101); A61K2039/505 (20130101); C07K2317/24 (20130101); C07K2317/76 (20130101); C07K2317/92 (20130101); C07K2319/30 (20130101); C12N5/10 (20130101); C12N15/10 (20130101); C12N15/63 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/336,713, filed Jun. 16, 2023, which is a continuation of U.S. patent application Ser. No. 16/824,034, filed Mar. 19, 2020, issued as U.S. Pat. No. 11,725,047 on Aug. 15, 2023, which is a continuation of U.S. patent application Ser. No. 15/655,263, filed Jul. 20, 2017, issued as U.S. Pat. No. 10,597,444 on Mar. 24, 2020, which is a continuation of U.S. patent application Ser. No. 14/863,870, filed Sep. 24, 2015, issued as U.S. Pat. No. 9,725,505 on Aug. 8, 2017, which is a divisional of U.S. patent application Ser. No. 14/137,415, filed Dec. 20, 2013, issued as U.S. Pat. No. 9,175,076 on Nov. 3, 2015, which claims the benefit of and priority to U.S. Provisional Patent Application No. 61/827,325, filed May 24, 2013, and U.S. Provisional Patent Application No. 61/745,508, filed Dec. 21, 2012, the entire disclosures of each of which are incorporated by reference herein in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ST.26 XML format and is hereby incorporated by reference in its entirety. Said ST.26 XML copy, created on Jun. 16, 2023, is named 406767_029USD1C3_SEQUENCE_LISTING_ST.26.XML and is 425,984 bytes in size.

FIELD OF THE INVENTION

[0003] The field of the invention is molecular biology, immunology, cachexia and cachexia-like disorders, and oncology. More particularly, the field is therapeutic antibodies.

BACKGROUND

[0004] Involuntary weight loss can be categorized into three primary etiologies that include, cachexia, sarcopenia and starvation. Cachexia is a debilitating metabolic syndrome associated with numerous diseases, including cancer, AIDS, chronic heart failure (also known as congestive heart failure), chronic obstructive pulmonary disease (COPD), chronic kidney disease, tuberculosis, sepsis and other forms of systemic inflammation. Cachexia varies in its manifestations, but generally involves involuntary loss of skeletal muscle mass and some form of underlying illness (Evans et al. (2008) CLIN. NUTR. 27:793-799). Cachexia is a wasting disorder involving involuntary weight loss and may be associated with systemic inflammation and/or an acute inflammatory response. Thomas (2007) CLIN. NUTRITION 26:389-399. Loss of fat mass as well as fat-free mass, such as muscle mass, often is a prominent clinical feature of cachexia. In many but not all cases, cachexia progresses through stages that have been designated precachexia, cachexia and refractory cachexia (Fearon et al. (2011) LANCET ONC. 12:489-495). Two different, but sometimes overlapping, processes appear to drive the development and progression of cachexia: (a)

metabolic processes that act directly on muscle, reducing its mass and function; and (b) reduced food intake, which leads to loss of both fat and muscle (Tsai et al. (2012) J. CACHEXIA SARCOPENIA MUSCLE 3:239-243).

[0005] Although cachexia is a complex and incompletely understood syndrome, it is clear that GDF15 (also known as MIC-1, PLAB, PDF and NAG-1), a member of the TGF- β superfamily, is an important mediator of cachexia in various diseases (Tsai et al., supra). At least some tumors over-express and secrete GDF15, and elevated serum GDF15 levels have been associated with various cancers (Johnen et al. (2007) NAT. MED. 13:1333-1340; Bauskin et al. (2006) CANCER RES. 66:4983-4986). Monoclonal antibodies against GDF15 have been recognized as potential anti-cachexia therapeutic agents. See, e.g., U.S. Pat. No. 8,192,735.

[0006] Weight loss resulting from cachexia is associated with poor prognosis in various diseases (Evans et al., supra), and cachexia and its consequences are considered to be the direct cause of death in about 20% of cancer deaths (Tisdale (2002) NAT. REV. CANCER 2:862-871). Cachexia is infrequently reversed by nutritional intervention, and currently this syndrome is seldom treated with drug therapy (Evans et al., supra).

[0007] Sarcopenia is a clinical condition related to cachexia that is characterized by loss of skeletal muscle mass and muscle strength. The decrease in muscle mass can lead to functional impairment, with loss of strength, increased likelihood of falls, and loss of autonomy. Respiratory function may also be impaired with a reduced vital capacity. During metabolic stress, muscle protein is rapidly mobilized in order to provide the immune system, liver and gut with amino acids, particularly glutamine. Sarcopenia often is a disease of the elderly; however, its development may also be associated with muscle disuse and malnutrition, and may coincide with cachexia. Sarcopenia can be diagnosed based upon functional observations such as low muscle weight and low gait speed. See, e.g., Muscaritoli et al. (2010) CLIN. NUTRITION 29:154-159.

[0008] Starvation typically results in a loss of body fat and non-fat mass due to inadequate diet and/or nutritional uptake (Thomas (2007) supra). The effects of starvation often are reversed by improving diet and nutritional, for example, protein, uptake.

[0009] Naturally occurring antibodies are multimeric proteins that contain four polypeptide chains (FIG. 1). Two of the polypeptide chains are called heavy chains (H chains), and two of the polypeptide chains are called light chains (L chains). The immunoglobulin heavy and light chains are connected by an interchain disulfide bond. The immunoglobulin heavy chains are connected by interchain disulfide bonds. A light chain consists of one variable region (V.sub.L in FIG. 1) and one constant region (C.sub.L in FIG. 1). The heavy chain consists of one variable region (V.sub.H in FIG. 1) and at least three constant regions (C.sub.H1, C.sub.H2 and C.sub.H3 in FIG. 1). The variable regions determine the specificity of the antibody. Each variable region comprises three hypervariable regions also known as complementarity determining regions (CDRs) flanked by four relatively conserved framework regions (FRs). The three CDRs, referred to as CDR.sub.1, CDR.sub.2, and CDR.sub.3, contribute to the antibody binding specificity. Naturally occurring antibodies have been used as starting material for engineered antibodies, such as chimeric antibodies and humanized antibodies.

[0010] There is a significant unmet need for effective therapeutic agents for treating cachexia and sarcopenia, including monoclonal antibodies targeting GDF15. Such therapeutic agents have the potential to play an important role in the treatment of various cancers and other life-threatening diseases.

SUMMARY

[0011] The invention is based, in part, upon the discovery of a family of antibodies that specifically bind human GDF15 (hGDF15). The antibodies contain hGDF15 binding sites based on the CDRs of the antibodies. The antibodies can be used as therapeutic agents. When used as therapeutic agents, the antibodies are engineered, e.g., humanized, to reduce or eliminate an immune response when administered to a human patient.

[0012] The disclosed antibodies prevent or inhibit the activity of (i.e., neutralize) hGDF15. When administered to a mammal, the antibodies can inhibit the loss of muscle mass, for example, the loss of muscle mass associated with an underlying disease. The underlying disease may be selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the loss of muscle mass may be accompanied by a loss of fat mass. The disclosed antibodies can also be used to inhibit involuntary weight loss in a mammal. In some embodiments, the disclosed antibodies may also be used to inhibit the loss of organ mass. Further, a method of treating cachexia and/or sarcopenia in a mammal comprising administering an effective amount of one of at least one of the disclosed antibodies to a mammal in need thereof is disclosed.

[0013] Also disclosed is a method for establishing a steady-state level of mature recombinant human GDF15 (rhGDF15) in plasma or serum in a mammal comprising administering a rhGDF15-immunoglobulin Fc (Fc-rhGDF15) fusion protein to the mammal. The Fc-rhGDF15 can be a mouse Fc mature recombinant human GDF15 (mFc-rhGDF15). In some embodiments, the mammal is a rodent, e.g., a mouse.

[0014] In another aspect, a method of treating obesity in a mammal, for example, a human, comprising administering a therapeutically effective amount of Fc-rhGDF15, e.g., a human Fc mature recombinant human GDF15 (hFc-rhGDF15), to the mammal in need thereof, is disclosed. Pharmaceutical compositions comprising an Fc-rhGDF15 fusion protein and a pharmaceutically acceptable carrier are also disclosed.

[0015] These and other aspects and advantages of the invention will become apparent upon consideration of the following figures, detailed description, and claims. As used herein, “including” means without limitation, and examples cited are non-limiting. As used herein, “antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11” means antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11, or humanized variants thereof.

Description

DESCRIPTION OF THE DRAWINGS

[0016] The invention can be more completely understood with reference to the following drawings.

[0017] FIG. 1 (prior art) is a schematic representation of a typical naturally-occurring antibody.

[0018] FIG. 2 is a graph representing results from an experiment to measure hGDF15 serum levels in naïve mice or mice bearing human xenograft tumors (Chago, RPMI7951, PC3, TOV21G, HT-1080, K-562, LS1034), as determined by ELISA.

[0019] FIG. 3 is a plot representing results from an experiment to determine the plasma pharmacokinetics (PK) of cleaved rhGDF15 administered by subcutaneous injection (1 µg/g) in naïve ICR-SCID mice, as determined by ELISA.

[0020] FIG. 4 is a graph summarizing results from an experiment to measure cachectic activity of cleaved rhGDF15 protein (.square-solid.) and negative control (PBS (.circle-solid.)) to induce body weight loss in immune-incompetent mice, ICR-SCID. Arrows indicate subcutaneous doses of 1 µg/g of rhGDF15.

[0021] FIGS. 5A and 5B are graphs summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (a mouse Fc fused to the amino terminus of a mature recombinant human GDF15; .square-solid.), rFc-rhGDF15 (a rabbit Fc fused to the amino terminus of a mature recombinant mouse GDF15; .box-tangle-solidup.), and negative control (PBS; .circle-solid.) to induce body weight loss in immune-competent Balb/C mice (FIG. 5A) and immune-incompetent CB17-SCID mice (FIG. 5B). Arrows indicate subcutaneous doses of 1 µg/g of recombinant protein.

[0022] FIGS. 6A-6E are graphs summarizing results from an experiment to demonstrate cachectic activity of mFc-rhGDF15 (.square-solid.) and negative control (PBS; .circle-solid.) to induce body

weight loss in immune-incompetent ICR-SCID mice (FIG. 6A; arrows indicate subcutaneous doses of 1 µg/g of mFc-rhGDF15); to induce loss of adipose tissue or gonadal fat mass (FIG. 6B); to induce loss of muscle mass of gastrocnemius muscle (FIG. 6C; Gastroc Mass); and to increase mRNA expression of muscle degradation molecular markers (mMuRF1 (FIG. 6D) and mAtrogin (FIG. 6E)).

[0023] FIG. 7 is a graph summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (.square-solid.) and negative control (PBS; .circle-solid.) to induce body weight loss in immune-incompetent Balb/C nude mice. Arrows indicate subcutaneous doses of 1.33 µg/g of mFc-rhGDF15.

[0024] FIG. 8 is a graph summarizing results from an experiment to measure serum levels of mFc-rhGDF15 in mice dosed with the recombinant protein. The presence of mFc-rhGDF15 was determined by Western Blot. Two positive bands corresponding to mFc-rhGDF15 and rhGDF15 (according to the appropriate molecular size) were quantified by Licor. The percentage of released-rhGDF15 versus mFc-rhGDF15 was calculated.

[0025] FIG. 9A is a graph summarizing results from an experiment to measure cachectic activity of mFc-rhGDF15 (0.1 µg/g (.square-solid.), 0.01 µg/g (Δ)) and negative control (mIgG 0.1p g/g (.circle-solid.)) to induce body weight loss in immune-incompetent ICR-SCID mice. Arrows indicate the intraperitoneal dose of the recombinant protein. FIG. 9B is a graph representing the total level of rhGDF15 in the plasma of mice dosed with mFc-rhGDF15 (0.1 µg/g (□), 0.01 µg/g (.square-solid.)) five days post dose, as determined by ELISA.

[0026] FIG. 10 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin heavy chain variable region of antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, and 17B11. The amino acid sequences for each antibody are aligned against one another, and CDR.sub.1, CDR.sub.2, and CDR.sub.3, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0027] FIG. 11 is a sequence alignment showing the CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the immunoglobulin heavy chain variable region sequences in FIG. 10.

[0028] FIG. 12 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin light chain variable region of antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, and 17B11. The amino acid sequences for each antibody are aligned against one another, and CDR.sub.1, CDR.sub.2, and CDR.sub.3, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0029] FIG. 13 is a sequence alignment showing the CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the immunoglobulin light chain variable region sequences in FIG. 12.

[0030] FIG. 14 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (.square-solid.), 03G05 (.box-tangle-solidup.), 04F08 (.Math.), 06C11 (◇), 14F11 (, custom-character), and 17B11 (□), and a murine IgG control (.circle-solid.; mIgG) dosed at 10 mg/kg in an mFc-rhGDF15 cachectic model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0031] FIG. 15 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (.box-tangle-solidup.), 03G05 (◇), 04F08 (∇), 06C11 (□), 08G01 (.square-solid.), 14F11 (.diamond-solid.), and 17B11 (, custom-character), and a murine IgG control (.circle-solid.; mIgG), dosed at 10 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrows indicate intra-peritoneal injection of antibody every three days.

[0032] FIGS. 16A-16E are graphs summarizing results from an experiment to demonstrate anti-

cachectic activity of anti-GDF15 antibody 01G06 (.square-solid.), dosed at 10 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody 01G06 reversed body weight loss (FIG. 16A); induced a significant increase in food consumption for up to three days post dose (FIG. 16B); induced a gain of gonadal fat mass (FIG. 16C); induced a gain of muscle mass of gastrocnemius muscle (FIG. 16D); and decreased mRNA expression of muscle degradation molecular markers (mMuRF1 and mAtrogin (FIG. 16E)) compared to negative control (murine IgG (.circle-solid.)). In FIG. 16A, the arrow indicates intra-peritoneal injection of antibody.

[0033] FIGS. 17A-17B are graphs summarizing results from an experiment to demonstrate anti-cachectic activity of anti-GDF15 antibody 01G06 (.square-solid.), dosed at 2 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody 01G06 reversed body weight loss compared to murine IgG (.circle-solid.) (FIG. 17A); and induced a gain of organ mass (liver, heart, spleen, kidney) and induced a gain of tissue mass (gonadal and gastrocnemius) (FIG. 17B) compared to negative control (murine IgG) and baseline (day 1). The arrows in FIG. 17A indicate intra-peritoneal injection of antibody.

[0034] FIG. 18 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 01G06 (.square-solid.), 03G05 (.box-tangle-solidup.), 04F08 (X), 06C11 (.diamond-solid.), 08G01 (°), 14F11 (□), and 17B11 (Δ), and a murine IgG control (.circle-solid.) dosed at 10 mg/kg in a K-562 leukemia tumor xenograft model in immune-incompetent (CB17SCRFMF) mice. The arrows indicate intra-peritoneal injection of antibody.

[0035] FIG. 19 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin heavy chain variable region of chimeric 01G06 variable region denoted as Ch01G06 Chimeric; humanized 01G06 heavy chain variable regions denoted as Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, and Hu01G06 IGHV1-69 F2; chimeric 06C11 denoted as Ch06C11 Chimeric; humanized 06C11 heavy chain variable regions denoted as HE LM 06C11 IGHV2-70, and Hu06C11 IGHV2-5; chimeric 14F11 denoted as Ch14F11 Chimeric; and humanized 14F11 heavy chain variable regions denoted as Sh14F11 IGHV2-5 and Sh14F11 IGHV2-70. The amino acid sequences for each antibody are aligned against one another, and CDR.sub.1, CDR.sub.2, and CDR.sub.3, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0036] FIG. 20 is a sequence alignment showing the CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the immunoglobulin heavy chain variable region sequences in FIG. 19.

[0037] FIG. 21 is a sequence alignment showing the amino acid sequence of the complete immunoglobulin light chain variable region of chimeric 01G06 denoted as Ch01G06 Chimeric; humanized 01G06 light chain variable regions denoted as Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, and Hu01G06 IGKV1-39 F2; chimeric 06C11 denoted as Ch06C11 Chimeric; humanized 06C11 light chain variable region denoted as Sh06C11 IGKV1-16; chimeric 14F11 denoted as Ch14F11 Chimeric; and humanized 14F11 light chain variable region denoted as Hu14F11 IGKV1-16. The amino acid sequences for each antibody are aligned against one another, and CDR.sub.1, CDR.sub.2, and CDR.sub.3, are identified in boxes. The unboxed sequences represent framework (FR) sequences. Alignment positioning (gaps) is based on Kabat numbering, rather than an alignment algorithm such as Clustal. Numbering above the sequences represents Kabat numbering.

[0038] FIG. 22 is a sequence alignment showing the CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the immunoglobulin light chain variable region sequences in FIG. 21.

[0039] FIG. 23 is a graph summarizing results from an experiment to measure cachectic inhibitory

activity of anti-GDF15 antibodies 01G06 (.square-solid.), Hu01G06-46 (.box-tangle-solidup.), and Hu01G06-52 (*), and a murine IgG control (.circle-solid.) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0040] FIG. 24 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 06C11 (.diamond-solid.), Hu06C11-27 (□), and Hu06C11-30 (.box-tangle-solidup.), and a murine IgG control (.circle-solid.) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0041] FIG. 25 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies 14F11 (.box-tangle-solidup.), Hu14F11-39 (□), and Hu14F11-47 (.diamond-solid.), and a murine IgG control (.circle-solid.) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0042] FIG. 26 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies Hu01G06-122 (.Math.), Hu01G06-127 (□), Hu01G06-135 (◇), Hu01G06-138 (.square-solid.), and Hu01G06-146 (*), and a human IgG control (.circle-solid.) dosed at 2 mg/kg in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0043] FIG. 27 is a graph summarizing results from an experiment to measure cachectic inhibitory activity of anti-GDF15 antibodies Hu01G06-122 (.Math.), Hu01G06-127 (□), Hu01G06-135 (◇), Hu01G06-138 (.square-solid.), and Hu01G06-146 (*), and a human IgG control (.circle-solid.) dosed at 2 mg/kg in an mFc-rhGDF15 cachectic model in ICR-SCID mice. The arrow indicates intra-peritoneal injection of antibody.

[0044] FIG. 28 is a graph summarizing results from an experiment to measure cachectic dose response inhibitory activity of anti-GDF15 antibodies Hu01G06-127 dosed at 20 mg/kg (□), 2 mg/kg (Δ), and 0.2 mg/kg (∇); Hu01G06-135 at 20 mg/kg (.square-solid.), 2 mg/kg (.box-tangle-solidup.), and 0.2 mg/kg (┐), and a human IgG control at 20 mg/kg (.circle-solid.) in an HT-1080 fibrosarcoma tumor xenograft model in ICR-SCID mice. The arrow indicates intravenous injection of antibody.

[0045] FIGS. 29A-29C are graphs summarizing results from an experiment to demonstrate anti-cachectic activity of anti-GDF15 antibodies Hu01G06-127 (.square-solid.), dosed at 10 mg/kg, in immune-incompetent mice (ICR-SCID) bearing an HT-1080 fibrosarcoma tumor xenograft model. Treatment with antibody Hu01G06-127 reversed body weight loss (FIG. 29A); induced a gain of gonadal fat mass (FIG. 29B); and induced a gain of muscle mass of gastrocnemius muscle (FIG. 29C) compared to negative control (hIgG (.circle-solid.); FIG. 29A) similar to levels found in non tumor bearing mice (SHAM (.box-tangle-solidup.); FIG. 29A). The arrows in FIG. 29A indicate intra-peritoneal injection of antibody.

DETAILED DESCRIPTION

[0046] The anti-GDF15 antibodies disclosed herein are based on the antigen binding sites of certain monoclonal antibodies that have been selected on the basis of binding and neutralization of human GDF15 (hGDF15). The antibodies contain immunoglobulin variable region CDR sequences that define a binding site for hGDF15.

[0047] By virtue of the neutralizing activity of these antibodies, they are useful for treating cachexia and/or sarcopenia. For use as therapeutic agents, the antibodies can be engineered to minimize or eliminate an immune response when administered to a human patient. Various features and aspects of the invention are discussed in more detail below.

[0048] As used herein, “cachexia” means a metabolic syndrome associated with underlying disease and characterized by involuntary loss of muscle mass. Cachexia is often accompanied by involuntary weight loss, loss of fat mass, anorexia, inflammation, insulin resistance, fatigue,

weakness, significant loss of appetite, and/or increased muscle protein breakdown. Cachexia is distinct from starvation, age-related loss of muscle mass, malabsorption, and hyperthyroidism. Underlying diseases associated with cachexia include cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.

[0049] As used herein, “sarcopenia” is understood to be a condition characterized primarily by loss of skeletal muscle mass and muscle strength. Sarcopenia is frequently associated with aging. See, Ruegg and Glass (2011) ANNUAL REV. PHARMACOL. TOXICOL. 51:373-395. In one approach, sarcopenia can be identified in a subject if a value of the appendicular skeletal muscle mass of a subject divided by the height of the subject in meters is more than two standard deviations below the young normal mean. (Thomas (2007) *supra*; see also Baumgartner et al. (1999) MECH. AGEING DEV. 147:755-763).

[0050] As used herein, unless otherwise indicated, “antibody” means an intact antibody (e.g., an intact monoclonal antibody) or antigen-binding fragment of an antibody, including an intact antibody or antigen-binding fragment that has been modified or engineered, or that is a human antibody. Examples of antibodies that have been modified or engineered are chimeric antibodies, humanized antibodies, and multispecific antibodies (e.g., bispecific antibodies). Examples of antigen-binding fragments include Fab, Fab', F(ab')₂, Fv, single chain antibodies (e.g., scFv), minibodies and diabodies.

I. Antibodies that Bind GDF15

[0051] The antibodies disclosed herein comprise: (a) an immunoglobulin heavy chain variable region comprising the structure CDR.sub.H1-CDR.sub.H2-CDR.sub.H3 and (b) an immunoglobulin light chain variable region comprising the structure CDR.sub.L1-CDR.sub.L2-CDR.sub.L3, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15 protein.

[0052] In some embodiments, the antibody comprises: (a) an immunoglobulin heavy chain variable region comprising the structure CDR.sub.H1-CDR.sub.H2-CDR.sub.H3 and (b) an immunoglobulin light chain variable region, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15. A CDR.sub.H1 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:1 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:2 (03G05), SEQ ID NO:3 (04F08), SEQ ID NO:4 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:6 (17B11); a CDR.sub.H2 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:8 (03G05), SEQ ID NO:9 (04F08, 06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), SEQ ID NO:10 (08G01), SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), SEQ ID NO:12 (17B11), SEQ ID NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and SEQ ID NO:14 (HE LM 06C11 IGHV2-70); and a CDR.sub.H3 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:15 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:16 (03G05), SEQ ID NO:17 (04F08), SEQ ID NO:18

(06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:20 (17B11). Throughout this specification, a particular SEQ ID NO. is followed in parentheses by the antibody that was the origin of that sequence. For example, "SEQ ID NO:2 (03G05)" means that SEQ ID NO:2 comes from antibody 03G05.

[0053] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-69 T30S I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0054] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:2 (03G05), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:8 (03G05), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:16 (03G05).

[0055] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:3 (04F08), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:9 (04F08), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:17 (04F08).

[0056] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:4 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:9 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:18 (06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5).

[0057] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (08G01), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:10 (08G01), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (08G01).

[0058] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70).

[0059] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:6 (17B11), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:12 (17B11), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:20 (17B11).

[0060] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1,

Hu01G06 IGHV1-69 F2).

[0061] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0062] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0063] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0064] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:1 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2).

[0065] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising the amino acid sequence of SEQ ID NO:4 (HE LM 06C11 IGHV2-70), a CDR.sub.H2 comprising the amino acid sequence of SEQ ID NO:14 (HE LM 06C11 IGHV2-70), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:18 (HE LM 06C11 IGHV2-70).

[0066] Preferably, the CDR.sub.H1, CDR.sub.H2, and CDR.sub.H3 sequences are interposed between fully human or humanized immunoglobulin FR sequences.

[0067] In some embodiments, the antibody comprises (a) an immunoglobulin light chain variable region comprising the structure CDR.sub.L1-CDR.sub.L2-CDR.sub.L3, and (b) an immunoglobulin heavy chain variable region, wherein the immunoglobulin light chain variable region and the immunoglobulin heavy chain variable region together define a single binding site for binding hGDF15. A CDR.sub.L1 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06

IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:22 (03G05), SEQ ID NO:23 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16, 14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), SEQ ID NO:24 (08G01), and SEQ ID NO:25 (17B11); a CDR.sub.L2 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:27 (03G05), SEQ ID NO:28 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:29 (08G01), SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:31 (17B11); and a CDR.sub.L3 comprises an amino acid sequence selected from the group consisting of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, 08G01, Hu01G06 IGKV1-39 F1), SEQ ID NO:244 (Hu01G06 IGKV1-39 F2), SEQ ID NO:33 (03G05), SEQ ID NO:34 (04F08), SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:37 (17B11).

[0068] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1).

[0069] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0070] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:22 (03G05), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:27 (03G05), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:33 (03G05).

[0071] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:23 (04F08), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:28 (04F08), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:34 (04F08).

[0072] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:23 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:28 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16).

[0073] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:24 (08G01), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:29 (08G01), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:32 (08G01).

[0074] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:23 (14F11,

Ch14F11 Chimeric, Hu14F11 IGKV1-16), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16).

[0075] In some embodiments, the antibody comprises an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:25 (17B11), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:31 (17B11), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:37 (17B11).

[0076] Preferably, the CDR.sub.L1, CDR.sub.L2, and CDR.sub.L3 sequences are interposed between fully human or humanized immunoglobulin FR sequences.

[0077] In some embodiments, the antibody comprises: (a) an immunoglobulin heavy chain variable region comprising the structure CDR.sub.H1-CDR.sub.H2-CDR.sub.H3 and (b) an immunoglobulin light chain variable region comprising the structure CDR.sub.L1-CDR.sub.L2-CDR.sub.L3, wherein the heavy chain variable region and the light chain variable region together define a single binding site for binding hGDF15. The CDR.sub.H1 is an amino acid sequence selected from the group consisting of SEQ ID NO:1 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:2 (03G05), SEQ ID NO:3 (04F08), SEQ ID NO:4 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:5 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:6 (17B11); the CDR.sub.H2 is an amino acid sequence selected from the group consisting of SEQ ID NO:7 (01G06, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:8 (03G05), SEQ ID NO:9 (04F08, 06C11, Ch06C11 Chimeric, Hu06C11 IGHV2-5), SEQ ID NO:10 (08G01), SEQ ID NO:11 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), SEQ ID NO:12 (17B11), SEQ ID NO:13 (Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:236 (Hu01G06 IGHV1-18 F1), SEQ ID NO:237 (Hu01G06 IGHV1-18 F2), SEQ ID NO:238 (Hu01G06 IGHV1-69 F1), SEQ ID NO:239 (Hu01G06 IGHV1-69 F2), and SEQ ID NO:14 (HE LM 06C11 IGHV2-70); and the CDR.sub.H3 is an amino acid sequence selected from the group consisting of SEQ ID NO:15 (01G06, 08G01, Ch01G06 Chimeric, Hu01G06 IGHV1-18, Hu01G06 IGHV1-69, Sh01G06 IGHV1-18 M69L, Sh01G06 IGHV1-18 M69L K64Q G44S, Sh01G06 IGHV1-18 M69L K64Q, Sh01G06 IGHV1-69 T30S I69L, Sh01G06 IGHV1-69 T30S K64Q I69L, Hu01G06 IGHV1-18 F1, Hu01G06 IGHV1-18 F2, Hu01G06 IGHV1-69 F1, Hu01G06 IGHV1-69 F2), SEQ ID NO:16 (03G05), SEQ ID NO:17 (04F08), SEQ ID NO:18 (06C11, Ch06C11 Chimeric, HE LM 06C11 IGHV2-70, Hu06C11 IGHV2-5), SEQ ID NO:19 (14F11, Ch14F11 Chimeric, Sh14F11 IGHV2-5, Sh14F11 IGHV2-70), and SEQ ID NO:20 (17B11). The CDR.sub.L1 is an amino acid sequence selected from the group consisting of SEQ ID NO:21 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:22 (03G05), SEQ ID NO:23 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16, 14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), SEQ ID NO:24 (08G01), and SEQ ID NO:25 (17B11); the CDR.sub.L2 is an amino acid sequence selected from the group consisting of SEQ ID NO:26 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, Hu01G06 IGKV1-39 F1, Hu01G06 IGKV1-39 F2), SEQ ID NO:27 (03G05), SEQ ID NO:28 (04F08, 06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:29 (08G01), SEQ ID NO:30 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:31 (17B11); and the CDR.sub.L3 is an amino acid sequence selected from the group consisting of SEQ ID NO:32 (01G06, Ch01G06 Chimeric, Hu01G06 IGKV1-39, Hu01G06 IGKV1-39 S43A V48I, Hu01G06 IGKV1-39 V48I, 08G01, Hu01G06 IGKV1-39 F1), SEQ ID

NO:244 (Hu01G06 IGKV1-39 F2), SEQ ID NO:33 (03G05), SEQ ID NO:34 (04F08), SEQ ID NO:35 (06C11, Ch06C11 Chimeric, Sh06C11 IGKV1-16), SEQ ID NO:36 (14F11, Ch14F11 Chimeric, Hu14F11 IGKV1-16), and SEQ ID NO:37 (17B11).

[0078] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:38 (Hu01G06 IGHV1-18 F1), a CDR.sub.H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:236 and SEQ ID NO:240 (Hu01G06 IGHV1-18 F1), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-18 F1); and an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0079] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:38 (Hu01G06 IGHV1-18 F2), a CDR.sub.H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:237 and SEQ ID NO:241 (Hu01G06 IGHV1-18 F2), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-18 F2); and an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F2), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F2), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0080] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F1), a CDR.sub.H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:238 and SEQ ID NO:241 (Hu01G06 IGHV1-69 F1), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F1); and an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0081] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:239 and SEQ ID NO:240 (Hu01G06 IGHV1-69 F2), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F2); and an immunoglobulin light chain variable region comprising a CDR.sub.L1 comprising the amino acid sequence of SEQ ID NO:21 (Hu01G06 IGKV1-39 F1), a CDR.sub.L2 comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F1), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:32 (Hu01G06 IGKV1-39 F1).

[0082] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising a CDR.sub.H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID NO:234 (Hu01G06 IGHV1-69 F2), a CDR.sub.H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:239 and SEQ ID NO:240 (Hu01G06 IGHV1-69 F2), and a CDR.sub.H3 comprising the amino acid sequence of SEQ ID NO:15 (Hu01G06 IGHV1-69 F2); and an immunoglobulin light chain variable region comprising a CDR.sub.Li comprising the amino acid sequence of SEQ ID NO:21

(Hu01G06 IGKV1-39 F2), a CDR.L2 comprising the amino acid sequence of SEQ ID NO:26 (Hu01G06 IGKV1-39 F2), and a CDR.sub.L3 comprising the amino acid sequence of SEQ ID NO:244 (Hu01G06 IGKV1-39 F2).

[0083] The antibodies disclosed herein comprise an immunoglobulin heavy chain variable region and an immunoglobulin light chain variable region. In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region selected from the group consisting of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70); and an immunoglobulin light chain variable region.

[0084] In other embodiments, the antibody comprises an immunoglobulin light chain variable region selected from the group consisting of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:96 (Sh06C11 IGKV1-16), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), and SEQ ID NO:98 (Hu14F11 IGKV1-16), and an immunoglobulin heavy chain variable region.

[0085] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region selected from the group consisting of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region selected from the group consisting of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:96 (Sh06C11 IGKV1-16), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), and SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0086] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (01G06, Ch01G06 Chimeric).

[0087] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:42 (03G05), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:78 (03G05).

[0088] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0089] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (06C11), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (06C11).

[0090] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:48 (08G01), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:84 (08G01).

[0091] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (14F11), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (14F11).

[0092] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:52 (17B11), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:88 (17B11).

[0093] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:54 (Hu01G06 IGHV1-18), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0094] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:56 (Hu01G06 IGHV1-69), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0095] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0096] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0097] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0098] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0099] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0100] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0101] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:54 (Hu01G06 IGHV1-18), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90

(Hu01G06 IGKV1-39).

[0102] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:56 (Hu01G06 IGHV1-69), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0103] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0104] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0105] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0106] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0107] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:90 (Hu01G06 IGKV1-39).

[0108] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0109] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:54 (Hu01G06 IGHV1-18), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0110] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:56 (Hu01G06 IGHV1-69), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0111] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0112] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0113] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0114] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable

region comprising the amino acid sequence of SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0115] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I).

[0116] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0117] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:54 (Hu01G06 IGHV1-18), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0118] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:56 (Hu01G06 IGHV1-69), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0119] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0120] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0121] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0122] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0123] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:94 (Hu01G06 IGKV1-39 V48I).

[0124] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 F1).

[0125] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:254 (Hu01G06 IGKV1-39 F2).

[0126] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92

(Hu01G06 IGKV1-39 F1).

[0127] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:92 (Hu01G06 IGKV1-39 F1).

[0128] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:254 (Hu01G06 IGKV1-39 F2).

[0129] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:68 (HE LM 06C11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0130] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:70 (Hu06C11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0131] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (Ch06C11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:96 (Sh06C11 IGKV1-16).

[0132] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:68 (HE LM 06C11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:96 (Sh06C11 IGKV1-16).

[0133] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:70 (Hu06C11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:96 (Sh06C11 IGKV1-16).

[0134] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:72 (Sh14F11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0135] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0136] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0137] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:72 (Sh14F11 IGHV2-5), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0138] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:74 (Sh14F11 IGHV2-70), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0139] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable

region comprising the amino acid sequence of SEQ ID NO:46 (Ch06C11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0140] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:80 (04F08).

[0141] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0142] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:50 (Ch14F11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:82 (Ch06C11 Chimeric).

[0143] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:44 (04F08), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0144] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:46 (Ch06C11 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:86 (Ch14F11 Chimeric).

[0145] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:48 (08G01), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:76 (Ch01G06 Chimeric).

[0146] In some embodiments, the antibody comprises an immunoglobulin heavy chain variable region comprising the amino acid sequence of SEQ ID NO:40 (Ch01G06 Chimeric), and an immunoglobulin light chain variable region comprising the amino acid sequence of SEQ ID NO:84 (08G01).

[0147] In certain embodiments, the antibodies disclosed herein comprise an immunoglobulin heavy chain and an immunoglobulin light chain. In some embodiments, the antibody comprises an immunoglobulin heavy chain selected from the group consisting of SEQ ID NO:100 (01G06), SEQ ID NO:104 (03G05), SEQ ID NO:108 (04F08), SEQ ID NO:112 (06C11), SEQ ID NO:116 (08G01), SEQ ID NO:120 (14F11), SEQ ID NO:124 (17B11), SEQ ID NO:176 (Ch01G06 Chimeric), SEQ ID NO:178 (Hu01G06 IGHV1-18), SEQ ID NO:180 (Hu01G06 IGHV1-69), SEQ ID NO:182 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:186 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:190 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), SEQ ID NO:258 (Hu01G06 IGHV1-18 F2), SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), SEQ ID NO:192 (Ch06C11 Chimeric), SEQ ID NO:194 (HE LM 06C11 IGHV2-70), SEQ ID NO:196 (Hu06C11 IGHV2-5), SEQ ID NO:198 (Ch14F11 Chimeric), SEQ ID NO:200 (Sh14F11 IGHV2-5), and SEQ ID NO:202 (Sh14F11 IGHV2-70); and an immunoglobulin light chain.

[0148] In other embodiments, the antibody comprises an immunoglobulin light chain selected from the group consisting of SEQ ID NO:102 (01G06), SEQ ID NO:106 (03G05), SEQ ID NO:110 (04F08), SEQ ID NO:114 (06C11), SEQ ID NO:118 (08G01), SEQ ID NO:122 (14F11), SEQ ID NO:126 (17B11), SEQ ID NO:204 (Ch01G06 Chimeric), SEQ ID NO:206 (Hu01G06 IGKV1-39), SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:210

(Hu01G06 IGKV1-39 V48I), SEQ ID NO:264 (Hu01G06 IGKV1-39 F2), SEQ ID NO:212 (Ch06C11 Chimeric), SEQ ID NO:214 (Sh06C11 IGKV1-16), SEQ ID NO:216 (Ch14F11 Chimeric), and SEQ ID NO:218 (Hu14F11 IGKV1-16), and an immunoglobulin heavy chain.

[0149] In some embodiments, the antibody comprises (i) an immunoglobulin heavy chain selected from the group consisting of SEQ ID NO:100 (01G06), SEQ ID NO:104 (03G05), SEQ ID NO:108 (04F08), SEQ ID NO:112 (06C11), SEQ ID NO:116 (08G01), SEQ ID NO:120 (14F11), SEQ ID NO:124 (17B11), SEQ ID NO:176 (Ch01G06 Chimeric), SEQ ID NO:178 (Hu01G06 IGHV1-18), SEQ ID NO:180 (Hu01G06 IGHV1-69), SEQ ID NO:182 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:186 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:190 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), SEQ ID NO:258 (Hu01G06 IGHV1-18 F2), SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), SEQ ID NO:192 (Ch06C11 Chimeric), SEQ ID NO:194 (HE LM 06C11 IGHV2-70), SEQ ID NO:196 (Hu06C11 IGHV2-5), SEQ ID NO:198 (Ch14F11 Chimeric), SEQ ID NO:200 (Sh14F11 IGHV2-5), and SEQ ID NO:202 (Sh14F11 IGHV2-70), and (ii) an immunoglobulin light chain selected from the group consisting of SEQ ID NO:102 (01G06), SEQ ID NO:106 (03G05), SEQ ID NO:110 (04F08), SEQ ID NO:114 (06C11), SEQ ID NO:118 (08G01), SEQ ID NO:122 (14F11), SEQ ID NO:126 (17B11), SEQ ID NO:204 (Ch01G06 Chimeric), SEQ ID NO:206 (Hu01G06 IGKV1-39), SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:264 (Hu01G06 IGKV1-39 F2), SEQ ID NO:212 (Ch06C11 Chimeric), SEQ ID NO:214 (Sh06C11 IGKV1-16), SEQ ID NO:216 (Ch14F11 Chimeric), and SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0150] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:176 (Ch01G06 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:204 (Ch01G06 Chimeric).

[0151] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:192 (Ch06C11 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:212 (Ch06C11 Chimeric).

[0152] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:198 (Ch14F11 Chimeric), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:216 (Ch14F11 Chimeric).

[0153] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:178 (Hu01G06 IGHV1-18), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:206 (Hu01G06 IGKV1-39).

[0154] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:180 (Hu01G06 IGHV1-69), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:206 (Hu01G06 IGKV1-39).

[0155] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I).

[0156] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:210 (Hu01G06 IGKV1-39 V48I).

[0157] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:184 (Sh01G06 IGHV1-18 M69L K64Q G44S), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I).

[0158] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising

the amino acid sequence of SEQ ID NO:188 (Sh01G06 IGHV1-69 T30S I69L), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 S43A V48I).

[0159] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:256 (Hu01G06 IGHV1-18 F1), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0160] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:258 (Hu01G06 IGHV1-18 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:264 (Hu01G06 IGKV1-39 F2).

[0161] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:260 (Hu01G06 IGHV1-69 F1), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0162] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:208 (Hu01G06 IGKV1-39 F1).

[0163] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:262 (Hu01G06 IGHV1-69 F2), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:264 (Hu01G06 IGKV1-39 F2).

[0164] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:194 (HE LM 06C11 IGHV2-70), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:214 (Sh06C11 IGKV1-16).

[0165] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:196 (Hu06C11 IGHV2-5), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:214 (Sh06C11 IGKV1-16).

[0166] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:200 (Sh14F11 IGHV2-5), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0167] In some embodiments, the antibody comprises an immunoglobulin heavy chain comprising the amino acid sequence of SEQ ID NO:202 (Sh14F11 IGHV2-70), and an immunoglobulin light chain comprising the amino acid sequence of SEQ ID NO:218 (Hu14F11 IGKV1-16).

[0168] In certain embodiments, an isolated antibody that binds hGDF15 comprises an immunoglobulin heavy chain variable region comprising an amino acid sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to the entire variable region or the FR sequence of SEQ ID NO:40 (01G06, Ch01G06 Chimeric), SEQ ID NO:42 (03G05), SEQ ID NO:44 (04F08), SEQ ID NO:46 (06C11, Ch06C11 Chimeric), SEQ ID NO:48 (08G01), SEQ ID NO:50 (14F11, Ch14F11 Chimeric), SEQ ID NO:52 (17B11), SEQ ID NO:54 (Hu01G06 IGHV1-18), SEQ ID NO:56 (Hu01G06 IGHV1-69), SEQ ID NO:58 (Sh01G06 IGHV1-18 M69L), SEQ ID NO:60 (Sh01G06 IGHV1-18 M69L K64Q G44S), SEQ ID NO:62 (Sh01G06 IGHV1-18 M69L K64Q), SEQ ID NO:64 (Sh01G06 IGHV1-69 T30S I69L), SEQ ID NO:66 (Sh01G06 IGHV1-69 T30S K64Q I69L), SEQ ID NO:246 (Hu01G06 IGHV1-18 F1), SEQ ID NO:248 (Hu01G06 IGHV1-18 F2), SEQ ID NO:250 (Hu01G06 IGHV1-69 F1), SEQ ID NO:252 (Hu01G06 IGHV1-69 F2), SEQ ID NO:68 (HE LM 06C11 IGHV2-70), SEQ ID NO:70 (Hu06C11 IGHV2-5), SEQ ID NO:72 (Sh14F11 IGHV2-5), and SEQ ID NO:74 (Sh14F11 IGHV2-70).

[0169] In certain embodiments, an isolated antibody that binds hGDF15 comprises an immunoglobulin light chain variable region comprising an amino acid sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to the entire variable region or the FR sequence of SEQ ID NO:76 (01G06, Ch01G06 Chimeric), SEQ ID NO:78 (03G05), SEQ ID NO:80 (04F08), SEQ ID NO:82 (06C11, Ch06C11 Chimeric), SEQ ID NO:84 (08G01), SEQ ID NO:86 (14F11, Ch14F11 Chimeric), SEQ ID NO:88 (17B11), SEQ ID NO:90 (Hu01G06 IGKV1-39), SEQ ID NO:92 (Hu01G06 IGKV1-39 S43A V48I or Hu01G06 IGKV1-39 F1), SEQ ID

NO:96 (Hu01G06 IGKV1-39 V48I), SEQ ID NO:254 (Hu01G06 IGKV1-39 F2), SEQ ID NO:96 (Sh06C11 IGKV1-16), and SEQ ID NO:98 (Hu14F11 IGKV1-16).

[0170] Sequence identity may be determined in various ways that are within the skill of a person skilled in the art, e.g., using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. BLAST (Basic Local Alignment Search Tool) analysis using the algorithm employed by the programs blastp, blastn, blastx, tblastn and tblastx (Karlin et al., (1990) PROC. NATL. ACAD. SCI. USA 87:2264-2268; Altschul, (1993) J. MOL. EVOL. 36:290-300; Altschul et al., (1997) NUCLEIC ACIDS RES. 25:3389-3402, incorporated by reference herein) are tailored for sequence similarity searching. For a discussion of basic issues in searching sequence databases see Altschul et al., (1994) NATURE GENETICS 6:119-129, which is fully incorporated by reference herein. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. The search parameters for histogram, descriptions, alignments, expect (i.e., the statistical significance threshold for reporting matches against database sequences), cutoff, matrix and filter are at the default settings. The default scoring matrix used by blastp, blastx, tblastn, and tblastx is the BLOSUM62 matrix (Henikoff et al., (1992) PROC. NATL. ACAD. SCI. USA 89:10915-10919, fully incorporated by reference herein). Four blastn parameters may be adjusted as follows: Q=10 (gap creation penalty); R=10 (gap extension penalty); wink=1 (generates word hits at every wink.sup.th position along the query); and gapw=16 (sets the window width within which gapped alignments are generated). The equivalent Blastp parameter settings may be Q=9; R=2; wink=1; and gapw=32. Searches may also be conducted using the NCBI (National Center for Biotechnology Information) BLAST Advanced Option parameter (e.g.: -G, Cost to open gap [Integer]: default=5 for nucleotides/11 for proteins; -E, Cost to extend gap [Integer]: default=2 for nucleotides/1 for proteins; -q, Penalty for nucleotide mismatch [Integer]: default=-3; -r, reward for nucleotide match [Integer]: default=1; -e, expect value [Real]: default=10; -W, wordsize [Integer]: default=11 for nucleotides/28 for megablast/3 for proteins; -y, Dropoff (X) for blast extensions in bits: default=20 for blastn/7 for others; -X, X dropoff value for gapped alignment (in bits): default=15 for all programs, not applicable to blastn; and -Z, final X dropoff value for gapped alignment (in bits): 50 for blastn, 25 for others). ClustalW for pairwise protein alignments may also be used (default parameters may include, e.g., Blosum62 matrix and Gap Opening Penalty=10 and Gap Extension Penalty=0.1). A Bestfit comparison between sequences, available in the GCG package version 10.0, uses DNA parameters GAP=50 (gap creation penalty) and LEN=3 (gap extension penalty). The equivalent settings in Bestfit protein comparisons are GAP=8 and LEN=2.

[0171] In each of the foregoing embodiments, it is contemplated herein that immunoglobulin heavy chain variable region sequences and/or light chain variable region sequences that together bind human GDF15 may contain amino acid alterations (e.g., at least 1, 2, 3, 4, 5, or 10 amino acid substitutions, deletions, or additions) in the framework regions of the heavy and/or light chain variable regions.

[0172] In some embodiments, the antibody binds hGDF15 with a K.sub.D of about 300 pM, 250 pM, 200 pM, 190 pM, 180 pM, 170 pM, 160 pM, 150 pM, 140 pM, 130 pM, 120 pM, 110 pM, 100 pM, 90 pM, 80 pM, 70 pM, 60 pM, 50 pM, 40 pM, 30 pM, 20 pM, or 10 pM, or lower. Unless otherwise specified, K.sub.D values are determined by surface plasmon resonance methods or biolayer interferometry under the conditions described in Examples 8, 14, and 15.

[0173] In some embodiments, a monoclonal antibody binds to the same epitope on hGDF15 (e.g., mature hGDF15 or cleaved rhGDF15) bound by one or more of the antibodies disclosed herein (e.g., antibodies 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11). In some embodiments, a monoclonal antibody competes for binding to hGDF15 with one or more of the antibodies disclosed herein (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11).

[0174] Competition assays for determining whether an antibody binds to the same epitope as, or

competes for binding with, an anti-GDF15 antibody disclosed herein are known in the art. Exemplary competition assays include immunoassays (e.g., ELISA assays, RIA assays), surface plasmon resonance analysis (e.g., using a BIAcore™ instrument), biolayer interferometry and flow cytometry.

[0175] Typically, a competition assay involves the use of an antigen (e.g., a hGDF15 protein or fragment thereof) bound to a solid surface or expressed on a cell surface, a test anti-GDF15-binding antibody and a reference antibody (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11). The reference antibody is labeled and the test antibody is unlabeled. Competitive inhibition is measured by determining the amount of labeled reference antibody bound to the solid surface or cells in the presence of the test antibody. Usually the test antibody is present in excess (e.g., 1×, 5×, 10×, 20× or 100×). Antibodies identified by competition assay (i.e., competing antibodies) include antibodies binding to the same epitope, or similar (e.g., overlapping) epitopes, as the reference antibody, and antibodies binding to an adjacent epitope sufficiently proximal to the epitope bound by the reference antibody for steric hindrance to occur.

[0176] In an exemplary competition assay, a reference anti-GDF15 antibody (e.g., antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) is biotinylated using commercially available reagents. The biotinylated reference antibody is mixed with serial dilutions of the test antibody or unlabeled reference antibody (self-competition control) resulting in a mixture of various molar ratios (e.g., 1×, 5×, 10×, 20× or 100×) of test antibody (or unlabeled reference antibody) to labeled reference antibody. The antibody mixture is added to a hGDF15 polypeptide coated-ELISA plate. The plate is then washed, and horseradish peroxidase (HRP)-streptavidin is added to the plate as the detection reagent. The amount of labeled reference antibody bound to the target antigen is detected following addition of a chromogenic substrate (e.g., TMB (3,3',5,5'-tetramethylbenzidine) or ABTS (2,2''-azino-di-(3-ethylbenzthiazoline-6-sulfonate)), which are known in the art. Optical density readings (OD units) are measured using a SpectraMax® M2 spectrometer (Molecular Devices). OD units corresponding to zero percent inhibition are determined from wells without any competing antibody. OD units corresponding to 100% inhibition, i.e., the assay background are determined from wells without any labeled reference antibody or test antibody. Percent inhibition of labeled reference antibody to GDF15 by the test antibody (or the unlabeled reference antibody) at each concentration is calculated as follows: % inhibition=(1-(OD units-100% inhibition)/(0% inhibition-100% inhibition))*100. Persons skilled in the art will appreciate that the competition assay can be performed using various detection systems known in the art.

[0177] A competition assay may be conducted in both directions to ensure that the presence of the label does not interfere or otherwise inhibit binding. For example, in the first direction the reference antibody is labeled and the test antibody is unlabeled, and in the second direction, the test antibody is labeled and the reference antibody is unlabeled.

[0178] A test antibody competes with the reference antibody for specific binding to the antigen if an excess of one antibody (e.g., 1×, 5×, 10×, 20× or 100×) inhibits binding of the other antibody, e.g., by at least 50%, 75%, 90%, 95% or 99%, as measured in a competitive binding assay.

[0179] Two antibodies bind to the same epitope if essentially all amino acid mutations in the antigen that reduce or eliminate binding of one antibody reduce or eliminate binding of the other. Two antibodies bind to overlapping epitopes if only a subset of the amino acid mutations that reduce or eliminate binding of one antibody reduce or eliminate binding of the other.

II. Production of Antibodies

[0180] Methods for producing antibodies, such as those disclosed herein, are known in the art. For example, DNA molecules encoding light chain variable regions and/or heavy chain variable regions can be chemically synthesized using the sequence information provided herein. Synthetic DNA molecules can be ligated to other appropriate nucleotide sequences, including, e.g., constant region coding sequences, and expression control sequences, to produce conventional gene expression constructs encoding the desired antibodies. Production of defined gene constructs is within routine

skill in the art. Alternatively, the sequences provided herein can be cloned out of hybridomas by conventional hybridization techniques or polymerase chain reaction (PCR) techniques, using synthetic nucleic acid probes whose sequences are based on sequence information provided herein, or prior art sequence information regarding genes encoding the heavy and light chains of murine antibodies in hybridoma cells.

[0181] Nucleic acids encoding desired antibodies can be incorporated (ligated) into expression vectors, which can be introduced into host cells through conventional transfection or transformation techniques. Exemplary host cells are *E. coli* cells, Chinese hamster ovary (CHO) cells, human embryonic kidney 293 (HEK 293) cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS), human hepatocellular carcinoma cells (e.g., Hep G2), and myeloma cells that do not otherwise produce IgG protein. Transformed host cells can be grown under conditions that permit the host cells to express the genes that encode the immunoglobulin light and/or heavy chain variable regions.

[0182] Specific expression and purification conditions will vary depending upon the expression system employed. For example, if a gene is to be expressed in *E. coli*, it is first cloned into an expression vector by positioning the engineered gene downstream from a suitable bacterial promoter, e.g., Trp or Tac, and a prokaryotic signal sequence. The expressed secreted protein accumulates in refractile or inclusion bodies, and can be harvested after disruption of the cells by French press or sonication. The refractile bodies then are solubilized, and the proteins refolded and cleaved by methods known in the art.

[0183] If the engineered gene is to be expressed in eukaryotic host cells, e.g., CHO cells, it is first inserted into an expression vector containing a suitable eukaryotic promoter, a secretion signal, a poly A sequence, and a stop codon. Optionally, the vector or gene construct may contain enhancers and introns. This expression vector optionally contains sequences encoding all or part of a constant region, enabling an entire, or a part of, a heavy or light chain to be expressed. The gene construct can be introduced into eukaryotic host cells using conventional techniques. The host cells express V.sub.L or V.sub.H fragments, V.sub.L-V.sub.H heterodimers, V.sub.H-V.sub.L or V.sub.L-V.sub.H single chain polypeptides, complete heavy or light immunoglobulin chains, or portions thereof, each of which may be attached to a moiety having another function (e.g., cytotoxicity). In some embodiments, a host cell is transfected with a single vector expressing a polypeptide expressing an entire, or part of, a heavy chain (e.g., a heavy chain variable region) or a light chain (e.g., a light chain variable region). In some embodiments, a host cell is transfected with a single vector encoding (a) a polypeptide comprising a heavy chain variable region and a polypeptide comprising a light chain variable region, or (b) an entire immunoglobulin heavy chain and an entire immunoglobulin light chain. In some embodiments, a host cell is co-transfected with more than one expression vector (e.g., one expression vector expressing a polypeptide comprising an entire, or part of, a heavy chain or heavy chain variable region, and another expression vector expressing a polypeptide comprising an entire, or part of, a light chain or light chain variable region).

[0184] A polypeptide comprising an immunoglobulin heavy chain variable region or light chain variable region can be produced by growing (culturing) a host cell transfected with an expression vector encoding such a variable region, under conditions that permit expression of the polypeptide. Following expression, the polypeptide can be harvested and purified or isolated using techniques known in the art, e.g., affinity tags such as glutathione-S-transferase (GST) or histidine tags.

[0185] A monoclonal antibody that binds hGDF15, or an antigen-binding fragment of the antibody, can be produced by growing (culturing) a host cell transfected with: (a) an expression vector that encodes a complete or partial immunoglobulin heavy chain, and a separate expression vector that encodes a complete or partial immunoglobulin light chain; or (b) a single expression vector that encodes both chains (e.g., complete or partial heavy and light chains), under conditions that permit expression of both chains. The intact antibody (or antigen-binding fragment) can be harvested and purified or isolated using techniques known in the art, e.g., Protein A, Protein G, affinity tags such

as glutathione-S-transferase (GST) or histidine tags. It is within ordinary skill in the art to express the heavy chain and the light chain from a single expression vector or from two separate expression vectors.

III. Antibody Modifications

[0186] Methods for reducing or eliminating the antigenicity of antibodies and antibody fragments are known in the art. When the antibodies are to be administered to a human, the antibodies preferably are “humanized” to reduce or eliminate antigenicity in humans. Preferably, each humanized antibody has the same or substantially the same affinity for the antigen as the non-humanized mouse antibody from which it was derived.

[0187] In one humanization approach, chimeric proteins are created in which mouse immunoglobulin constant regions are replaced with human immunoglobulin constant regions. See, e.g., Morrison et al., 1984, PROC. NAT. ACAD. SCI. 81:6851-6855, Neuberger et al., 1984, NATURE 312:604-608; U.S. Pat. No. 6,893,625 (Robinson); U.S. Pat. No. 5,500,362 (Robinson); and U.S. Pat. No. 4,816,567 (Cabilly).

[0188] In an approach known as CDR grafting, the CDRs of the light and heavy chain variable regions are grafted into frameworks from another species. For example, murine CDRs can be grafted into human FRs. In some embodiments, the CDRs of the light and heavy chain variable regions of an anti-GDF15 antibody are grafted into human FRs or consensus human FRs. To create consensus human FRs, FRs from several human heavy chain or light chain amino acid sequences are aligned to identify a consensus amino acid sequence. CDR grafting is described in U.S. Pat. No. 7,022,500 (Queen); U.S. Pat. No. 6,982,321 (Winter); U.S. Pat. No. 6,180,370 (Queen); U.S. Pat. No. 6,054,297 (Carter); U.S. Pat. No. 5,693,762 (Queen); U.S. Pat. No. 5,859,205 (Adair); U.S. Pat. No. 5,693,761 (Queen); U.S. Pat. No. 5,565,332 (Hoogenboom); U.S. Pat. No. 5,585,089 (Queen); U.S. Pat. No. 5,530,101 (Queen); Jones et al. (1986) NATURE 321: 522-525; Riechmann et al. (1988) NATURE 332: 323-327; Verhoeven et al. (1988) SCIENCE 239: 1534-1536; and Winter (1998) FEBS LETT 430: 92-94.

[0189] In an approach called “SUPERHUMANIZATION™,” human CDR sequences are chosen from human germline genes, based on the structural similarity of the human CDRs to those of the mouse antibody to be humanized. See, e.g., U.S. Pat. No. 6,881,557 (Foote); and Tan et al., 2002, J. IMMUNOL. 169:1119-1125.

[0190] Other methods to reduce immunogenicity include “reshaping,” “hyperchimerization,” and “veneering/resurfacing.” See, e.g., Vaswami et al., 1998, ANNALS OF ALLERGY, ASTHMA, & IMMUNOL. 81:105; Roguska et al., 1996, PROT. ENGINEER 9:895-904; and U.S. Pat. No. 6,072,035 (Hardman). In the veneering/resurfacing approach, the surface accessible amino acid residues in the murine antibody are replaced by amino acid residues more frequently found at the same positions in a human antibody. This type of antibody resurfacing is described, e.g., in U.S. Pat. No. 5,639,641 (Pedersen).

[0191] Another approach for converting a mouse antibody into a form suitable for medical use in humans is known as ACTIVMAB™ technology (Vaccinex, Inc., Rochester, NY), which involves a vaccinia virus-based vector to express antibodies in mammalian cells. High levels of combinatorial diversity of IgG heavy and light chains are said to be produced. See, e.g., U.S. Pat. No. 6,706,477 (Zauderer); U.S. Pat. No. 6,800,442 (Zauderer); and U.S. Pat. No. 6,872,518 (Zauderer).

[0192] Another approach for converting a mouse antibody into a form suitable for use in humans is technology practiced commercially by KaloBios Pharmaceuticals, Inc. (Palo Alto, CA). This technology involves the use of a proprietary human “acceptor” library to produce an “epitope focused” library for antibody selection.

[0193] Another approach for modifying a mouse antibody into a form suitable for medical use in humans is HUMAN ENGINEERING™ technology, which is practiced commercially by XOMA (US) LLC. See, e.g., PCT Publication No. WO 93/11794 and U.S. Pat. No. 5,766,886 (Studnicka); U.S. Pat. No. 5,770,196 (Studnicka); U.S. Pat. No. 5,821,123 (Studnicka); and U.S. Pat. No.

5,869,619 (Studnicka).

[0194] Any suitable approach, including any of the above approaches, can be used to reduce or eliminate human immunogenicity of an antibody.

[0195] In addition, it is possible to create fully human antibodies in mice. Fully human mAbs lacking any non-human sequences can be prepared from human immunoglobulin transgenic mice by techniques referenced in, e.g., Lonberg et al., NATURE 368:856-859, 1994; Fishwild et al., NATURE BIOTECHNOLOGY 14:845-851, 1996; and Mendez et al., NATURE GENETICS 15:146-156, 1997. Fully human mAbs can also be prepared and optimized from phage display libraries by techniques referenced in, e.g., Knappik et al., J. MOL. BIOL. 296:57-86, 2000; and Krebs et al., J. Immunol. Meth. 254:67-84 2001).

IV. Therapeutic Uses

[0196] The antibodies disclosed herein can be used to treat a variety of disorders, for example, cachexia and/or sarcopenia. In some embodiments, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to inhibit the loss of muscle mass, for example, the loss of muscle mass associated with an underlying disease. Underlying diseases associated with cachexia include, but are not limited to, cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the disclosed antibodies inhibit loss of muscle mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%.

[0197] In some embodiments, a loss of muscle mass is accompanied by a loss of fat mass. The antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) may inhibit loss of fat mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%.

[0198] In other embodiments, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to treat one or more features accompanying cachexia and/or sarcopenia, e.g., involuntary body weight loss. In some embodiments, the antibodies revert involuntary body weight loss by at least 2%, 5%, 10%, 15%, 20%, 25%, 30% or 35%.

[0199] In another embodiment, the antibodies disclosed herein (e.g., 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) are used to inhibit loss of organ mass, for example, loss of organ mass associated with an underlying disease. Underlying diseases associated with cachexia include, but are not limited to, cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. In some embodiments, the disclosed antibodies inhibit loss of organ mass by at least 40%, 50%, 60%, 70%, 80%, 90%, 95%, 98%, 99%, or 100%. In some embodiments, loss of organ mass is observed in heart, liver, kidney, and/or spleen. In some embodiments, the loss of organ mass is accompanied by a loss of muscle mass, a loss of fat mass and/or involuntary weight loss.

[0200] Antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 can be used in therapy. For example, antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 can be used to treat cachexia and/or sarcopenia. Use of antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11 to treat cachexia and/or sarcopenia in a mammal comprises administering to the mammal a therapeutically effective amount of the antibody.

[0201] Sarcopenia, muscle wasting disorders and significant muscle weight loss may occur in the absence of cachexia, decreased appetite or body weight loss. In certain embodiments, therefore, one or more of the anti-GDF antibodies of the invention (for example, antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11, or 17B11) can be used to treat a subject suffering from, or who has been diagnosed with, sarcopenia, a muscle wasting disorder and/or significant muscle weight loss, whether or not the subject has, or has been diagnosed with, cachexia or decreased appetite. Such a method comprises administering a therapeutically effective amount of one or more antibodies of the invention to the subject in need thereof.

[0202] The Fc-rhGDF15 fusion proteins disclosed herein can be used to treat obesity. In some embodiments, the hFc-rhGDF15 fusion proteins disclosed herein are used to inhibit weight gain or

to reduce body weight by at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45% or 50%. Use of an hFc-hGDF15 fusion protein to treat obesity in a mammal comprises administering to the mammal a therapeutically effective amount of the fusion protein.

[0203] As used herein, “treat,” “treating” and “treatment” mean the treatment of a disease in a mammal, e.g., in a human. This includes: (a) inhibiting the disease, i.e., arresting its development; and (b) relieving the disease, i.e., causing regression of the disease state.

[0204] Generally, a therapeutically effective amount of an active component (e.g., an antibody or a fusion protein) is in the range of 0.1 mg/kg to 100 mg/kg, e.g., 1 mg/kg to 100 mg/kg, e.g., 1 mg/kg to 10 mg/kg, e.g., 2.0 mg/kg to 10 mg/kg. The amount administered will depend on variables such as the type and extent of disease or indication to be treated, the overall health of the patient, the in vivo potency of the antibody or fusion protein, the pharmaceutical formulation, the serum half-life of the antibody or fusion protein, and the route of administration. The initial dosage can be increased beyond the upper level in order to rapidly achieve the desired blood-level or tissue level. Alternatively, the initial dosage can be smaller than the optimum, and the dosage may be progressively increased during the course of treatment. Human dosage can be optimized, e.g., in a conventional Phase I dose escalation study designed to run from 0.5 mg/kg to 20 mg/kg. Dosing frequency can vary, depending on factors such as route of administration, dosage amount, serum half-life of the antibody or fusion protein, and the disease being treated. Exemplary dosing frequencies are once per day, once per week and once every two weeks. In some embodiments, dosing is once every two weeks. A preferred route of administration is parenteral, e.g., intravenous infusion. Formulation of monoclonal antibody-based drugs and fusion protein-based drugs are within ordinary skill in the art. In some embodiments, the antibody or fusion protein is lyophilized, and then reconstituted in buffered saline, at the time of administration. The effective amount of a second active agent, for example, an anti-cancer agent or the other agents discussed below, will also follow the principles discussed hereinabove and will be chosen so as to elicit the required therapeutic benefit in the patient.

[0205] For therapeutic use, an antibody preferably is combined with a pharmaceutically acceptable carrier. As used herein, “pharmaceutically acceptable carrier” means buffers, carriers, and excipients suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio. The carrier(s) should be “acceptable” in the sense of being compatible with the other ingredients of the formulations and not deleterious to the recipient. Pharmaceutically acceptable carriers include buffers, solvents, dispersion media, coatings, isotonic and absorption delaying agents, and the like, that are compatible with pharmaceutical administration. The use of such media and agents for pharmaceutically active substances is known in the art.

[0206] Pharmaceutical compositions containing antibodies or fusion proteins, such as those disclosed herein, can be presented in a dosage unit form and can be prepared by any suitable method. A pharmaceutical composition should be formulated to be compatible with its intended route of administration. Examples of routes of administration are intravenous (IV), intradermal, inhalation, transdermal, topical, transmucosal, and rectal administration. A preferred route of administration for monoclonal antibodies is IV infusion. Useful formulations can be prepared by methods known in the pharmaceutical art. For example, see *Remington's Pharmaceutical Sciences*, 18th ed. (Mack Publishing Company, 1990). Formulation components suitable for parenteral administration include a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl paraben; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as EDTA; buffers such as acetates, citrates or phosphates; and agents for the adjustment of tonicity such as sodium chloride or dextrose.

[0207] For intravenous administration, suitable carriers include physiological saline, bacteriostatic

water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). The carrier should be stable under the conditions of manufacture and storage, and should be preserved against microorganisms. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol), and suitable mixtures thereof.

[0208] Pharmaceutical formulations preferably are sterile. Sterilization can be accomplished, for example, by filtration through sterile filtration membranes. Where the composition is lyophilized, filter sterilization can be conducted prior to or following lyophilization and reconstitution.

[0209] In addition to the GDF15 (i.e., MIC-1/PLAB/PDF/NAG-1) pathway, other cytokines implicated in cachexia include Activin A and IL-6. Increased activin levels have been associated with cancer-associated cachexia and gonadal tumors. See, e.g., Marino et al. (2013) CYTOKINE & GROWTH FACTOR REV. 24:477-484. Activin A is a member of the TGF-beta family, and is a ligand of the activin type 2 receptor, ActRIIB. See, e.g., Zhou et al. (2010) CELL 142:531-543. Circulating levels of IL-6 have been shown to correlate with weight loss in cancer patients, as well as with reduced survival. See, e.g., Fearon et al. (2012) CELL METABOLISM 16:153-166.

[0210] Accordingly, in certain embodiments of the present invention, one or more inhibitors of Activin-A or the Activin-A receptor, ActRIIB, IL-6 or the IL-6 receptor (IL-6R), may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the present invention that inhibits GDF-15 activity. Exemplary inhibitors of Activin A or ActRIIB, include, for example, an anti-Activin-A antibody or an antigen binding fragment thereof, an anti-ActRIIB antibody or an antigen binding fragment thereof, a small molecule inhibitor of Activin-A, a small molecule inhibitor of ActRIIB, and a 'decoy' receptor of ActRIIB, such as a soluble ActRIIB receptor and a fusion of the soluble ActRIIB receptor with an Fc molecule (ActRIIB-Fc). See, for example, Zhou et al. (2010), supra. Suitable inhibitors of IL-6 or IL-6R, include an anti-IL-6 antibody or an antigen binding fragment thereof, an anti-IL-6R antibody or an antigen binding fragment thereof, a small molecule inhibitor of IL-6, a small molecule inhibitor of IL-6R, and a 'decoy' receptor of IL-6R, such as a soluble IL-6 receptor and a fusion of the soluble IL-6 receptor with an Fc molecule (IL6R-Fc). See, e.g., Enomoto et al. (2004) BIOCHEM. AND BIOPHYS. RES. COMM. 323:1096-1102; Argiles et al. (2011) EUR. J. PHARMACOL. 668:S81-S86; Tuca et al. (2013) ONCOLOGY/HEMATOLOGY 88:625-636. Suitable inhibitors of IL-6 or IL-6R may include, for example, Tocilizumab (Actemra®, Hoffmann-LaRoche), a humanized anti-IL-6R monoclonal antibody approved for treatment of rheumatoid arthritis, and Sarilumab/REGN88 (Regeneron), a humanized anti-IL6R antibody in clinical development for treatment of rheumatoid arthritis; and Selumetinib/AZD6244 (AstraZeneca), an allosteric inhibitor of MEK, which has been shown to inhibit IL-6 production. Prado et al. (2012) BRITISH J. CANCER 106:1583-1586.

[0211] TNF α and IL-1 are cytokines known to be involved in mediation of the proinflammatory response, which are also implicated in muscle depletion, anorexia and cachexia. Increased circulating levels of TNF α appear to inhibit myogenesis. TNF α , also known as "cachectin," stimulates interleukin-1 secretion and is implicated in the induction of cachexia. IL-1 is a potent trigger of the acute-phase inflammatory response, and it has been shown that infusion of IL-1 can lead to marked weight loss and appetite loss. IL-1 has been shown to contribute to the initiation of cancer cachexia in mice bearing a murine colon-26 adenocarcinoma (Strassmann et al. (1993) J. IMMUNOL. 150:2341). See also, Mathys and Billiau (1997) NUTRITION 13:763-770; Fong et al. (1989) AM. J. PHYSIOL.—REGULATORY, INTEGRATIVE AND COMPARATIVE PHYSIOL., 256:R659-R665. Thus, TNF α inhibitors and IL-1 inhibitors that are used in the treatment of rheumatoid arthritis may also be useful in the treatment of cachexia.

[0212] Accordingly, in certain embodiments of the present invention, one or more inhibitors of TNF α or IL-1 may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the present invention that

inhibits GDF-15 activity. Suitable inhibitors of TNF α or IL-1 include an anti-TNF α antibody or an antigen binding fragment thereof, an anti-IL-1 antibody or an antigen binding fragment thereof, a small molecule inhibitor of TNF α or IL-1, and a 'decoy' receptor of TNF α or IL-1, such as a soluble TNF α or IL-1 receptor and a fusion of the soluble form of TNF α or IL-1 with an Fc molecule. Suitable inhibitors of TNF α include for example, etanercept (Enbrel®, Pfizer/Amgen), infliximab (Remicade®, Janssen Biotech), adalimumab (Humira®, Abbvie), golimumab (Simponi®, Johnson and Johnson/Merck), and certolizumab pegol (Cimzia®, UCB). Suitable IL-1 inhibitors include, for example, Xilonix® antibody that targets IL-1 α (XBiotech), anakinra (Kinaret®, Amgen), canakinumab (Ilaris®, Novartis), and rilonacept (Arcalyst®, Regeneron). In certain embodiments, the TNF α inhibitor or IL-1 inhibitor, which is typically administered systemically for the treatment of rheumatoid arthritis may be administered locally and directly to the tumor site.

[0213] Myostatin, also known as GDF-8, is a member of the TGF- β family of peptides that is a negative regulator of muscle mass, as shown by increased muscle mass in myostatin deficient mammals. Myostatin is a ligand of the activin type 2 receptor, ActRIIB. Accordingly, in certain embodiments of the present invention, one or more inhibitors of myostatin or its receptor may be administered in combination with (for example, administered at the same time as, administered before, or administered after) an antibody of the invention that inhibits GDF-15 activity. Suitable inhibitors of myostatin or ActRIIB, include an anti-myostatin antibody or an antigen binding fragment thereof, an anti-ActRIIB antibody or an antigen binding fragment thereof, a small molecule inhibitor of myostatin, a small molecule inhibitor of ActRIIB, and a 'decoy' receptor of GDF-8, such as a soluble ActRIIB and a fusion of the soluble form of ActRIIB with an Fc molecule. See, e.g., Lokireddy et al. (2012) BIOCHEM. J. 446(1):23-26. Myostatin inhibitors that may be suitable for the present invention include REGN1033 (Regeneron); see Bauerlein et al. (2013) J. CACHEXIA SARCOPENIA MUSCLE: Abstracts of the 7.sup.th Cachexia Conference, Kobe/Osaka, Japan, Dec. 9-11, 2013, Abstract 4-06; LY2495655 (Lilly), a humanized anti-myostatin antibody in clinical development by Eli Lilly; see also "A PHASE 2 STUDY OF LY2495655 IN PARTICIPANTS WITH PANCREATIC CANCER," available on the world wide web at clinicaltrials.gov/ct2/NCT01505530; NML identifier: NCT01505530; ACE-031 (Acceleron Pharma); and stamulumab (Pfizer).

[0214] Agents such as Ghrelin or ghrelin mimetics, or other growth hormone secretagogues (GHS) which are able to activate the GHS receptor (GHS-R1a), also known as the ghrelin receptor, may be useful for increasing food intake and body weight in humans. See Guillory et al. (2013) in VITAMINS AND HORMONES vol. 92, chap. 3; and Steinman and DeBoer (2013) VITAMINS AND HORMONES vol. 92, chap. 8. Suitable ghrelin mimetics include anamorelin (Helsinn, Lugano, CH); See Temel et al. (2013) J. CACHEXIA SARCOPENIA MUSCLE: Abstracts of the 7.sup.th Cachexia Conference, Kobe/Osaka, Japan, Dec. 9-11, 2013, Abstract 5-01. Other suitable GHS molecules can be identified, for example, using the growth hormone secretagogue receptor Ghrelin competition assay described in PCT Publication Nos. WO2011/117254 and WO2012/113103.

[0215] Agonists of the androgen receptor, including small molecules and other selective androgen receptor modulators (SARMs) may be useful in treating cachexia and/or sarcopenia. See, e.g., Mohler et al. (2009) J. MED. CHEM. 52:3597-3617; Nagata et al. (2011) BIOORGANIC AND MED. CHEM. LETTERS 21:1744-1747; and Chen et al. (2005) MOL. INTERV. 5:173-188. Ideally, SARMs should act as full agonists, like testosterone, in anabolic target tissues, such as muscle and bone, but should demonstrate only partial or pure androgen receptor antagonistic activities on prostate tissue. See, e.g., Bovee et al. (2010) J. STEROID BIOCHEM. & MOL. BIOL. 118:85-92. Suitable SARMs can be identified, for example, by use of the methods and assays described in Zhang et al. (2006) BIOORG. MED. CHEM. LETT. 16:5763-5766; and Zhang et al. (2007) BIOORG. MED. CHEM. LETT. 17:439-443. Suitable SARMs include, for example, GTx-

024 (enobosarm, Ostearine®, GTx, Inc.), a SARM in phase II clinical development by GTx, Inc. See also, Dalton et al. (2011) J. CACHEXIA SARCOPENIA MUSCLE 2:153-161. Other suitable SARMs include 2-(2,2,2)-trifluoroethyl-benzimidazoles (Ng et al. (2007) BIOORG. MED. CHEM. LETT. 17:1784-1787) and JNJ-26146900 (Allan et al. (2007) J. STEROID BIOCHEM. & MOL. BIOL. 103:76-83).

[0216] β -adrenergic receptor blockers, or beta-blockers, have been studied for their effect on body weight in cachexic subjects, and have been associated with partial reversal of cachexia in patients with congestive heart failure. See, e.g., Hryniewicz et al. (2003) J. CARDIAC FAILURE 9:464-468. Beta-blocker MT-102 (PsiOxus Therapeutics, Ltd.) has been evaluated in a phase 2 clinical trial for subjects with cancer cachexia. See Coats et al. (2011) J. CACHEXIA SARCOPENIA MUSCLE 2:201-207.

[0217] Melanocortin receptor-knockout mice with a genetic defect in melanocortin signaling exhibit a phenotype opposite that of cachexia: increased appetite, increased lean body mass, and decreased metabolism. Thus, melanocortin antagonism has emerged as a potential treatment for cachexia associated with chronic disease (DeBoer and Marks (2006) TRENDS IN ENDOCRINOLOGY AND METABOLISM 17:199-204). Accordingly, in certain embodiments of the present invention, one or more inhibitors of a melanocortin peptide or a melanocortin receptor may be administered in combination (for example, administered at the same time as, administered before, or administered after) with an antibody of the invention that inhibits GDF-15 activity. Suitable inhibitors of melanocortins or melanocortin receptors include an anti-melanocortin peptide antibody or an antigen binding fragment thereof, an anti-melanocortin receptor antibody or an antigen binding fragment thereof, a small molecule inhibitor of a melanocortin peptide, a small molecule inhibitor of a melanocortin receptor, and a 'decoy' receptor of a melanocortin receptor, such as soluble melanocortin receptor and a fusion of a soluble melanocortin receptor with an Fc molecule. Suitable melanocortin receptor inhibitors include, for example, the melanocortin receptor antagonist agouti-related peptide (AgRP(83-132)), which has been demonstrated to prevent cachexia-related symptoms in a mouse model of cancer-related cachexia (Joppa et al. (2007) PEPTIDES 28:636-642).

[0218] Anti-cancer agents, especially those that can cause cachexia and elevate GDF-15 levels, such as cisplatin, may be used in methods of the present invention in combination with (for example, administered at the same time as, administered before, or administered after) an anti-GDF-15 antibody of the invention. Many cancer patients are weakened by harsh courses of radio- and/or chemotherapy, which can limit the ability of the patient to tolerate such therapies, and hence restrict the dosage regimen. Certain cancer agents themselves, such as fluorouracil, Adriamycin, methotrexate and cisplatin, may contribute to cachexia, for example by inducing severe gastrointestinal complications. See, e.g., Inui (2002) CANCER J. FOR CLINICIANS 52:72-91. By the methods of the present invention, in which an anti-cancer agent is administered in combination with an anti-GDF-15 antibody of the invention, it is possible to decrease the incidence and/or severity of cachexia, and ultimately increase the maximum tolerated dose of such an anti-cancer agent. Accordingly, efficacy of treatment with anti-cancer agents that may cause cachexia can be improved by reducing the incidence of cachexia as a dose-limiting adverse effect, and by allowing administration of higher doses of a given anti-cancer agent.

[0219] Thus, the present invention includes pharmaceutical compositions comprising an anti-GDF-15 antibody of the present invention in combination with an agent selected from the group consisting of: an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, and an anti-cancer agent. The present invention also includes methods of treating, preventing or minimizing cachexia and/or sarcopenia in a mammal comprising administering to a mammal in need thereof a pharmaceutical composition or compositions

comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor.

[0220] In another embodiment, the invention comprises a method of inhibiting loss of muscle mass associated with an underlying disease comprising administering to a mammal in need thereof a pharmaceutical composition or compositions comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor to prevent or reduce loss of muscle mass. The underlying disease may be selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis. Additionally, in certain embodiments, the loss of muscle mass is accompanied by a loss of fat mass.

[0221] In yet further embodiments, the present invention comprises a method of inhibiting or reducing involuntary weight loss in a mammal comprising administering to a mammal in need thereof a pharmaceutical composition or pharmaceutical compositions comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with an effective amount of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6 or an inhibitor of IL-6R, a ghrelin, a ghrelin mimetic or a GHS-R1a agonist, a SARM, a TNF α inhibitor, a IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker, a melanocortin peptide inhibitor, or a melanocortin receptor inhibitor.

[0222] Certain anti-cancer agents, such as cisplatin, have one or more undesirable adverse effects that involve causing or increasing one or more syndromes such as cachexia, sarcopenia, muscle wasting, bone wasting or involuntary body weight loss. Accordingly, in certain embodiments, the present invention comprises a method of treating cancer, while preventing, minimizing or reducing the occurrence, frequency or severity of cachexia, sarcopenia, or muscle wasting, bone wasting or involuntary loss of body weight in a mammal, comprising administering to a mammal in need thereof a pharmaceutical composition comprising an effective amount of an anti-GDF-15 antibody of the present invention in combination with with one or more anti-cancer agents. In particular embodiments, the invention comprises a method of treating cancer, while preventing, minimizing or reducing the occurrence, frequency or severity of cachexia, sarcopenia or muscle wasting, bone wasting or involuntary loss of body weight in a mammal, comprising administering to a mammal in need thereof a pharmaceutical composition comprising an effective amount of an anti-GDF-15 antibody of the invention in combination with one or more anti-cancer agents known to cause or increase the occurrence, frequency or severity of cachexia, sarcopenia, or muscle wasting, bone wasting or involuntary loss of body weight in a mammal.

EXAMPLES

[0223] The following Examples are merely illustrative and are not intended to limit the scope or content of the invention in any way.

Example 1: Human GDF15 Serum Levels in Mouse Xenograft Tumor Models

[0224] In this example, the amount of hGDF15 in the serum of mice bearing various xenograft tumors was measured. Serum was collected from three mice for each of the following tumor xenograft models: Chago, RPMI7951, PC3, TOV21G, HT-1080, K-562, and LS1034. Serum was also collected from three naïve mice as a control. Human GDF15 serum levels were determined by ELISA (R&D Systems, Cat. No. DY957E). Mice bearing human xenograft tumors that induce cachexia had serum levels of hGDF15 above 2 ng/mL, while mice bearing human xenograft tumors that do not induce cachexia had serum levels of hGDF15 below 1 ng/mL (FIG. 2). Naïve mice had

no detectable hGDF15 (control). These results indicate that a serum level of approximately 2 ng/mL GDF15 is a threshold for inducing cachexia in this mouse model. Similar levels of hGDF15 were also observed in plasma when determined by ELISA.

Example 2: Non-Tumor Bearing Mouse Model of Cachexia

[0225] An existing non-tumor bearing mouse model of cachexia is based on the injection of mature rhGDF15 into a mouse (Johnen et al. (1997) NAT. MED. 13:1333-1340). Mature rhGDF15 corresponds to amino acids 197 to 308 of hGDF15 protein. Mature rhGDF15 can be produced in the yeast *Pichia pastoris* as described in Fairlie et al. (2000) GENE 254:67-76). Cleaved-rhGDF15 corresponds to amino acids 197 to 308 of hGDF15 protein released from an Fc-rhGDF15 fusion protein. In FIGS. 3-4 described below, cleaved-rhGDF15 was produced by enzymatic digestion of mFc-rhGDF15 fusion protein with Factor Xa, and subsequent purification, prior to injection in mice.

[0226] To investigate the half-life of cleaved-rhGDF15, plasma was collected from a group of three mice after single dose of cleaved-rhGDF15 (1 µg/g) at different time points (2, 5, 8, 11, and 23 hours). Human GDF15 plasma levels were determined by ELISA (R&D Systems, Cat. No. DY957E). As shown in FIG. 3, cleaved-rhGDF15 was rapidly cleared from the plasma following injection. Eleven hours post-injection, the amount of cleaved-rhGDF15 in the plasma was below 10 ng/mL, and, within 23-hours, cleaved-rhGDF15 was almost completely cleared from the plasma.

[0227] The rapid clearance of cleaved-rhGDF15 in non-tumor bearing mice was further investigated. Eight-week old female ICR-SCID mice were randomized into two groups of ten mice each. Mice were dosed subcutaneously into the flank every eight hours for three days (a total of nine doses) with one of the following treatments: PBS (control) or cleaved-rhGDF15 at 1 µg/g. Body weight was measured daily. Statistical analyses were performed using a two-way ANOVA.

[0228] As shown in FIG. 4, cleaved-rhGDF15 induced body weight loss. After nine doses over a three day period, percent body weight dropped to 88% at day 4 ($p<0.001$), but approximately 24 hours after the last dose the mice began to gain weight. On day 6, the last day of the experiment, percent body weight increased to 94.8 percent ($p<0.001$). These results indicate that weight loss induced by cleaved-rhGDF15 is not sustained over long periods of time. The activity observed with cleaved-rhGDF15 described herein was similar to that observed with mature rhGDF15 in the existing mouse model (Johnen et al., supra).

[0229] The existing non-tumor bearing mouse model for cachexia relies on the injection of large amounts of mature rhGDF15 delivered in multiple doses per day to induce muscle loss and body weight loss (Johnen et al., supra). It appears that if mature rhGDF15 or cleaved-rhGDF15 is used, the mice do not sustain muscle weight loss or body weight loss for long periods of time without continuous dosing. This limits the usefulness of such models. Moreover, repeated dosing requires frequent handling of these mice which introduces stress that can compromise the reliability of body weight loss measurements. For example, as shown in FIG. 4, mice treated with multiple doses of PBS demonstrated a body weight drop due to the stress of repeated dosing and handling.

Example 3: GDF15 Fusion Proteins

[0230] In view of the large amounts of mature rhGDF15 (or cleaved-rhGDF15) and the labor intensity required to induce non-tumor bearing cachexia mouse models (as well as the resulting limitations of these models), we investigated alternate forms of rhGDF15 to induce a cachectic phenotype in mice. This Example describes the construction and production of two fusion proteins consisting of GDF15 and an immunoglobulin Fc fragment, designated mFc-rhGDF15 (mouse IgG1 Fc fused to the amino terminus of mature human GDF15) and rFc-rmGDF15 (rabbit IgG1 Fc fused to the amino terminus of mature mouse GDF15). The GDF15 fusion proteins were designed using methods known in the art. The mFc-rhGDF15 DNA sequences were constructed from fragments using overlap extension PCR to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, mouse IgG1 Fc, Factor Xa cleavage site, a polypeptide linker (GGGGS) (SEQ ID NO: 139), mature hGDF15, stop codon, and a 3' EcoRI

restriction site. The rFc-rmGDF15 amino acid sequences were converted to codon-optimized DNA sequences and synthesized to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, rabbit IgG1 Fc, a polypeptide linker (GGGG) (SEQ ID NO: 265), mature mouse GDF15, stop codon, and a 3' EcoRI restriction site.

[0231] The GDF15 fusion proteins were subcloned into the mammalian expression vector pEE14.4 (Lonza, Basel, Switzerland) via HindIII and EcoRI sites using In-Fusion™ PCR cloning (Clontech, Mountain View, CA). GDF15 fusion proteins were stably expressed in CHOK1SV cells using the GS System™ (Lonza Biologics) in order to produce large quantities of purified protein. Each expression vector was linearized and transfected into CHOK1SV cells. Stable clones were selected in the presence of methionine sulfoximine. Secreted proteins produced by CHOK1SV stably transfected cell lines were purified by Protein A and size exclusion chromatography.

[0232] The nucleic acid sequence and the encoded protein sequence defining the mouse IgG1 Fc-mature human GDF15 fusion protein (mFc-rhGDF15) are shown below. mFc-rhGDF15 contains mouse IgG1 Fc from amino acids 1-222, Factor Xa cleavage site from amino acids 223-228, an artificial linker sequence from amino acids 229-233, and mature hGDF15 from amino acids 234-345.

Nucleic Acid Sequence Encoding the Mouse IgG1 Fc—Mature Human GDF15 Fusion Protein (mFc-rhGDF15) (SEQ ID NO:219)

TABLE-US-00001 1 gggtgtaaac cctgcatctg cacggtgccg gaggtgtcct ccgtctttat cttccctccc 61
aaaccaagg atgtgctgac aatcactttg actccaaaag tcacatgcgt agtcgtggac 121 atctcgaaag
acgacccgga agtgcagttc tcgtggtttg ttgatgatgt agaagtgcac 181 accgctcaaa cccagccgag
ggaagaacag ttaacagca cgttaggag tgtgtcggaa 241 ctgccatta tgcaccagga ttggcttaat
gggaaggagt tcaaatgtcg cgtgaatagt 301 gcggcggtcc cagcccctat tgaaaagact atttcaaaa
cgaagggtcg gcccaaagct 361 cccaagtat acacaatccc tccgccgaaa gaacaaatgg caaaagacaa
agtgaatttg 421 acgtgcatga tcacggactt tttccggag gatatcaccg tcgaatggca atggaatggg 481
caacctgccg aaaactacaa gaatacaciaa cccattatgg ataccgatgg atcgtatttc 541 gtctactcaa
agttgaacgt acagaagtca aattgggagg cagggaatac gttcacttgc 601 agtgttttgc acgaaggcct
ccataaccac catacggaaa agtcactgtc gcactccccg 661 ggaaaaatcg agggcagaat ggatgggtga
ggagggtcgg cgcgcaacgg ggaccactgt 721 ccgctcgggc ccgggcgttg ctgccgtctg cacacgggtc
gcgcgtcgct ggaagacctg 781 ggctgggccc attgggtgct gtcgccacgg gaggtgcaag tgaccatgtg
catcggcgcg 841 tgcccgagcc agttccgggc ggcaaactg cacgcgcaga tcaagacgag cctgcaccgc 901
ctgaagcccc acacggtgcc agcgccttgc tgcgtgcccc ccagctacaa tcccatggtg 961 ctcattcaaa
agaccgacac cgggggtgtc ctccagacct atgatgactt gtagccaaa 1021 gactgccact gcata

Protein Sequence Defining the Mouse IgG1 Fc—Mature Human GDF15 Fusion Protein (mFc-rhGDF15) (SEQ ID NO:220)

TABLE-US-00002 1 gckpcictvp evssvfifpp kpkdvlitl tpkvtcvvvd iskddpevqf swfvddvevh
61 taqtqpreeq fnstfrsvse lpimhqdwln gkefkcrvns aafpapiekt isktkgrpka 121 pqvytipppk
eqmakdkvsl tcmittffpe ditviewqwnq qpaenykntq pimtdtgsyf 181 vysklnvqks
nweagntftc svlhheghnh htekslshsp gkiegrmdgg ggsarngdhc 241 plgpgrcrl htvrasledl
gwadwvlspr evqvtmciga cpsqfraanm haqiktslhr 301 lkpdtpapc cvpasynpmv liqktdtgv
lqtyddllak dchci

[0233] The nucleic acid sequence and the encoded protein sequence defining the rabbit IgG1 Fc-mature mouse GDF15 fusion protein (rFc-rmGDF15) are shown below. rFc-rmGDF15 contains rabbit IgG1 Fc from amino acids 1-223, an artificial linker sequence from amino acids 224-227, and mature mouse GDF15 from amino acids 228-342.

Nucleic Acid Sequence Encoding the Rabbit IgG1 Fc—Mature Mouse GDF15 Fusion Protein (rFc-rmGDF15) (SEQ ID NO:221)

TABLE-US-00003 1 tcgaaacca cttgccctcc tccggagctg ttgggcggac cctccgtgtt tatctttccc 61
ccgaagccga aagataccct tatgatctca cggacgccgg aggtcacttg cgtagtagtg 121 gatgtgtcgg
aggatgacct cgaagtccag ttcacctggg atatcaataa cgagcaagtg 181 aggacagcga ggccccact

tagggagcag cagttcaact ccacaattcg ggtcgtcagc 241 actttgccca tcgctcatga ggactggctc
cgcggaaaag agttcaagt taaggtgcat 301 aacaaggcat tgccagcgcc tattgaaaag acaatctcga
aggcgcgagg gcagccgctc 361 gagcccaaag tgtatacgat gggacccccg agggagaagaat tgtcgtcgcg
ctcagtaagc 421 cttacgtgca tgattaacgg tttctaccct agcgacatca gcgtagagtg ggaaaagaat 481
ggaaaggcgg aggataacta caagacgact cccgcggtgc tggattcgga tgggtcgtac 541 tttctgtata
gcaaattgtc agtcccagacc tcagaatggc agagggggtga cgtgttcacg 601 tgctccgtga tgcacgaagc
acttcacaat cactacaccc agaaatcaat ctgcggtcc 661 ccaggcaaag gtggaggagg gtcggctcac
gcccaccctc gcgattcgtg tccgctgggg 721 cctggtagat gctgtcatct cgagacagtc caggccacgc
tggaggacct cgggtggtca 781 gactgggtcc tgtccccacg acaactgcag ctttcgatgt gcgtggggga
atgtccgcac 841 ttgtacagat cggcgaatac ccacgctcag attaaggcac gactccatgg ttgcagcca 901
gataaagtcc ccgcaccttg ctgtgtcccc agctcatata ctctgtcgt actcatgcat 961 cggacagaca
gcggcgtgtc gcttcaaacg tatgacgacc tcgtagcgag aggatgtcat 1021 tgcgcc

Protein Sequence Defining the Rabbit IgG1 Fc—Mature Mouse GDF15 Fusion Protein (rFc-rmGDF15) (SEQ ID NO:222)

TABLE-US-00004 1 skptcpppel lggpsvfifp pkpkdtlmis rpevtcvvv dvseddpevq
ftwyinneqv 61 rtarppreq qfnstirvvs tlpiahedwl rgkefkckvh nkalpapie kiskargqpl 121
epkvytmgpp reelsrsvs ltcmingfyp sdisvewekn gkaednyktt pavldsdsy 181 flysklsvpt
sewqrgdvft csvmhealhn hytqksirs pgkggggsah ahprdsclp 241 pgrcchletv qatledlgws
dwvlsprqlq lsmcvgecph lyrsanthaq ikarlhlglp 301 dkvpapccvp ssytpvvlmh rtdsgvslqt
yddlvargch ca

[0234] The following sequences represent exemplary protein sequences for human IgG1 Fc-mature human GDF15 fusion proteins (hFc-rhGDF15). hFc-rhGDF15 Xa consists of human IgG1 Fc from amino acids 1-227, Factor Xa cleavage site from amino acids 228-233, an artificial linker sequence from amino acids 234-238, and mature hGDF15 from amino acids 239-350. hFc-rhGDF15 consists of human IgG1 Fc from amino acids 1-227, an artificial linker sequence from amino acids 228-232, and mature hGDF15 from amino acids 233-344.

Protein Sequence Defining the Human IgG1 Fc—Mature Human GDF15 Fusion Protein with Xa Cleavage Site (hFc-hGDF15 Xa) (SEQ ID NO:223)

TABLE-US-00005 1 dkthtppcp apellggpsv flfppkpkdt lmisrtpevt cvvvdvshed
pevkfnwyvd 61 gvevhnatk preeqynsty rvsvltvlh qdwlngkeyk ckvsnkappa piektiskak 121
gqprepvyt lppsreemtk nqvsltclvk gfypsdiave wesngqpenn ykttppvlds 181 dgsfflyskl
tvdksrwqqg nvfscsvmhe alhnhytqks lslspgkieg rmdggggsar 241 ngdhcplpgg rccrlhtvra
sleldgwadw vlsprevqvt mcigacpsqf raanmhaqik 301 tslhrkpdt vpapccvpas ynpmvliqkt
dtgvsllqtd dllakdchci

Protein Sequence Defining the Human IgG1 Fc—Mature Human GDF15 Fusion Protein with (hFc-hGDF15) (SEQ ID NO:224)

TABLE-US-00006 1 dkthtppcp apellggpsv flfppkpkdt lmisrtpevt cvvvdvshed
pevkfnwyvd 61 gvevhnatk preeqynsty rvsvltvlh qdwlngkeyk ckvsnkappa piektiskak 121
gqprepvyt lppsreemtk nqvsltclvk gfypsdiave wesngqpenn ykttppvlds 181 dgsfflyskl
tvdksrwqqg nvfscsvmhe alhnhytqks lslspgkggg gsarngdhc 241 lpggrcrlh tvasledlg
wadwvlspre vqvtmcigac psqfraanmh aqiktshrl 301 kpdtvpapcc vpsynpmvl iqktdtgvsll
qtyddllakd chci

Example 4: Fc-rhGDF15 Induced Cachexia Model

[0235] This Example describes the generation of an Fc-GDF15-induced cachexia model in mice. Immune-competent (Balb/C) and immune-incompetent (CB17-Scid) mice were randomized into three groups of ten mice each. Each group received one of the following treatments: PBS (control), mFc-rhGDF15 (as described in Example 3), or rFc-rmGDF15 (as described in Example 3) at 1 µg/g. Eight-week old female mice were dosed subcutaneously into the flank for three days (Balb/C) or once (CB17-Scid). Body weight was measured daily.

[0236] As shown in FIG. 5A and FIG. 5B, administration of mFc-rhGDF15 or rFc-rmGDF15

induced body weight loss in immune-competent mice (FIG. 5A) and immune-incompetent mice (FIG. 5B). These results indicate that a steady-state level of active rhGDF15 was achieved, because regardless of dose (one vs. three doses), both mFc-rhGDF15 and rFc-rmGDF15 induced sustained weight loss over the measured time course (7 days).

[0237] The fusion proteins, mFc-rhGDF15 and rFc-rmGDF15, were further tested in additional immune-competent (C57BL6, Swiss Webster) and immune-incompetent (ICR-SCID) mouse strains. In each tested mouse strain, the administration of mFc-rhGDF15 or rFc-rmGDF15 induced cachexia, as measured by body weight loss. Similar results were obtained regardless of whether mFc-rhGDF15 was dosed subcutaneously or intraperitoneally.

[0238] It was also investigated whether mFc-rhGDF15 induced weight loss regardless of the age of the mice treated with fusion protein. Swiss Webster (immune-competent) female mice of different ages (7, 13 and 25 weeks old) were divided into two groups of ten and treated with three doses per day of mFc-rhGDF15 or PBS (0.8 µg/g, 7 week old mice; 0.6 µg/g, 13 week old mice; or 0.4 µg/g, 25 week old mice). mFc-rhGDF15-induced weight loss was observed in all three mice age populations. In each age population, the mice lost approximately 10% of their body weight following treatment with mFc-rhGDF15 measured at ten days post treatment.

[0239] In another experiment, mFc-rhGDF15 induction of cachexia was investigated by measuring the loss of body weight, the loss of muscle mass, the loss of fat mass, and the expression levels of two molecular markers indicative of muscle degradation (i.e., mMuRF1 and mAtrogin). MuRF1 and Atrogin are E3-ubiquitin ligases that are upregulated in multiple models of muscle atrophy and cachexia (Glass, D. (2010) CURR. OPIN. CLIN. NUTR. MET. CARE 13:225-229).

[0240] Eight-week old female ICR-SCID mice were randomly divided into ten groups of ten mice each. Five groups (ten mice each) were dosed subcutaneously in the flank with PBS (control) and five groups (ten mice each) were dosed subcutaneously in the flank with mFc-rhGDF15 at 1.6 µg/g on day one. Body weight was measured daily for up to 17 days. One control group and one treatment group were sacrificed at different time points (0, 1, 3, 7 and 16 days post dose). Gonadal fat and gastrocnemius muscles were removed surgically from each group of mice at the indicated sacrifice time, and weighed. Tissues were snap frozen in liquid nitrogen, and RNA was isolated from the gastrocnemius muscle samples. Levels of mMuRF1 and mAtrogin mRNA were measured by qRT-PCR in samples corresponding to groups collected after 1, 7, and 16 days post dose. Statistical analyses were performed using a two-way ANOVA.

[0241] As shown in FIG. 6A, mFc-rhGDF15 induced body weight loss in ICR-SCID mice. Percent body weight was 79.4 percent when measured after 16 days following one dose of mFc-rhGDF15 ($p < 0.001$). mFc-rhGDF15 also induced loss of fat (adipose tissue), as observed by the loss of gonadal fat (FIG. 6B; $p < 0.01$ at day 7 and $p < 0.001$ at day 16) and loss of muscle, as observed by the loss of gastrocnemius muscle (FIG. 6C; $p < 0.05$ at days 1 and 3, and $p < 0.0001$ at days 7 and 16). Administration of mFc-rhGDF15 also elevated gene expression of two enzymes associated with muscle degradation and cachexia, mMuRF1 (FIG. 6D; $p < 0.001$ at days 1, 7, and 16) and mAtrogin (FIG. 6E; $p < 0.001$ at days 1 and 7, and $p < 0.01$ at day 16).

[0242] These results indicated that mFc-rhGDF15 induces cachexia in mice.

Example 5: mFc-rhGDF15 Induces Cachexia with a Longer GDF15 Half-Life in Serum

[0243] In this Example, the serum hGDF15 levels were measured following administration of mFc-rhGDF15, to determine the half-life of rhGDF15 in this model. Eight-week old female Balb/C nude mice were randomly divided into two groups of twelve mice each. Mice were dosed subcutaneously in the flank every twelve hours for three days (a total of six doses) with one of the following treatments: PBS (control) or mFc-rhGDF15 at 1.33 µg/g. Body weight was measured daily. As shown in FIG. 7, mFc-rhGDF15 induced sustained body weight loss for at least one week after the final injection.

[0244] In this experiment, hGDF15 serum levels were measured 0.2, 5, and 8 days after the last dose of mFc-rhGDF15. Mice were sacrificed at the indicated time, and sera were collected. Human

GDF15 serum levels were determined by ELISA (R&D Systems, Cat. No. DY957E). Table 1 provides the serum levels ($\mu\text{g/mL}$) for each mouse in the study.

TABLE-US-00007 TABLE 1 Days post Mouse Treatment Serum GDF15 last dose # Agent $\mu\text{g/g}$ ($\mu\text{g/mL}$); ELISA 0.2 1 mFc-rhGDF15 1.33 10.02 0.2 2 mFc-rhGDF15 1.33 9.54 0.2 3 mFc-rhGDF15 1.33 9.36 5 4 mFc-rhGDF15 1.33 8.24 5 5 mFc-rhGDF15 1.33 8.01 5 6 mFc-rhGDF15 1.33 6.59 8 7 mFc-rhGDF15 1.33 5.60 8 8 mFc-rhGDF15 1.33 5.52 8 9 mFc-rhGDF15 1.33 5.57

[0245] The results in Table 1 reveal that strong, sustained levels of hGDF15 are present in the serum at least eight days after the last dose of mFc-rhGDF15.

[0246] Serum samples from day 0.2 and day 5 after the last dose were also analyzed by Western blot (reducing gel; blot with an antibody against hGDF15 (R&D Systems, Cat. No AF957)) and quantified by Licor to determine the stability of mFc-rhGDF15 in the serum. Unexpectedly, two bands were observed. The upper band was approximately 40 kDa, and appeared to be mFc-rhGDF15. The lower band was approximately 15 kDa, and appeared to be cleaved mature rhGDF15. This indicated that mature rhGDF15 was released from mFc-rhGDF15 in the serum. Quantification of the two bands showed that about 90% of the rhGDF15 present in the serum was in the form of mFc-rhGDF15, with about 10% of the total rhGDF15 in the serum being present as the cleaved mature form (FIG. 8). Quantification showed a slight decrease in mFc-rhGDF15 in the serum samples collected five days after the last dose, but, surprisingly, a constant level of mature rhGDF15 remained in the serum. The ratio of mature rhGDF15 to mFc-rhGDF15 slightly increased over time, as a result of a decrease in mFc-rhGDF15 in the serum. Similar results were observed when rFc-rmGDF15 was injected into mice.

[0247] The results presented in FIGS. 7-8 and Table 1 were unexpected. The expectation was that very little, if any, mature rhGDF15 would be cleaved (released) from the mFc-rhGDF15 by day 0.2, and that any cleaved rhGDF15 would be rapidly cleared from the serum, as had been previously observed. For example, in FIG. 4, a series of nine doses at 1 $\mu\text{g/g}$ per dose (for a total of 9 $\mu\text{g/g}$) of cleaved-rhGDF15 was required to induce significant body weight loss in mice. These mice gained weight, almost immediately when dosing stopped. In contrast, a single dose of mFc-rhGDF15 at 0.1 $\mu\text{g/g}$ was sufficient to induce significant body weight loss for at least eight days (FIG. 9A; ten ICR-SCID mice dosed intraperitoneally with 0.1 $\mu\text{g/g}$ on day 1). The data in Table 1 revealed that rhGDF15 serum levels were stable for at least eight days, when rhGDF15 was administered as an mFc-rhGDF15 fusion protein.

[0248] To determine the source of activity resulting in sustained body weight loss, we investigated whether the observed rhGDF15 activity was attributable to the mFc-rhGDF15 fusion protein, the released mature rhGDF15 form, or both. As shown in FIG. 9A, a low dose of mFc-rhGDF15 (0.1 $\mu\text{g/g}$) resulted in body weight loss continuing for at least eight days. A lower dose of mFc-rhGDF15 (0.01 $\mu\text{g/g}$) also induced body weight loss, but the effect was not sustained for longer than 3 days post dose.

[0249] In this experiment, plasma was collected from three mice each dosed with 0.1 $\mu\text{g/g}$ or 0.01 $\mu\text{g/g}$ at 5 days post dose. Total rhGDF15 was measured by ELISA as described above. Total rhGDF15 plasma levels in the mice dosed with 0.1 $\mu\text{g/g}$ were above 70 ng/mL, consistent with the observation that these mice had significant weight loss (FIG. 9B). Total rhGDF15 plasma levels in mice dosed with 0.01 $\mu\text{g/g}$ were approximately 3.3 ng/mL, but it was observed that these mice were gaining weight (FIG. 9A and FIG. 9B). As described in FIG. 2, the threshold for hGDF15 to induce cachexia in tumor bearing mice is approximately 2 ng/mL. Thus, if both forms of rhGDF15 were active (i.e., mFc-rhGDF15 and released mature rhGDF15), then these mice should be losing weight, not gaining weight (i.e., 3.3 ng/mL total rhGDF15 is above the threshold of approximately 2 ng/mL hGDF15).

[0250] To determine which form was the active form (i.e., either mFc-rhGDF15 or released mature rhGDF15), we considered the data from FIG. 8 which showed that approximately 90% of the total rhGDF15 in serum was in the mFc-rhGDF15 form, and the remaining 10% was the released mature

form. Based on this extrapolation, approximately 3.0 ng/mL of rhGDF15 in the plasma was in the mFc-rhGDF15 form (i.e., 90% of 3.3 ng/mL). Once again, if mFc-rhGDF15 were active, these mice would be losing weight, not gaining weight because 3.3 ng/mL mFc-rhGDF15 is above the threshold of approximately 2 ng/mL hGDF15. The mice dosed with 0.1 µg/g mFc-rhGDF15 served as an internal control, because these mice had sustained body weight loss indicating that at least one of the two forms must be active. A calculation of 10% of 70 ng/mL total rhGDF15 in these mice is 7 ng/mL released mature rhGDF15. This amount is consistent with inducing the observed body weight loss and the threshold observed in FIG. 2. Thus, the data indicate that the mFc-rhGDF15 is not an active form of the protein, and only the mature rhGDF15 is active. These results were unexpected, because: (a) there was no reason to predict that the Fc fusion protein (mFc-rhGDF15) would be inactive; and (b) there was no reason to predict that the Fc fusion protein would release mature rhGDF15 at the observed rate.

[0251] These results indicate that mFc-rhGDF15 sustains a cachectic phenotype by slowly releasing mature rhGDF15 into the serum. These results further indicate that a steady state level of mature rhGDF15 in the plasma or serum can be achieved in a non-tumor bearing mouse by administering mFc-rhGDF15 to the mouse. Therefore, administration of mFc-rhGDF15 to non-tumor bearing mice is particularly useful as a mouse model of cachexia with a robust and sustained loss of muscle mass, loss of fat mass, and body weight loss (see FIGS. 6A-C).

Example 6: Anti-GDF15 Antibodies

[0252] This Example describes the production of anti-GDF15 monoclonal antibodies.

Immunizations, fusions, and primary screens were conducted using conventional methods following the Repetitive Immunization Multiple Sites (RIMMS) protocol. Five AJ mice and five Balb/c mice were immunized with 6×His (SEQ ID NO: 266) tagged recombinant human GDF15 (His-rhGDF15) (R&D Systems, Inc., Minneapolis, MN). Two Balb/c mice with sera displaying the highest anti-GDF15 activity by Enzyme Linked Immunosorbent Assay (ELISA) were chosen for subsequent fusion. Spleens and lymph nodes from the appropriate mice were harvested. B-cells were harvested and fused with a myeloma line. Fusion products were serially diluted onto forty 96-well plates to near clonality. Two AJ mice with sera displaying the highest anti-GDF15 activity by ELISA were chosen for subsequent fusion. Spleens and lymph nodes from the appropriate mice were harvested. B-cells were harvested and fused with a myeloma line. Fusion products were serially diluted onto forty 96-well plates to near clonality.

[0253] Approximately 3,840 supernatants from the cell fusions were screened by ELISA for binding to rhGDF15. A total of 172 supernatants containing antibodies against GDF15 were further characterized in vitro. A panel of hybridomas was selected, subcloned and expanded. Antibodies were expressed and subsequently purified by affinity chromatography on Protein G resin, under standard conditions.

Example 7: Antibody Sequence Analysis

[0254] The light chain isotype and heavy chain isotype of each monoclonal antibody in Example 6 was determined using the IsoStrip™ Mouse Monoclonal Antibody Isotyping Kit according to the kit vendor's instructions (Roche Applied Science, Indianapolis, IN). All antibodies were found to be kappa light chain, and IgG1 or IgG2b heavy chain.

[0255] The heavy and light chain variable regions of the mouse monoclonal antibodies were sequenced using 5' RACE (Rapid Amplification of cDNA Ends). Total RNA was extracted from each monoclonal hybridoma cell line using the RNeasy® Miniprep kit according to the kit vendor's instructions (Qiagen, Valencia, CA). Full-length first strand cDNA containing 5' ends was generated using the SMARTer™ RACE cDNA Amplification Kit (Clontech, Mountain View, CA) according to the kit vendor's instructions for 5' RACE.

[0256] The variable regions of the light (kappa) and heavy (IgG1 or IgG2b) chains were amplified by PCR using KOD Hot Start Polymerase (EMD Chemicals, Gibbstown, NJ) according to the kit vendor's instructions. For amplification of 5' cDNA ends in conjunction with the SMARTer™

RACE cDNA Amplification Kit, the Universal Primer Mix A primer (Clontech), a mix of: 5' CTAATACGACTCACTATAGGGCAAGCAGTGGTATCAACGCAGAGT 3' (SEQ ID NO:233) and 5' CTAATACGACTCACTATAGGGC 3' (SEQ ID NO:225), was used as a 5' primer. Heavy chain variable regions were amplified using the above 5' primers and a 3' IgG1 constant region specific primer, 5' TATGCAAGGCTTACAACCACA 3' (SEQ ID NO:226), or a 3' IgG2b constant region specific primer, 5' AGGACAGGGGTTGATTGTTGA 3' (SEQ ID NO:227). Kappa chain variable regions were first amplified with the above 5' primers and a 3' kappa constant region specific primer, 5' CTCATTCCTGTTGAAGCTCTTGACAAT 3' (SEQ ID NO:228). The light chains were subjected to a second, nested, round of PCR using the Nested Universal Primer A (Clontech) 5' primer, 5' AAGCAGTGGTATCAACGCAGAGT 3' (SEQ ID NO:229) and a nested 3' kappa constant region specific primer, 5' CGACTGAGGCACCTCCAGATGTT 3' (SEQ ID NO:230). Individual PCR products were either purified using the Qiaquick® PCR Purification kit or isolated by agarose gel electrophoresis and purified using the Qiaquick® Gel Purification kit according to the kit vendor's instructions (Qiagen). The PCR products were subsequently cloned into the pCR®4Blunt plasmid using the Zero Blunt® TOPO® PCR Cloning according to the kit vendor's instructions (Invitrogen) and transformed into DH5-α bacteria (Invitrogen) through standard molecular biology techniques. Plasmid DNA isolated from transformed bacterial clones was sequenced using M13 Forward (5' GTAAAACGACGGCCAGT 3') (SEQ ID NO:231) and M13 Reverse primers (5' CAGGAAACAGCTATGACC 3') (SEQ ID NO:232) by Beckman Genomics (Danvers, MA), using standard dideoxy DNA sequencing methods to identify the sequence of the variable region sequences. The sequences were analyzed using Vector NTI software (Invitrogen) and the IMGT/V-Quest web server (imgt.cines.fr) to identify and confirm variable region sequences.

[0257] The nucleic acid sequences encoding and the protein sequences defining variable regions of the murine monoclonal antibodies are shown below (amino terminal signal peptide sequences are not shown). CDR sequences (Kabat definition) are indicated by bold font and underlining in the amino acid sequences.

Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 01G06 Antibody (SEQ ID NO:39)

TABLE-US-00008 1 gaggtcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaagata
61 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc 121 catggaaaga
gccttgagtg gattggacaa attaactcta acaatggtgg tattttcttc 181 aaccagaagt tcaagggcaa
ggccacattg actgtagaca agtcctccaa tacagccttc 241 atggaggtcc gcagcctgac atctgaggac
actgcagtct attactgtgc aagagaggca 301 attactacgg taggcgctat ggactactgg ggtcaaggaa
cctcagtcac cgtctcctca

Protein Sequence Defining the Heavy Chain Variable Region of the 01G06 Antibody (SEQ ID NO:40)

TABLE-US-00009 1 evllqqsgpe lvkpgasvki pckasgytft **dynmd**wvkqs hgkslewigq
inpnnggiff 61 **nqkfk**gkatl tvdkssntaf mevrsltsed tavyycare**a** **ittvgamdy**w gqgtsvtvss

Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 01G06 Antibody (SEQ ID NO:75)

TABLE-US-00010 1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga
aactgtcacc 61 atcacatgtc gaacaagtga gaatcttcac aattatttag catggtatca gcagaaacag 121
ggaaaatctc ctgagctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca 181 aggttcagtg
gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct 241 gaagattttg ggagttatta
ctgtcaacat ttttgagta gtccttacac gttcggaggg 301 gggaccaagc tggaaataaa a

Protein Sequence Defining the Kappa Chain Variable Region of the 01G06 Antibody (SEQ ID NO:76)

TABLE-US-00011 1 diqmtqspas lsavgetvt it**crtsenlh** **nylawyqqkq** gkspqllvyd
aktladgvps 61 rfsgsgsgtq yslkinslqp edfgsyyc**qh** **fwsspyt**fgg gtkleik

Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 03G05 Antibody (SEQ ID NO:41)

TABLE-US-00012 1 caggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc
agtgaagctg 61 tcctgcaagg cttctggcta caccttcacc agctactgga ttactgggt gaaccagagg 121
cctggacaag gccttgagtg gattggagac attaactcta gcaacggccg tagtaagtat 181 aatgagaagt
tcaagaacaa ggccacaatg actgcagaca aatcctcaa cacagcctac 241 atgcaactca gcagcctgac
atctgaggac tctgcggtct attactgtgc aagagagggt 301 ctggatgggt ctatggacta ctgggggtcaa
ggaacctcag tcaccgtctc ctca

Protein Sequence Defining the Heavy Chain Variable Region of the 03G05 Antibody (SEQ ID NO:42)

TABLE-US-00013 1 qvqlqppgae lvkpgasvkl sckasgytft sywihwvnqr pgqglewigd
inpsngrsky 61 nekfknkatm tadkssntay mqlssltsed savyycarev ldgamdywgq gtsvtvss

Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 03G05 Antibody (SEQ ID NO:77)

TABLE-US-00014 1 gacatttgtt tgaccaatc tccagcttct ttggcttgtt ctctagggca
gagggccacc 61 atctcttgca gagccagcga aagtgttgat aattatggca ttagttttat gaactgggtc 121
caacagaaac caggacagcc acccaaactc ctcatctatg ctgcatccaa ccaaggctcc 181 ggggtccctg
ccaggtttag tggcagtggg tctgggacag acttcagcct caacatccat 241 cctatggagg aggatgatac
tgcaatgtat ttctgtcagc aaagtaagga gggtccgtgg 301 acgttcggtg gaggtccaa gctggaaatc aaa

Protein Sequence Defining the Kappa Chain Variable Region of the 03G05 Antibody (SEQ ID NO:78)

TABLE-US-00015 1 divltqspas lavslgqrat iscrasesvd nygisfmnwf qqkpgqppkl
liyaasnqgs 61 gvparfsgsg sgtdfslnih pmeeddtamy fcqqskevpw tfgggsklei k

Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 04F08 Antibody (SEQ ID NO:43)

TABLE-US-00016 1 caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttactgagc acttatggta tgggtgtgac ctggattcgt 121
cagccttcag gaaagggctt ggagtggctg gcacacattt actgggatga tgacaagcgc 181 tataacctat
ccctgaagag ccggctcaca atctcaagg atacctcaa caaccaggta 241 ttcctcaaga tcaccagtgt
ggacactgca gatactgcca cactactg tgctcaaacg 301 gggtatagta acttgtttgc ttactggggc
caagggactc tggtcactgt ctctgca

Protein Sequence Defining the Heavy Chain Variable Region of the 04F08 Antibody (SEQ ID NO:44)

TABLE-US-00017 1 qvtlkespgg ilqpsqtlls tcsfsgfsls tygmgtwir qpsgkglewl
ahiywdddkr 61 ynpslksrlt iskdtstnnqv flkitsvdta dtatycaqt gysnlfaywg qgtlvtvsa

Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 04F08 Antibody (SEQ ID NO:79)

TABLE-US-00018 1 gacatttgta tgaccagtc tcaaaaattc atgtccacat cagtaggaga
cagggtcagc 61 gtcacctgca aggccagtca gaatgtgggt actaatgtag cctgggtatca acagaaatta 121
ggacaatctc ctaaaact gattactcg gcacctacc ggtacagtgg agtcctgat 181 cgcttcacag
gcagtggatc tgggacagat ttactctca ccatcagcaa tgtgcagtct 241 gaagacttgg cagagtattt
ctgtcagcaa tataacagct atccgtacac gttcggaggg 301 gggaccaagc tggaaataaa a

Protein Sequence Defining the Kappa Chain Variable Region of the 04F08 Antibody (SEQ ID NO:80)

TABLE-US-00019 1 divmtsqskf mstsvgdrvs vtckasqnv tnvawyqqkl gqspktliys
asyrysgvpd 61 rftgsgsgtd fltisnvqs edlaeyfcq ynsypytfgg gtleik

Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 06C11 Antibody (SEQ ID NO:45)

TABLE-US-00020 1 caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttactgaac acttatggta tgggtgtgag ctggattcgt 121

cagccttcag gaaagggtct ggagtggctg gcacacattt actgggatga tgacaagcgc 181 tataacccat
ccctgaagag ccggctcaca atctccaagg atgcctccaa caaccgggtc 241 ttctcaaga tcaccagtgt
ggacactgca gatactgcca catactactg tgctcaaaga 301 gggtatgatg attactgggg ttactggggc
caagggactc tggtcactat ctctgca

Protein Sequence Defining the Heavy Chain Variable Region of the 06C11 Antibody (SEQ ID NO:46)

TABLE-US-00021 1 qvtlkesgpg ilqpsqtlsi tcsfsgfsln **tygm~~g~~vswir** qpsgkglewl
ahiywdddkr 61 **ynpslks**rlt iskdasnnrv flkitsvda dtatyycar **gyddywgywg** qgtltvisa
Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 06C11 Antibody (SEQ ID NO:81)

TABLE-US-00022 1 gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga
cagggtcagc 61 gtcacctgca aggccagtca gaattgggt actaatgtag cctggtttca acagaaacca 121
ggatcaatctc ctaaagcact gatttactcg gcatcttacc ggtacagtgg agtcctgat 181 cgcttcacag
gcagtggatc tgggacagat ttattctca ccatcagcaa tgtgcagtct 241 gaagacctgg cagagtattt
ctgtcagcaa tataacaact atcctctcac gtccggtgct 301 gggaccaagc tggagctgaa a

Protein Sequence Defining the Kappa Chain Variable Region of the 06C11 Antibody (SEQ ID NO:82)

TABLE-US-00023 1 divmtsqkf mstsvgdrvs vt**ckasqnv~~g~~** **tnvawfqqkp** gqspkaliys
asyrysgvpd 61 rftgsgsgtd fildisnvqs edlaeyfc**q~~q~~** **ynnypltfga** gtleik
Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 08G01 Antibody (SEQ ID NO:47)

TABLE-US-00024 1 gaggtcctgc tgcaacagtc tggacctgag gtggtgaagc ctggggcttc
agtgaagata 61 ccctgcaagg ctctggata cacattcact gactacaaca tggactgggt gaagcagagc 121
catggaaaga gccttgagt gattggagag attaactcta acaatggtgg tactttctac 181 aaccagaagt
tcaagggcaa ggccacattg actgtagaca agtcctccag cacagcctac 241 atggagctcc gcagcctgac
atctgaggac actgcagtct attactgtc aagagaggca 301 attactacgg taggcgctat ggactactgg
ggatcaaggaa cctcagtcac cgtctctca

Protein Sequence Defining the Heavy Chain Variable Region of the 08G01 Antibody (SEQ ID NO:48)

TABLE-US-00025 1 evllqqsgpe vvkpgasvki pckasgytft **dynmdwvkqs** hgkslewige
inpnnggtfy 61 **nqkfk**gkatl tvdkssstay melrsltsed tavyycare**a** **ittvgamdyw** gqgtsvtvss
Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 08G01 Antibody (SEQ ID NO:83)

TABLE-US-00026 1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga
aactgtcacc 61 atcacatgtc gagcaagtgg gaatttcac aattatttag catggtatca gcagaaacag 121
ggaaaatctc ctgagctct ggtctataat gcaaaaacct tagcagatgg tgtgccatca 181 aggttcagt
gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct 241 gaagattttg ggagttatta
ctgtcaacat tttggagtt ctcttacac gtccggaggg 301 gggaccaagc tggaaataaa a

Protein Sequence Defining the Kappa Chain Variable Region of the 08G01 Antibody (SEQ ID NO:84)

TABLE-US-00027 1 diqmtqspas lsasvgetvt it**crasgni~~h~~** **nylawyqqkq** gkspqllvyn
aktladgvps 61 rfsgsgsgtq yslkinslqp edfgsyyc**qh** **fwsspytf~~g~~g** gtleik
Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 14F11 Antibody (SEQ ID NO:49)

TABLE-US-00028 1 caggttactc tgaaagagtc tggccctgga atattgcagc cctccagac
cctcagtctg 61 acttgttctt tctctgggtt ttactgagc acttatggta tgggtgtagg ctggattcgt 121
cagccttcag gaaagggtct agagtggctg gcagacattt ggtgggatga cgataagtac 181 tataacccat
ccctgaagag ccggctcaca atctccaagg atacctccag caatgaggta 241 ttctcaaga tcgccattgt
ggacactgca gatactgcca ctactactg tgctcgaaga 301 ggctactact ctgctatgga ctactgggg
caaggaaacct cagtcaccgt ctctca

Protein Sequence Defining the Heavy Chain Variable Region of the 14F11 Antibody (SEQ ID NO:50)

TABLE-US-00029 1 qvtlkesgpg ilqpsqtls tcsfsgfsls **tygmgygwir** qpsgkglewl
adiwwdddky 61 **ynpslksrlt** iskdtssnev flkiaivdta dtatyycarr **ghysamdywg** qgtsvtvss
Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 14F11 Antibody (SEQ ID NO:85)

TABLE-US-00030 1 gacattgtaa tgaccagtc tcaaaaattc atgtccacat cagtaggaga cagggtcagc
61 gtcacctgca aggcagtc gaattgggt actaatgtag cctggatatca acagaaacca 121
gggcaatctc ctaaagcact gattactcg ccatactacc ggtacagtgg agtcctgat 181 cgcttcacag
gcagtggatc tgggacagat ttactctca ccatacagaa tgtcagctc 241 gaagacttgg cagaatattt
ctgtcagcaa tataacagct atctcacac gttcggaggg 301 gggaccaagc tggaaatgaa a
Protein Sequence Defining the Kappa Chain Variable Region of the 14F11 Antibody (SEQ ID NO:86)

TABLE-US-00031 1 divmtqsqkf mstsvgdrvs vt**kasqnvg** **tnvawyqqk** gqspkaliys
psyrysgvpd 61 rftgsgsgtd fltisnvqs edlaeyfc**qg** **ynsyphftgg** gtklemk
Nucleic Acid Sequence Encoding the Heavy Chain Variable Region of the 17B11 Antibody (SEQ ID NO:51)

TABLE-US-00032 1 caggttactc tgaaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttactgagc acttctggta tgggtgtgag ttggattcgt 121
cagccttcag gaaaggtct ggagtggctg gcacacaatg actgggatga tgacaagcgc 181 tataagtcac
ccctgaagag ccggctcaca atatccaagg atacctccag aaaccaggta 241 ttcctcaaga tcaccagtgt
ggacactgca gatactgcca catactactg tgctcgaaga 301 gttgggggat tagagggcta ttttgattac
tgggggccaag gcaccactct cacagtctcc 361 tca

Protein Sequence Defining the Heavy Chain Variable Region of the 17B11 Antibody (SEQ ID NO:52)

TABLE-US-00033 1 qvtlkesgpg ilqpsqtls tcsfsgfsls **tsgmgvswir** qpsgkglewl
ahndwdddkr 61 **yksllksrlt** iskdtstrnv flkitsvda dtatyycarr **vgglegfydy** wgqgttlts 121
s

Nucleic Acid Sequence Encoding the Kappa Chain Variable Region of the 17B11 Antibody (SEQ ID NO:87)

TABLE-US-00034 1 gacattgtgc tgacacagtc tctgtctcc ttagctgtat ctctggggca
gagggccacc 61 atctcatgca gggccagcca aagtgtcagt acatctaggt ttagttatat gcactgggtc 121
caacagaaac caggacaggc acccaaactc ctatcaagt atgcatccaa cctagaatct 181 ggggtccctg
ccaggttcag tggcagtggg tctgggacag acttcacct caacatccat 241 cctgtggagg gggaggatac
tgcaacatat tactgtcagc acagttggga gattccgtac 301 acgttcggag gggggaccaa gctggaaata aaa
Protein Sequence Defining the Kappa Chain Variable Region of the 17B11 Antibody (SEQ ID NO:88)

TABLE-US-00035 1 divltqspas layslgqrat iscr**asqsvs** **tsrfsymh**wf qqkpgqapkl
liky**asnles** 61 gvparfsgsg sgtdftlnih pvegedtaty yc**qhsweipy** **tfgggtklei** k

[0258] The amino acid sequences defining the immunoglobulin heavy chain variable regions for the antibodies produced in Example 6 are aligned in FIG. 10. Amino terminal signal peptide sequences (for expression/secretion) are not shown. CDR.sub.1, CDR.sub.2, and CDR.sub.3 (Kabat definition) are identified by boxes. FIG. 11 shows an alignment of the separate CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each antibody.

[0259] The amino acid sequences defining the immunoglobulin light chain variable regions of the antibodies in Example 6 are aligned in FIG. 12. Amino terminal signal peptide sequences (for expression/secretion) are not shown. CDR.sub.1, CDR.sub.2 and CDR.sub.3 are identified by boxes. FIG. 13 shows an alignment of the separate CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each antibody.

[0260] Table 2 shows the SEQ ID NO. of each sequence discussed in this Example.

TABLE-US-00036 TABLE 2 SEQ. ID NO. Nucleic Acid or Protein 39 01G06 Heavy Chain Variable Region-nucleic acid 40 01G06 Heavy Chain Variable Region-protein 75 01G06 Light (kappa) Chain Variable Region-nucleic acid 76 01G06 Light (kappa) Chain Variable Region-protein 1 01G06 Heavy Chain CDR.sub.1 7 01G06 Heavy Chain CDR.sub.2 15 01G06 Heavy Chain CDR.sub.3 21 01G06 Light (kappa) Chain CDR.sub.1 26 01G06 Light (kappa) Chain CDR.sub.2 32 01G06 Light (kappa) Chain CDR.sub.3 41 03G05 Heavy Chain Variable Region-nucleic acid 42 03G05 Heavy Chain Variable Region-protein 77 03G05 Light (kappa) Chain Variable Region-nucleic acid 78 03G05 Light (kappa) Chain Variable Region-protein 2 03G05 Heavy Chain CDR.sub.1 8 03G05 Heavy Chain CDR.sub.2 16 03G05 Heavy Chain CDR.sub.3 22 03G05 Light (kappa) Chain CDR.sub.1 27 03G05 Light (kappa) Chain CDR.sub.2 33 03G05 Light (kappa) Chain CDR.sub.3 43 04F08 Heavy Chain Variable Region-nucleic acid 44 04F08 Heavy Chain Variable Region-protein 79 04F08 Light (kappa) Chain Variable Region-nucleic acid 80 04F08 Light (kappa) Chain Variable Region-protein 3 04F08 Heavy Chain CDR.sub.1 9 04F08 Heavy Chain CDR.sub.2 17 04F08 Heavy Chain CDR.sub.3 23 04F08 Light (kappa) Chain CDR.sub.1 28 04F08 Light (kappa) Chain CDR.sub.2 34 04F08 Light (kappa) Chain CDR.sub.3 45 06C11 Heavy Chain Variable Region-nucleic acid 46 06C11 Heavy Chain Variable Region-protein 81 06C11 Light (kappa) Chain Variable Region-nucleic acid 82 06C11 Light (kappa) Chain Variable Region-protein 4 06C11 Heavy Chain CDR.sub.1 9 06C11 Heavy Chain CDR.sub.2 18 06C11 Heavy Chain CDR.sub.3 23 06C11 Light (kappa) Chain CDR.sub.1 28 06C11 Light (kappa) Chain CDR.sub.2 35 06C11 Light (kappa) Chain CDR.sub.3 47 08G01 Heavy Chain Variable Region-nucleic acid 48 08G01 Heavy Chain Variable Region-protein 83 08G01 Light (kappa) Chain Variable Region-nucleic acid 84 08G01 Light (kappa) Chain Variable Region-protein 1 08G01 Heavy Chain CDR.sub.1 10 08G01 Heavy Chain CDR.sub.2 15 08G01 Heavy Chain CDR.sub.3 24 08G01 Light (kappa) Chain CDR.sub.1 29 08G01 Light (kappa) Chain CDR.sub.2 32 08G01 Light (kappa) Chain CDR.sub.3 49 14F11 Heavy Chain Variable Region-nucleic acid 50 14F11 Heavy Chain Variable Region-protein 85 14F11 Light (kappa) Chain Variable Region-nucleic acid 86 14F11 Light (kappa) Chain Variable Region-protein 5 14F11 Heavy Chain CDR.sub.1 11 14F11 Heavy Chain CDR.sub.2 19 14F11 Heavy Chain CDR.sub.3 23 14F11 Light (kappa) Chain CDR.sub.1 30 14F11 Light (kappa) Chain CDR.sub.2 36 14F11 Light (kappa) Chain CDR.sub.3 51 17B11 Heavy Chain Variable Region-nucleic acid 52 17B11 Heavy Chain Variable Region-protein 87 17B11 Light (kappa) Chain Variable Region-nucleic acid 88 17B11 Light (kappa) Chain Variable Region-protein 6 17B11 Heavy Chain CDR.sub.1 12 17B11 Heavy Chain CDR.sub.2 20 17B11 Heavy Chain CDR.sub.3 25 17B11 Light (kappa) Chain CDR.sub.1 31 17B11 Light (kappa) Chain CDR.sub.2 37 17B11 Light (kappa) Chain CDR.sub.3 [0261] Mouse monoclonal antibody heavy chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 3.

TABLE-US-00037 TABLE 3 Kabat Variable Region CDR1 CDR2 CDR3 SEQ ID NO
 01G06 DYNMD QINPNNGGIFFNQKFKG EAITTVGAMDY 40 (SEQ ID NO: 1) (SEQ ID NO: 7) (SEQ ID NO: 15) 03G05 SYWIH DINPSNGRSKYNEKFKN EVLDGAMDY 42 (SEQ ID NO: 2) (SEQ ID NO: 8) (SEQ ID NO: 16) 04F08 TYGMGVT HIYWDDDKRYNPSLKS TGYSNLFAY 44 (SEQ ID NO: 3) (SEQ ID NO: 9) (SEQ ID NO: 17) 06C11 TYGMGVS HIYWDDDKRYNPSLKS RGYDDYWGY 46 (SEQ ID NO: 4) (SEQ ID NO: 9) (SEQ ID NO: 18) 08G01 DYNMD EINPNNGGTFYNQKFKG EAITTVGAMDY 48 (SEQ ID NO: 1) (SEQ ID NO: 10) (SEQ ID NO: 15) 14F11 TYGMGVG DIWWDDDKYYNPSLKS RGHYSAMDY 50 (SEQ ID NO: 5) (SEQ ID NO: 11) (SEQ ID NO: 19) 17B11 TSGMGVS HNDWDDDKRYKSSLKS RVGGLEGYFDY 52 (SEQ ID NO: 6) (SEQ ID NO: 12) (SEQ ID NO: 20) Chothia Variable Region CDR1 CDR2 CDR3 SEQ ID NO 01G06 GYTFTDY NPNNGG EAITTVGAMDY 40 (SEQ ID NO: 38) (SEQ ID NO: 143) (SEQ ID NO: 15) 03G05 GYTFTSY NPSNGR EVLDGAMDY 42 (SEQ ID NO: 128)

(SEQ ID NO: 144) (SEQ ID NO: 16) 04F08 GFSLSTYGM YWDDD TGYSNLFAY 44
 (SEQ ID NO: 130) (SEQ ID NO: 145) (SEQ ID NO: 17) 06C11 GFSLNTYGM
 YWDDD RGYDDYWGY 46 (SEQ ID NO: 132) (SEQ ID NO: 145) (SEQ ID
 NO: 18) 08G01 GYTFTDY NPNNGG EAITTVGAMDY 48 (SEQ ID NO: 38) (SEQ
 ID NO: 143) (SEQ ID NO: 15) 14F11 GFSLSTYGM WWDDD RGHYSAMDY 50
 (SEQ ID NO: 130) (SEQ ID NO: 146) (SEQ ID NO: 19) 17B11 GFSLSTSGM
 DWDDD RVGGLEGYFDY 52 (SEQ ID NO: 134) (SEQ ID NO: 147) (SEQ ID
 NO: 20) IMGT Variable Region CDR1 CDR2 CDR3 SEQ ID NO 01G06 GYTFTDYN
 INPNNGGI AREAITTVGAMDY 40 (SEQ ID NO: 136) (SEQ ID NO: 148) (SEQ
 ID NO: 154) 03G05 GYTFTSYW INPSNGRS AREVLDGAMDY 42 (SEQ ID NO: 138)
 (SEQ ID NO: 149) (SEQ ID NO: 155) 04F08 GFSLSTYGMG IYWDDDDK
 AQTGYSNLFAY 44 (SEQ ID NO: 140) (SEQ ID NO: 150) (SEQ ID NO: 156)
 06C11 GFSLNTYGMG IYWDDDDK AQRGYDDYWGY 46 (SEQ ID NO: 141) (SEQ
 ID NO: 150) (SEQ ID NO: 157) 08G01 GYTFTDYN INPNNGGT AREAITTVGAMDY
 48 (SEQ ID NO: 136) (SEQ ID NO: 151) (SEQ ID NO: 154) 14F11
 GFSLSTYGMG IYWDDDDK ARRGHYSAMDY 50 (SEQ ID NO: 140) (SEQ ID NO:
 152) (SEQ ID NO: 158) 17B11 GFSLSTSGMG NDWDDDDK ARRGGLEGYFDY 52
 (SEQ ID NO: 142) (SEQ ID NO: 153) (SEQ ID NO: 159)

[0262] Mouse monoclonal antibody Kappa light chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 4.

TABLE-US-00038 TABLE 4 Kabat/Chothia Variable Region CDR1 CDR2 CDR3 SEQ ID
 NO 01G06 RTSENHNYLA DAKTLAD QHFWSSPYT 76 (SEQ ID NO: 21) (SEQ ID
 NO: 26) (SEQ ID NO: 32) 03G05 RASEVDNYGISFMN AASNQGS QQSKEVPWT 78
 (SEQ ID NO: 22) (SEQ ID NO: 27) (SEQ ID NO: 33) 04F08 KASQNVGTNVA
 SASYRYS QQYNSYPYT 80 (SEQ ID NO: 23) (SEQ ID NO: 28) (SEQ ID NO:
 34) 06C11 KASQNVGTNVA SASYRYS QQYNNYPLT 82 (SEQ ID NO: 23) (SEQ ID
 NO: 28) (SEQ ID NO: 35) 08G01 RASGNIHNYLA NAKTLAD QHFWSSPYT 84
 (SEQ ID NO: 24) (SEQ ID NO: 29) (SEQ ID NO: 32) 14F11 KASQNVGTNVA
 SPSYRYS QQYNSYPHT 86 (SEQ ID NO: 23) (SEQ ID NO: 30) (SEQ ID NO:
 36) 17B11 RASQSVSTSRFSYMH YASNLES QHSWEIPYT 88 (SEQ ID NO: 25) (SEQ
 ID NO: 31) (SEQ ID NO: 37) IMGT Variable Region CDR1 CDR2 CDR3 SEQ ID
 NO 01G06 ENLHNY DAK QHFWSSPYT 76 (SEQ ID NO: 160) (SEQ ID NO: 32)
 03G05 ESVDNYGISF AAS QQSKEVPWT 78 (SEQ ID NO: 161) (SEQ ID NO: 33)
 04F08 QNVGTN SAS QQYNSYPYT 80 (SEQ ID NO: 162) (SEQ ID NO: 34) 06C11
 QNVGTN SAS QQYNNYPLT 82 (SEQ ID NO: 162) (SEQ ID NO: 35) 08G01
 GNIHNY NAK QHFWSSPYT 84 (SEQ ID NO: 163) (SEQ ID NO: 32) 14F11
 QNVGTN SPS QQYNSYPHT 86 (SEQ ID NO: 162) (SEQ ID NO: 36) 17B11
 QSVSTSRFSY YAS QHSWEIPYT 88 (SEQ ID NO: 164) (SEQ ID NO: 37)

[0263] To create the complete heavy or kappa chain antibody sequences, each variable sequence above is combined with its respective constant region. For example, a complete heavy chain comprises a heavy variable sequence followed by the murine IgG1 or IgG2b heavy chain constant sequence, and a complete kappa chain comprises a kappa variable sequence followed by the murine kappa light chain constant sequence.

Nucleic Acid Sequence Encoding the Murine IgG1 Heavy Chain Constant Region (SEQ ID NO: 165)

TABLE-US-00039 1 gccaaaacga ccccccatc tgtctatcca ctggcccctg gatctgctgc
 ccaaactaac 61 tccatggtga ccctgggatg cctgggtcaag ggctatttcc ctgagccagt gacagtgacc 121
 tggaactctg gatccctgtc cagcgggtgtg cacaccttcc cagctgtcct gcagtctgac 181 ctctacactc
 tgagcagctc agtgactgtc ccctccagca cctggcccag cgagaccgtc 241 acctgcaacg ttgccacccc
 ggccagcagc accaaggtgg acaagaaaat tgtgcccagg 301 gattgtggtt gtaagccttg catatgtaca

gtccagaag tatcatctgt cttcatcttc 361 ccccaaaagc ccaaggatgt gctcaccatt actctgactc
ctaaggtcac gtgtgtgttg 421 gtagacatca gcaaggatga tcccagggtc cagttcagct gggtttaga
tgatgtggag 481 gtgcacacag ctacagacgca accccgggag gagcagttca acagcacttt ccgctcagtc 541
agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttaa atgcagggtc 601 aacagtgcag
ctttccctgc ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg 661 aaggctccac aggtgtacac
cattccacct cccaaggagc agatggccaa ggataaagtc 721 agtctgacct gcatgataac agactcttc
cctgaagaca ttactgtgga gtggcagtgg 781 aatgggcagc cagcggagaa ctacaagaac actcagccca
tcatggacac agatggctct 841 tacttctgt acagcaagct caatgtgcag aagagcaact gggaggcagg
aaatactttc 901 acctgctctg tgttatcatga gggcctgcac aaccaccata ctgagaagag cctctccac 961
tctctggtg aa

Protein Sequence Defining the Murine IgG1 Heavy Chain Constant Region (SEQ ID NO:166)

TABLE-US-00040 1 aktppsyp lapgsaaqtn smvtlgclvk gyfpepvtvt wnsghlssgv
htfpavlqsd 61 lytlsssvtv psstwpsetv tcnvahpass tkvdkkiivr dgcckpcict vpevssvfif 121
ppkpkdvlti tltpkvtcvv vdiskddpev qfswfvddve vhtaqtqpre eqfnstfrsv 181 selpimhqdw
lngkefkcrv nsaafpapie ktisktkgrp kapqvtyipp pkeqmakdkv 241 sltcmidff peditviewqw
ngqpaenykn tqpimtdgs yfvysklvq ksnweagntf 301 tcsvlheglh nhhtekslsh spgk

Nucleic Acid Sequence Encoding the Murine IgG2b Heavy Chain Constant Region (SEQ ID NO:167)

TABLE-US-00041 1 gccaaaacaa ccccccatc agtctatcca ctggcccctg ggtgtggaga
tacaactggt 61 tcctccgtga ctctgggatg cctggtaag ggctacttcc ctgagtcagt gactgtgact 121
tggaactctg gatccctgtc cagcagtgtg cacaccttcc cagctctctt gcagtctgga 181 ctctacacta
tgagcagctc agtgactgtc ccctccagca cctggccaag tcagaccgtc 241 acctgcagcg ttgctcacc
agccagcagc accacgggtgg aaaaaaact tgagcccagc 301 gggcccattt caacaatcaa ccctgtctct
ccatgcaagg agtgcacaa atgccagct 361 cctaacctcg aggggtggacc atccgtcttc atcttccctc
caaatatcaa ggatgtactc 421 atgatctccc tgacacccaa ggtcacgtgt gtggtgggtg atgtgagcga
ggatgacca 481 gacgtccaga tcagctggtt tgtgaacaac gtggaagtac acacagctca gacacaaacc 541
catagagagg attacaacag tactatccgg gtggtcagca cctccccat ccagcaccag 601 gactggatga
gtggcaagga gttcaaatgc aaggtaaca acaaagacct cccatcacc 661 atcgagagaa ccatctcaaa
aattaaagg ctagttagag ctccacaagt atacatcttg 721 ccgccaccag cagagcagtt gtccaggaaa
gatgtcagtc tcttgcct ggctgtgggc 781 ttcaacctg gagacatcag tgtggagtgg accagcaatg
ggcatacaga ggagaactac 841 aaggacaccg caccagctct agactctgac ggttcttact tcatatatag
caagctcaat 901 atgaaaacaa gcaagtggga gaaaacagat tccttctcat gcaacgtgag acacaggggt 961
ctgaaaaatt actacctgaa gaagaccatc tcccggtctc cgggtaaa

Protein Sequence Defining the Murine IgG2b Heavy Chain Constant Region (SEQ ID NO:168)

TABLE-US-00042 1 aktppsyp lapgcgdtg ssvtlgclvk gyfpesvtvt wnsghlsssv
htfpallqsg 61 lytmsssvtv psstwpsqtv tcsvahpass ttvdkkleps gpistinpcp pckechkcpa 121
pnleggpsvf ifppnikdvl misltpkvtc vvvdvseddp dvqiswfvnn vevhtaqtqt 181 hredynstir
vvstlpihq dwmsgkefk kvnnkdldsp iertiskikg lvrpqvyil 241 pppaeqlsrk dvsltclvvg
fnpgdisvew tsnghteny kdtapvldsd gsyfiyskln 301 mktskwektd sfscnvrheg lknyylkkti
srspgk

Nucleic Acid Sequence Encoding the Murine Kappa Light Chain Constant Region (SEQ ID NO:169)

TABLE-US-00043 1 cgggctgatg ctgcaccaac tgtatccatc ttcccacat ccagttagca
gttaacatct 61 ggaggtgcct cagtcgtgtg ctcttgaac aacttctacc ccaaagacat caatgtcaag 121
tggaagattg atggcagtga acgacaaaat ggcgtcctga acagtggac tgatcaggac 181 agcaaagaca
gcacctacag catgagcagc accctcacgt tgaccaagga cgagtatgaa 241 cgacataaca gctatactg
tgaggccact cacaagacat caacttacc cattgtcaag 301 agcttcaaca ggaatgagt t

Protein Sequence Defining the Murine Kappa Light Chain Constant Region (SEQ ID NO:170)

TABLE-US-00044 1 radaaptvsi fppsseqlts ggasvvcfln nfypkdinvk wkidgserqn
gvlnswtdqd 61 skdstysmss tltltkdeye rhnsytceat hktstspivk sfrnec

[0264] The following sequences represent the actual or contemplated full length heavy and light chain sequence (i.e., containing both the variable and constant regions sequences) for each antibody described in this Example. Signal sequences for proper secretion of the antibodies (e.g., signal sequences at the 5' end of the DNA sequences or the amino terminal end of the protein sequences) are not shown in the full length heavy and light chain sequences disclosed herein and are not included in the final secreted protein. Also not shown are stop codons for termination of translation required at the 3' end of the DNA sequences. It is within ordinary skill in the art to select a signal sequence and/or a stop codon for expression of the disclosed full length immunoglobulin heavy chain and light chain sequences. It is also contemplated that the variable region sequences can be ligated to other constant region sequences to produce active full length immunoglobulin heavy and light chains.

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 01G06 (SEQ ID NO:99)

TABLE-US-00045 1 gaggtcctgc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc
agtgaagata 61 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121 catggaaaga gccttgagtg gattggacaa attaatccta acaatggtgg tattttcttc 181 aaccagaagt
tcaagggcaa ggccacattg acttagacaa agtcctccaa tacagccttc 241 atggaggtcc gcagcctgac
atctgaggac actgcagtct attactgtgc aagagaggca 301 attactacgg taggcgctat ggactactgg
ggtcaaggaa cctcagtcac cgtctcctca 361 gccaaaacga caccctcatc tgtctatcca ctggcccctg
gatctgctgc ccaaactaac 421 tccatggtga ccctgggatg cctggccaag ggctatttcc ctgagccagt
gacagtgacc 481 tggaactctg gatccctgtc cagcgggtgtg cacaccttc cagctgtcct gcagtctgac
541 ctctacactc tgagcagctc agtgactgtc ccctccagca cctggcccag cgagaccgtc 601
acctgcaacg tgccacccc ggccagcagc accaaggtgg acaagaaaat tgtgccagg 661 gattgtggtt
gtaagccttg catatgtaca gtcccagaag tatcatctgt cttcatcttc 721 ccccaaagc ccaaggatgt
gtccaccatt actctgactc ctaaggtcac gtgtgtgtg 781 gtagacatca gcaaggatga tcccagggtc
cagttcagct ggttttaga tgatgtggag 841 gtgcacacag ctcagacgca acccggggag gagcagttca
acagcacttt ccgctcagtc 901 agtgaacttc ccatcatgca ccaggactgg ctcaatggca aggagttcaa
atgcagggtc 961 aacagtgcag ctttccctgc ccccatcgag aaaaccatct ccaaaaccaa aggcagaccg
1021 aaggctccac aggtgtacac cattccacct cccaaggagc agatggccaa ggataaagtc 1081
agtctgacct gcatgataac agacttcttc cctgaagaca ttactgtgga gtggcagtgg 1141 aatgggcagc
cagcggagaa ctacaagaac actcagccca tcatggacac agatggctct 1201 tacttcgtct acagcaagct
caatgtgcag aagagcaact gggaggcagg aaatacttct 1261 acctgctctg tgttacatga gggcctgcac
aaccaccata ctgagaagag cctctcccac 1321 tctcctggtg aa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 01G06 (SEQ ID NO:100)

TABLE-US-00046 1 evllqqsgpe lvkpgasvki pckasgytft dynmdwvkqs hgkslewigq
inpnnggiff 61 nqkfkkgatl tvdkssntaf mevrsltsed tavyyarea ittvgamdyw ggqtsvtvss 121
akttppsvyp lapgsaaqtn smvtlgclvk gyfpepvtvt wnsgslssgv htfpavlqsd 181 lytlsssvtv
psstwpsetv tcnvahpass tkvdkkivpr dcgckpcict vpevssvfif 241 ppkpkdvlti tltpkvtcvv
vdisckddpev qfswfvddve vhtaqtqpre eqfnstfrsv 301 selpimhqdw lngkefkcrv nsaafpapie
ktisktkgrp kapqvytipp pkeqmakdkv 361 sltcmittff peditviewqw ngqpaenykn tqpimtdtgs
yfvysklnvq ksnweagntf 421 tcsvlheglh nhhtekslsh spgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 01G06 (SEQ ID NO:101)

TABLE-US-00047 1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga
aactgtcacc 61 atcacatgtc gaacaagtga gaatcttcac aattatttag catggtatca gcagaaacag 121
ggaaaatctc ctcagctcct ggtctatgat gcaaaaacct tagcagatgg tgtgccatca 181 aggttcagtg
gcagtggatc aggaacacaa tattctctca agatcaacag cctgcagcct 241 gaagattttg ggagttatta
ctgtcaacat ttttggagta gtccttacac gttcggaggg 301 gggaccaagc tggaataaaa acgggctgat
gctgcaccaa ctgtatccat cttcccacca 361 tccagtgagc agttaacatc tggaggtgcc tcagtcgtgt

gcttcttgaa caacttctac 421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaa
tggcgtcctg 481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg 541
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 601 tcaactcac
ccattgtcaa gagcttcaac aggaatgagt gt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 01G06 (SEQ ID NO:102)

TABLE-US-00048 1 diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd
aktladgvps 61 rfsghsgsgtg yslkinslqp edfgsyycqh fwsspytfgg gtleikrad aaptvsifpp 121
sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdskd stysmsstlt 181 ltkdeyerhn
sytceathkt stspivksfn rnec

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 03G05 (SEQ ID NO:103)

TABLE-US-00049 1 caggtccaac tgcagcagcc tggggctgaa ctggtgaagc ctggggcttc
agtgaagctg 61 tcctgcaagg cttctggcta caccctcacc agctactgga ttactgggt gaaccagagg
121 cctggacaag gccttgagtg gattggagac attaactcta gcaacggccg tagtaagtat 181
aatgagaagt tcaagaacaa ggccacaatg actgcagaca aatcctcaa cacagcctac 241 atgcaactca
gcagcctgac atctgaggac tctgcggtct attactgtgc aagagagggt 301 ctggatggtg ctatggacta
ctgggggtcaa ggaacctcag tcaccgtctc ctacgcaaaa 361 acgacacccc catctgtcta tccactggcc
cctggatctg ctgcccacac taactccatg 421 gtgaccctgg gatgcctggt caagggctat ttccctgagc
cagtgcagtg gacctggaac 481 tctggatccc tgtccagcgg tgtgcacacc ttcccagctg tcttcgagtc
tgacctctac 541 actctgagca gctcagtgac tgtcccctcc agcacctggc ccagcgagac cgtcacctgc
601 aacgttgccc acccgccag cagcaccaag gtggacaaga aaattgtgcc cagggattgt 661
gggtgtaagc cttgcatatg tacagtcca gaagtatcat ctgtcttcat cttccccca 721 aagcccaagg
atgtgctcac cattactctg actcctaagg tcacgtgtgt tgtggtagac 781 atcagcaagg atgatccga
gggtccagttc agctggtttg tagatgatgt ggaggtgcac 841 acagctcaga cgcaaccccg ggaggagcag
ttcaacagca ctttcgctc agtcagtga 901 cttcccatca tgcaccagga ctggctcaat ggcaaggagt
tcaaatgcag ggtcaacagt 961 gcagcttcc ctgcccccat cgagaaaacc atctcaaaa ccaaaggcag
accgaaggct 1021 ccacaggtgt acaccattcc acctccaag gagcagatgg ccaaggataa agtcagtctg
1081 acctgcatga taacagactt cttccctgaa gacattactg tggagtggca gtggaatggg 1141
cagccagcgg agaactacaa gaacactcag cccatcatgg acacagatgg ctcttacttc 1201 gtctacagca
agctcaatgt gcagaagagc aactgggagg caggaaatac tttcacctgc 1261 tctgtgttac atgagggcct
gcacaaccac catactgaga agagcctctc ccactctct 1321 ggtaaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 03G05 (SEQ ID NO:104)

TABLE-US-00050 1 qvqlqppgae lvkpgasvkl sckasgyft sywihwvnqr pgqglewigd
inpsngrsky 61 nekfkknatm tadsstntay mqlsslted savyycarev ldgamdywgq gtsvtvssak
121 ttpsvypla pgsaaqtnsm vtlgclvkgy fpepvtvtwn sgsllsgvht fpavlqsdly 181 tlsssvtvp
stwpsetvtc nvahpasstk vdkkivprdc gckpcictvp evssvfifpp 241 kpkdvlitl tpkvtcvvvd
iskddpevqf swfvddvevh taqtqpreeq fnstfrsvse 301 lpimhqdwln gkefkcrvns aafpapiekt
isktkgrpka pqvytipppk eqmakdkvsl 361 tcmtdffpe ditviewqwnq qpaenykntq pimtdgtsyf
vysklnvqks nweagntftc 421 svlheglhnh htekslshsp gk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 03G05 (SEQ ID NO:105)

TABLE-US-00051 1 gacattgtgt tgaccaatc tccagcttct ttggctgtgt ctctagggca
gagggccacc 61 atctctgca gagccagcga aagtgtgat aattatggca ttagtttat gaactggtc 121
caacagaaac caggacagcc accaaactc ctcatctatg ctgcatcaa ccaaggctcc 181 ggggtccctg
ccaggtttag tggcagtggtg tctgggacag acttcagcct caacatccat 241 cctatggagg aggatgatac
tgcaatgtat ttctgtcagc aaagtaagga ggttccgtgg 301 acgttcggtg gaggtccaa gctggaaatc
aaacgggctg atgctgcacc aactgtatcc 361 atcttccac catccagtga gcagttaaca tctggaggtg
cctcagtcgt gtgcttctg 421 aacaactct accccaaaga catcaatgtc aagtggaga ttgatggcag

tgaacgacaa 481 aatggcgtcc tgaacagttg gactgatcag gacagcaaag acagcaccta cagcatgagc 541
agcaccctca cgttgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc 601 actcacaaga
catcaacttc acccattgtc aagagcttca acaggaatga gtgt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 03G05 (SEQ ID NO:106)

TABLE-US-00052 1 divltqspas lavslgqrat iscrasesvd nygisfmnwf qqkpgqppkl
liyaasnqgs 61 gvparfsgsg sgtdfslnih pmeeddtamy fcqqskevpw tfgggsklei kradaaptvs
121 ifppsseqlt sggasvvcfl nnfypkdivn kwkidgserq ngvlswtdq dskdstysms 181
stlttkdey erhnsytcea thktstspiv ksfnrnec

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 04F08 (SEQ ID NO:107)

TABLE-US-00053 1 caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttcactgagc acttatggta tgggtgtgac ctggattcgt 121
cagccttcag gaaagggctt ggagtggctg gcacacattt actgggatga tgacaagcgc 181 tataacctat
ccctgaagag ccggctcaca atctccaagg atacctcaa caaccaggta 241 ttcctcaaga tcaccagtgt
ggacactgca gatactgcca catactactg tgctcaaacg 301 gggtatagta acttgtttgc ttactggggc
caagggactc tggtcactgt ctctgcagcc 361 aaaacgacac ccccatctgt ctatccactg gcccttggat
ctgtgcccc aactaactcc 421 atggtgacct tgggatgcct ggtcaagggc tatttcctg agccagtgc
agtgcactgg 481 aactctggat ccctgtccag cggtgtgcac acctccag ctgtctgca gtctgacctc
541 tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc 601
tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gccagggat 661 tgtggttga
agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc 721 ccaaagccca aggatgtgct
caccattact ctgactcta aggtcacgtg tgtgtggta 781 gacatcagca aggatgatcc cgaggccag
ttcagctggt tttagatga tgtggaggtg 841 cacacagctc agacgcaacc ccgggaggag cagtcaaca
gcatttccg ctgactcagt 901 gaacttccca tcatgcacca ggactggctc aatggcaagg agttcaaatg
cagggtcaac 961 agtgcagctt tccctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag
1021 gctccacagg gtacacat tccacctcc aaggagcaga tggccaagga taaagtcagt 1081
ctgacctgca tgataacaga ctcttccct gaagacatta ctgtggagtg gcagtggat 1141 gggcagccag
cggagaacta caagaacact cagcccatca tggacacaga tggctcttac 1201 ttcgtctaca gcaagctcaa
tgtgcagaag agcaactggg aggcaggaaa tactttcacc 1261 tgctctgtgt tacatgaggg cctgcacaac
caccatactg agaagagcct ctccactct 1321 cctggtaaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 04F08 (SEQ ID NO:108)

TABLE-US-00054 1 qvtlkespg ilqpsqtls tcsfsgfsls tygmgtwir qpsgkglewl
ahiywdddkr 61 ynpslksrlt iskdtstnnqv flkitsvda dtatycaqt gysnlfaywg qgtlvtvsaa 121
kttppsvypl apgsaaqtns mvtlgclvkg yfpepvtvtw nsgslssgvh tfpavlqsd 181 ytlsssvtvp
sstwpsetvt cnvahpasst kvdkkivprd cgckpcictv pevssvfifp 241 pkpkdvltit ltpkvtcvvv
diskddpevq fswfvdddev htaqtqpre qfnstfrsvs 301 elpmhqdw ngkefkcrvn saafpapie
tisktkgrp apqvytippp keqmakdkvs 361 ltcmitdffp editvewqwn gqpaenyknt qpimtdgsy
fvysklvqk snweagntft 421 csvlheglhn hhtekslshs pgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 04F08 (SEQ ID NO:109)

TABLE-US-00055 1 gacattgtga tgaccagtc tcaaaaattc atgtccacat cagtaggaga
cagggtcagc 61 gtcacctgca aggccagtca gaatgtgggt actaatgtag cctggtatca acagaaatta 121
ggacaatctc ctaaaacact gatttactcg gcatcctacc ggtacagtgg agtcctgat 181 cgcttcacag
gcagtggatc tgggacagat ttactctca ccatcagcaa tgtgcagtct 241 gaagacttgg cagagtattt
ctgtcagcaa tataacagct atccgtacac gttcggaggg 301 gggaccaagc tggaaataaa acgggctgat
gctgcaccaa ctgtatccat ctccacca 361 tccagtgcag agttaacatc tggaggtgcc tcagtcgtgt
gcttcttgaa caacttctac 421 cccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa
tggcgtcctg 481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg 541

ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 601 tcaacttcac
ccattgtcaa gagcttcaac aggaatgagt gt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 04F08 (SEQ ID NO:110)

TABLE-US-00056 1 divmtsqskf mstsvgdrvs vtckasqnvgt tnwavyqqkl gqspkqliys
asyrysgvpd 61 rftgsgsgtd ftltsnvqs edlaeyfcqq ynsypytfgg gtleikrad aaptvsifpp 121
sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdsd stysmsstlt 181 ltkdeyerhn
sytceathkt stspivksfn mec

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 06C11 (SEQ ID NO:111)

TABLE-US-00057 1 caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttcactgaac acttatggta tgggtgtgag ctggattcgt 121
cagccttcag gaaagggctt ggagtggctg gcacacattt actgggatga tgacaagcgc 181 tataacccat
ccctgaagag ccggctcaca atctccaagg atgcctcaa caaccgggtc 241 ttctcaaga tcaccagtgt
ggacactgca gatactgcca catactactg tgctcaaaga 301 gggtatgatg attactgggg ttactggggc
caagggactc tggtcactat ctctgcagcc 361 aaaacgacac ccccatctgt ctatccactg gcccttgat
ctgtgcccc aactaactcc 421 atggtgacct tgggatgcct ggtcaagggc tatttcctg agccagtgc
agtgcactgg 481 aactctggat cctgtccag cggtgtgcac acctccag ctgtctgca gtctgacct
541 tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc 601
tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gccagggat 661 tgtggttga
agccttgcac atgtacagtc ccagaagtat catctgtctt catctcccc 721 ccaagccca aggatgtgt
caccattact ctgactcta aggtcacgtg tgttggtga 781 gacatcagca aggatgatcc cgaggccag
ttcagctgtt ttgtagatga tgtggaggtg 841 cacacagctc agacgaacc ccgggaggag cagtcaaca
gcactttccg ctgactcagt 901 gaactccca tcatgcacca ggactggctc aatggcaagg agttcaaatg
cagggtcaac 961 agtgcagctt tcctgcccc catcgagaaa accatctcca aaacaaagg cagaccgaag
1021 gctccacagg tgtacacat tccacctccc aaggagcaga tggccaagga taaagtcagt 1081
ctgacctgca tgataacaga cttcttcct gaagacatta ctgtggagtg gcagtggaat 1141 gggcagccag
cggagaacta caagaacact cagcccatca tggacacaga tggctcttac 1201 ttcgtctaca gcaagctcaa
tgtgcagaag agcaactggg aggcaggaaa tactttcacc 1261 tgctctgtgt tacatgaggg cctgcacaac
caccatactg agaagagcct ctcccactct 1321 cctggtaaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 06C11 (SEQ ID NO:112)

TABLE-US-00058 1 qvtlkesgpg ilqpsqtlsl tcsfsgfsln tygmvgswir qpsgkglewl
ahiywdddkr 61 ynpslksrlt iskdasnrv flkitsvda dtatycaqr gyddywgywg qgtlvtsaa 121
kttpsvypl apgsaaqtns mvtlglclvg yfpepvtvtw nsgslssgvh tfpavlqsd 181 ytlsssvtvp
sstwpsetvt cnvahpasst kvdkkivprd cgckpcictv pevssvfifp 241 pkpkdvlit ltpkvtcvvv
diskddpevq fswfvdddev htaqtqpre qfnstfrsvs 301 elpimhqdlw ngkefkcrvn saafpapie
tisktkgrpk apqvtytipp keqmakdkvs 361 ltcmitdffp editvewqwn gqpaenyknt qpimtdg
fvysklvqk snweagntft 421 csvlheglhn hhtekslshs pgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 06C11 (SEQ ID NO:113)

TABLE-US-00059 1 gacattgtga tgaccagtc taaaaattc atgtccacat cagtaggaga
cagggtcagc 61 gtcacctgca aggccagtca gaatgtgggt actaatgtag cctggttca acagaaacca 121
ggatcaatctc cttaaagcact gatttactcg gcattctacc ggtacagtgg agtccctgat 181 cgcttcacag
gcagtggatc tgggacagat ttattctca ccatcagcaa tgtgcagtct 241 gaagacctgg cagagtattt
ctgtcagcaa tataacaact atcctctcac gtccgggtgct 301 gggaccaagc tggagctgaa acgggctgat
gctgcacaa ctgtatccat ctcccacca 361 tccagtgcag agttaacatc tggaggtgcc tcagtcgtgt
gcttcttgaa caactctac 421 ccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa
tggcgtcctg 481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg 541
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 601 tcaacttcac

ccattgtcaa gagcttcaag aggaatgagt gt
Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 06C11 (SEQ ID NO:114)

TABLE-US-00060 1 divmtsqskf mstsvgdrvs vtckasqnv g tnvwfqqkp gqspkaliys
asyrysgvpd 61 rftgsgsgtd filtisnvqs edlaeyfcq y nnypltfga gtklelkrad aaptvsifpp 121
sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdsd stysmsstlt 181 ltkdeyerhn
sytceathkt stspivksfn rne

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG2b Constant Region) of 08G01 (SEQ ID NO:115)

TABLE-US-00061 1 gaggtcctgc tgcaacagtc tggacctgag gtgggtgaagc ctggggcttc
agtgaagata 61 ccctgcaagg cttctggata cacattcact gactacaaca tggactgggt gaagcagagc
121 catggaaaga gccttgagtg gattggagag attaataccta acaatgggtg tactttctac 181 aaccagaagt
tcaagggcaa ggccacattg actgtagaca agtctccag cacagcctac 241 atggagctcc gcagcctgac
atctgaggac actgcagtct attactgtgc aagagaggca 301 attactacgg taggcgctat ggactactgg
ggtaagga cctcagtcac cgtctcctca 361 gccaaaacaa ccccccatc agtctatcca ctggcccctg
gggtgtggaga tacaactgg 421 tctcctgta ctctgggatg cctgggtcaag ggctacttcc ctgagtcagt
gactgtgact 481 tggaactctg gatccctgtc cagcagtgtg cacaccttcc cagctctcct gcagctctgga 541
ctctacacta tgagcagctc agtgactgtc cctccagca cctggccaag tcagaccgtc 601 acctgcagcg
ttgtcaccc agccagcagc accacgggtg aaaaaaact tgagccagc 661 gggcccattt caacaatcaa
cccctgtcct ccatgcaagg agtgtcacia atgccagct 721 cctaacctcg aggggtggacc atccgtcttc
atcttcctc caaatatcaa ggatgtactc 781 atgatctccc tgacacccaa ggtcacgtgt gtgggtgggtg
atgtgagcga ggatgaccca 841 gacgtccaga tcagctggtt tgtgaacaac gtggaagtac acacagctca
gacacaaacc 901 catagagagg attacaacag tactatccgg gtggtcagca cctccccc ccagcaccag
961 gactggatga gtggcaagga gttcaaatgc aaggtcaaca acaaagacct cccatcacc 1021
atcgagagaa ccattcctcaa aattaaagg ctatgcagag ctccacaagt atacatcttg 1081 ccgccaccag
cagagcagtt gtccaggaaa gatgtcagtc tcaattgcct ggctgtgggc 1141 ttcaaccctg gagacatcag
tgtggagtgg accagcaatg ggcatcacga ggagaactac 1201 aaggacaccg caccagtcct agactctgac
ggttcttact tcatatatag caagctcaat 1261 atgaaaacaa gcaagtggga gaaaacagat tccttctcat
gcaacgtgag acacgagggt 1321 ctgaaaaatt actacctgaa gaagaccatc tcccggtctc cgggtaaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG2b Constant Region) of 08G01 (SEQ ID NO:116)

TABLE-US-00062 1 evllqqsgpe vvkpgasvki pckasgyft dynmdwvkqs hgkslewise
inpnnggtfy 61 nqkfkkgatl tvdkssstay melrsltsed tavyycarea ittvgamdyw gqgtsvtvss 121
akttppsvyp lapgcgdtg ssvtlgclvk gyfpesvtvt wnsghlsssv htfpallqsg 181 lytmsssvtv
psstwpsqtv tcsvahpass ttvdkkleps gpistinpcp pckechkcpa 241 pnleggpsvf ifppnikdvl
misltpkvtc vvvdvseddp dvqiswfvnn vevhtaqtqt 301 hredynstir vvstlpihqh dwmsgkefk
kvnnkdldsp iertiskikg lvrappvyil 361 pppaeqlsrk dvsltclvvg fnpgdisvew tsnghteeny
kdtapvldsd gsyfiyskln 421 mktskwektd sfscnvrhcg lknyylkkti srspgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 08G01 (SEQ ID NO:117)

TABLE-US-00063 1 gacatccaga tgactcagtc tccagcctcc ctatctgcat ctgtgggaga
aactgtcacc 61 atcatatgct gagcaagtgg gaatattcac aattatttag catggtatca gcagaaacag 121
ggaaaatctc ctgagctcct ggtctataat gcaaaaacct tagcagatgg tgtgccatca 181 aggttcagtg
gcagtggtatc aggaacacaa tattctctca agatcaacag cctgcagcct 241 gaagattttg ggagttatta
ctgtcaacat ttttgaggtt ctcttacac gttcggaggg 301 gggaccaagc tggaaataaa acgggctgat
gctgcaccaa ctgtatccat ctccaccca 361 tccagtgcagc agttaacatc tggaggtgcc tcagtcgtgt
gcttcttgaa caacttctac 421 ccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa
tggcgtcctg 481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg 541
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 601 tcaactcac
ccattgtcaa gagcttcaac aggaatgagt gt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 08G01 (SEQ ID NO:118)

TABLE-US-00064 1 diqmtqspas lsasvgetvt itcrasgnih nylawyqqkq gkspqllvyn
aktladgvps 61 rfsqsgsgtq yslkinslqp edfgsyycqh fwsspytfgg gtleikrad aaptvsifpp 121
sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdskd stysmsstlt 181 ltkdeyerhn
sytceathkt stspivksfn r nec

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 14F11 (SEQ ID NO:119)

TABLE-US-00065 1 caggttactc tgaagagtc tggccctgga atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttcactgagc acttatggta tgggtgtagg ctggattcgt 121
cagccttcag gaaagggctc agagtggctg gcagacattt ggtgggatga cgataagtac 181 tataacccat
ccctgaagag ccggctcaca atctccaagg atacctccag caatgaggta 241 ttcctcaaga tcgccattgt
ggacactgca gatactgcca ctactactg tgctcgaaga 301 ggtcactact ctgctatgga ctactgggggt
caaggaacct cagtcaccgt ctctcagcc 361 aaaacgacac ccccatctgt ctatccactg gcccctggat
ctgctgccc aactaactcc 421 atgggtgacc tgggatgcct ggtcaagggc tatttcctg agccagtgc
agtgcactgg 481 aactctggat ccctgtccag cggtgtgcac acctcccag ctgtctgca gtctgacct
541 tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc 601
tgcaacgttg cccaccggc cagcagcacc aaggtggaca agaaaattgt gccagggat 661 tgtggttgta
agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc 721 ccaaagccca aggatgtgct
caccattact ctgactcta aggtcacgtg tgttggtgta 781 gacatcagca aggatgatcc cgaggctccag
ttcagctggt tttagatga tgtggaggtg 841 cacacagctc agacgcaacc ccgggaggag cagttaaca
gcatttccg ctgactcagt 901 gaacttcca tcatgcacca ggactggctc aatggcaagg agttcaaag
cagggtcaac 961 agtgcagctt tcctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag
1021 gctccacagg tgtaacacat tccacctcc aaggagcaga tggccaagga taaagtcagt 1081
ctgacctgca tgataacaga ctcttccct gaagacatta ctgtggagtg gcagtggat 1141 gggcagccag
cggagaacta caagaacact cagcccatca tggacacaga tggctcttac 1201 ttcgtctaca gcaagctcaa
tgtgcagaag agcaactggg aggcaggaaa tactttcacc 1261 tgctctgtgt tacatgaggg cctgcacaac
caccatactg agaagagcct ctcccactct 1321 cctaataaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 14F11 (SEQ ID NO:120)

TABLE-US-00066 1 qvtlkespg ilqpsqtls tcsfsgfsls tygmvgwir qpsgkglewl
adiwwdddkey 61 ynpslksrlt iskdtssnev flkiaivda dtatyycarr ghysamdywg qgtsvtvssa 121
kttpsvypl apgsaaqtns mvtlgclvkg yfpepvtvtw nsgslssgvh tfpavlqsd 181 ytlsssvtvp
sstwpsetvt cnvahpasst kvdkkivprd cgckpcictv pevssvfifp 241 pkpkdvltit ltpkvctvvv
diskddpevq fswfvdddev htaqtqpre qfnstfrsvs 301 elpmhqdlw ngkefkcrvn saafpapie
tisktkgrp apqvytippp keqmakdkvs 361 ltcmitdffp editvewqwn gqpaenyknt qpimtdgtsy
fvysklvqk snweagntft 421 csvlheglhn hhtekslshs pgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 14F11 (SEQ ID NO:121)

TABLE-US-00067 1 gacattgtaa tgaccagtc tcaaaaattc atgtccacat cagtaggaga
cagggtcagc 61 gtcacctgca aggccagtca gaattgggt actaatgtag cctggtatca acagaaacca 121
gggcaatctc ctaagcact gatttactcg ccactctacc ggtacagtgg agtcctgat 181 cgcttcacag
gcagtggatc tgggacagat ttcactctca ccactcagca tgtgcagtct 241 gaagacttgg cagaatattt
ctgtcagcaa tataacagct atctcagac gttcggaggg 301 gggaccaagc tggaaatgaa acgggctgat
gctgcaccaa ctgtatccat ctccacca 361 tccagtgc agttaacatc tggaggtgcc tcagtcgtgt
gcttctgaa caacttctac 421 ccaaagaca tcaatgtcaa gtggaagatt gatggcagtg aacgacaaaa
tggcgtcctg 481 aacagttgga ctgatcagga cagcaaagac agcacctaca gcatgagcag caccctcacg 541
ttgaccaagg acgagtatga acgacataac agctatacct gtgaggccac tcacaagaca 601 tcaactcac
ccattgtcaa gagcttcaac aggaatgagt gt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region

and Constant Region) of 14F11 (SEQ ID NO:122)

TABLE-US-00068 1 divmtqsqkf mstsvgdrvs vtckasqnvq tnwavyqqkp gqspkaliys
psyrysgvpd 61 rftgsgsgtd fltisnvqs edlaeyfcqq ynsyphftgg gtklemkrad aaptvsifpp 121
sseqltsgga svvcflnnfy pkdinvkwki dgserqngvl nswtdqdsd stysmsstlt 181 ltkdeyerhn
sytceathkt stspivksfn rnc

Nucleic Acid Sequence Encoding the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 17B11 (SEQ ID NO:123)

TABLE-US-00069 1 caggttactc tgaagagtc tggccctggg atattgcagc cctcccagac
cctcagtctg 61 acttgttctt tctctgggtt ttcactgagc acttctggta tgggtgtgag ttggattcgt 121
cagccttcag gaaagggctc ggagtggctg gcacacaatg actgggatga tgacaagcgc 181 tataagtcac
ccctgaagag ccggctcaca atatccaagg atacctccag aaaccaggta 241 ttcctcaaga tcaccagtgt
ggacactgca gatactgcca catactactg tgctcgaaga 301 gttgggggat tagagggtta ttttgattac
tggggccaag gcaccactct cacagtctcc 361 tcagccaaaa cgacaccccc atctgtctat ccactggccc
ctggatctgc tgcccaaact 421 aactccatgg tgaccctggg atgcctggtc aagggtatt tccctgagcc
agtgcagtgc 481 acctggaact ctggatccct gtccagcggg gtgcacacct tccagctgt cctgcagtct
541 gacctctaca ctctgagcag ctcagtact gtccctcca gcacctggcc cagcgagacc 601
gtcacctgca acgttgccca cccggccagc agcaccaagg tggacaagaa aattgtgccc 661 agggattgtg
gttgtaagcc ttgcatatgt acagtcccag aagtatcatc tgtctcatc 721 tccccccaa agcccaagga
tgtgtcacc attactctga ctctaaggc cacgtgtgtt 781 gtggtagaca tcagcaagga tgatcccag
gtccagttca gctggtttgt agatgatgtg 841 gaggtgcaca cagctcagac gcaacccccg gaggagcagt
tcaacagcac ttccgctca 901 gtcagtgaac ttccatcat gcaccaggac tggctcaatg gcaaggagtt
caaatgcagg 961 gtcaacagtgc cagctttccc tgccccatc gagaaaacca tctccaaaac caaaggcaga
1021 ccgaaggctc cacaggtgta caccattcca cctcccaagg agcagatggc caaggataaa 1081
gtcagtctga cctgcatgat aacagacttc ttcctgaag acattactgt ggagtggcag 1141 tggaatgggc
agccagcggg gaactacaag aacactcagc ccatcatgga cacagatggc 1201 tcttacttcg tctacagca
gctcaatgtg cagaagagca actgggaggc aggaataact 1261 ttcacctgct ctgtgttaca tgagggcctg
cacaaccacc atactgagaa gagcctctcc 1321 cactctctg gtaaa

Protein Sequence Defining the Full Length Heavy Chain Sequence (Heavy Chain Variable Region and IgG1 Constant Region) of 17B11 (SEQ ID NO:124)

TABLE-US-00070 1 qvtlkespg ilqpsqtlsl tcsfsgfsls tsgmgvswir qpsgkglewl
ahndwdddkr 61 yksslksrlt iskdtsrnqv flkitsvda dtatyycarr vgglegfydy wgqgtltvs 121
sakttppsvy plapgsaaqt nsmvtlgclv kgyfpepvtv twngslssg vhtfpavlqs 181 dlytlsssvt
vpsstwpset vtcnvahpas stkvdckivp rdcgckpcic tvpevssvfi 241 fppkpkdvlt itltpkvte
vvdiskddpe vqfswfvddv evhtaqtqpr eeqfnstfrs 301 vselpimhq d wlngkefkcr vnsaafpapi
ektisktkgr pkapqvtyip ppkeqmakdk 361 vsltcmidf fpeditviewq wngqpaenyk
ntqpimdtg syfvysklv qksnweagnt 421 ftcsvlhegl hnhhteksls hspgk

Nucleic Acid Sequence Encoding the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 17B11 (SEQ ID NO:125)

TABLE-US-00071 1 gacattgtgc tgacacagtc tctgtcttc ttagctgtat ctctggggca
gagggccacc 61 atctcatgca gggccagcca aagtgtcagt acatctaggt ttagttatat gcactgggtc 121
caacagaaac caggacaggc acccaaactc ctcatcaagt atgcatcaa cctagaatct 181 ggggtccctg
ccaggttcag tggcagtggg tctgggacag acttcacct caacatccat 241 cctgtggagg gggaggatac
tgcaacatat tactgtcagc acagttggga gattccgtac 301 acgttcggag gggggaccaa gctggaaata
aaacgggctg atgctgcacc aactgtatcc 361 atctccac catccagtga gcagtaaca tctggaggtg
cctcagtctg gtgcttctg 421 aacaacttct accccaaaga catcaatgtc aagtggaaga ttgatggcag
tgaacgacaa 481 aatggcgtcc tgaacagttg gactgatcag gacagcaaag acagcaccta cagcatgagc 541
agcaccctca cgttgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc 601 actcacaaga
catcaacttc acccattgtc aagagcttca acaggaatga gtgt

Protein Sequence Defining the Full Length Light Chain Sequence (Kappa Chain Variable Region and Constant Region) of 17B11 (SEQ ID NO:126)

TABLE-US-00072 1 divltqspas lavslgqrat iscrasqsvs tsrfsymhwf qqkpgqapkl
 likyasnles 61 gvparfsgsg sgtdftlnih pvegedtaty ycqhswiepy tfggggtklei kradaaptvs 121
 ifppsseqlt sggasvvcfl nnfypkdivn kwkidgserq ngvlmswtdq dskdstysms 181 stltltkdey
 erhnsytcea thktstspiv ksfmrnec

[0265] Table 5 shows the correspondence between the full-length sequences of the antibodies discussed in this Example with those presented in the Sequence Listing.

TABLE-US-00073 TABLE 5 SEQ ID NO. Nucleic Acid or Protein 99 01G06_Heavy Variable + IgG1 Constant-nucleic acid 100 01G06_Heavy Variable + IgG1 Constant-protein 101 01G06_Kappa Variable + Constant-nucleic acid 102 01G06_Kappa Variable + Constant-protein 103 03G05 Heavy Variable + IgG1 Constant-nucleic acid 104 03G05 Heavy Variable + IgG1 Constant-protein 105 03G05 Kappa Variable + Constant-nucleic acid 106 03G05 Kappa Variable + Constant-protein 107 04F08 Heavy Variable + IgG1 Constant-nucleic acid 108 04F08 Heavy Variable + IgG1 Constant-protein 109 04F08 Kappa Variable + Constant-nucleic acid 110 04F08 Kappa Variable + Constant-protein 111 06C11 Heavy Variable + IgG1 Constant-nucleic acid 112 06C11 Heavy Variable + IgG1 Constant-protein 113 06C11 Kappa Variable + Constant-nucleic acid 114 06C11 Kappa Variable + Constant-protein 115 08G01 Heavy Variable + IgG2b Constant-nucleic acid 116 08G01 Heavy Variable + IgG2b Constant-protein 117 08G01 Kappa Variable + Constant-nucleic acid 118 08G01 Kappa Variable + Constant-protein 119 14F11 Heavy Variable + IgG1 Constant-nucleic acid 120 14F11 Heavy Variable + IgG1 Constant-protein 121 14F11 Kappa Variable + Constant-nucleic acid 122 14F11 Kappa Variable + Constant-protein 123 17B11 Heavy Variable + IgG1 Constant-nucleic acid 124 17B11 Heavy Variable + IgG1 Constant-protein 125 17B11 Kappa Variable + Constant-nucleic acid 126 17B11 Kappa Variable + Constant-protein

Example 8: Binding Affinities

[0266] The binding affinities and kinetics of binding of antibodies to 6×His tagged (SEQ ID NO: 266) recombinant human GDF15 (His-rhGDF15 (R&D Systems, Inc.)), untagged recombinant human GDF15 (rhGDF15 (Peprotech, Rocky Hill, NJ)), and recombinant human GDF15 produced as either mouse Fc fused to human GDF15 (mFc-rhGDF15) or a version in which the Fc was enzymatically removed (cleaved-rhGDF15) were measured by surface plasmon resonance, using a Biacore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0267] Rabbit anti-mouse IgGs (GE Healthcare) were immobilized on carboxymethylated dextran CM4 sensor chips (GE Healthcare) by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 µL/minute. Injection time was varied for each antibody to yield an R_{max} between 30 and 60 RU. 250 µg/mL mouse Fc (Jackson ImmunoResearch, West Grove, PA) was injected at 30 µL/minute for 120 seconds to block non-specific binding of capture antibodies to mouse Fc portion of the recombinant GDF15 protein when needed. Buffer, mFc-rhGDF15, cleaved-rhGDF15, His-rhGDF15, or rhGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60 µL/minute. The dissociation phase was monitored for up to 1500 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 1.7, at a flow rate of 30 µL/minute. The GDF15 concentration range tested was 30 nM to 0.625 nM.

[0268] Kinetic parameters were determined using the kinetic function of the BIAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters for each antibody, k_{sub.a} (association rate constant), k_{sub.d} (dissociation rate constant), and K_{sub.D} (equilibrium dissociation constant) were determined. Kinetic values of the monoclonal antibodies on mFc-rhGDF15, cleaved rhGDF15, His-rhGDF15, or rhGDF15 are summarized in Tables 6, 7, 8, and 9, respectively.

TABLE-US-00074 TABLE 6 Antibody Binding to mFc-rhGDF15 Antibody k_{sub.a} (1/Ms) k_{sub.d} (1/s) K_{sub.D} (M) n 01G06 5.6E+06 7.0E−04 2.1E−10 7 03G05 1.0E+07 6.4E−04 6.9E−11 3

04F08 3.6E+06 6.4E-04 1.9E-10 3 06C11 4.5E+06 6.8E-04 1.7E-10 5 08G01 6.0E+06 1.1E-03 1.9E-10 4 14F11 1.7E+06 3.3E-04 2.2E-10 4 17B11 3.7E+06 5.1E-04 1.4E-10 3

[0269] The data in Table 6 demonstrate that antibodies bind mFc-rhGDF15 with a K.sub.D of about 250 pM or less, 200 pM or less, 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

[0270] Kinetic values of the monoclonal antibodies on cleaved-rhGDF15 are summarized in Table 7.

TABLE-US-00075 TABLE 7 Antibody Binding to Cleaved-rhGDF15 Antibody k.sub.a (1/Ms) k.sub.d (1/s) K.sub.D (M) n 01G06 7.5E+06 8.6E-04 1.1E-10 1 06C11 1.2E+07 2.0E-03 1.7E-10 2 14F11 5.7E+06 6.0E-04 1.1E-10 1

[0271] The data in Table 7 demonstrate that antibodies 01G06, 06C11 and 14F11 bind cleaved-rhGDF15 with a K.sub.D of about 200 pM or less, 150 pM or less, or 100 pM or less.

[0272] Kinetic values of the monoclonal antibodies on His-rhGDF15 are summarized in Table 8.

TABLE-US-00076 TABLE 8 Antibody Binding to His-rhGDF15 Antibody k.sub.a (1/Ms) k.sub.d (1/s) K.sub.D (M) n 01G06 1.4E+07 1.1E-03 8.1E-11 2 06C11 2.9E+07 1.5E-03 5.1E-11 2 14F11 4.4E+06 4.2E-04 9.6E-11 1

[0273] The data in Table 8 demonstrate that antibodies 01G06, 06C11 and 14F11 bind His-rhGDF15 with a K.sub.D of about 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

[0274] Kinetic values of the monoclonal antibodies on rhGDF15 are summarized in Table 9.

TABLE-US-00077 TABLE 9 Antibody Binding to rhGDF15 Antibody k.sub.a (1/Ms) k.sub.d (1/s) K.sub.D (M) n 01G06 2.1E+07 1.9E-03 9.3E-11 1 06C11 2.2E+07 4.6E-03 2.1E-10 1 14F11 3.1E+07 2.2E-03 7.1E-11 1

[0275] The data in Table 9 demonstrate that antibodies 01G06, 06C11 and 14F11 bind rhGDF15 with a K.sub.D of about 250 pM or less, 200 pM or less, 150 pM or less, 100 pM or less, 75 pM or less, or 50 pM or less.

Example 9: Reversal of Cachexia in an mFc-rhGDF15-Induced Model

[0276] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 in an mFc-rhGDF15-induced cachexia model. mFc-rhGDF15 (2 µg/g) was administered subcutaneously into the flank of 8-week old female ICR-SCID mice. Body weight was measured daily. When body weight reached 93%, the mice were randomized into seven groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 at 10 mg/kg. Treatment was administered once by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 14F11, or 17B11 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 14 and Table 10).

TABLE-US-00078 TABLE 10 Treatment % Body ANOVA Analysis Gr. Agent mg/kg Weight (compared to mIgG) 1 mIgG 10 77.1 NA 2 01G06 10 94.1 p < 0.001 3 03G05 10 95.1 p < 0.001 4 04F08 10 95.8 p < 0.001 5 06C11 10 93.8 p < 0.001 7 14F11 10 95.4 p < 0.001 8 17B11 10 92.8 p < 0.001

[0277] The data in FIG. 14 and Table 10 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced mouse model (i.e., a non-tumor bearing mouse model).

Example 10: Reversal of Cachexia in an HT-1080 Xenograft Tumor Model

[0278] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO.sub.2, using Eagle's Minimum Essential Medium (ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10.sup.6 cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 93%, the mice were randomized into eight groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11,

08G01, 14F11 or 17B11 at 10 mg/kg. Treatment was administered every three days by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 15 and Table 11).

TABLE-US-00079 TABLE 11 ANOVA ANALYSIS TREATMENT % BODY (COMPARED GR. AGENT MG/KG WEIGHT TO MIGG) 1 MIGG 10 81.4 NA 2 01G06 10 103.3 $P < 0.001$ 3 03G05 10 106.1 $p < 0.001$ 4 04F08 10 104.3 $p < 0.001$ 5 06C11 10 106.6 $p < 0.001$ 6 08G01 10 105.3 $p < 0.001$ 7 14F11 10 99.6 $p < 0.001$ 8 17B11 10 103.7 $p < 0.001$

[0279] The data in FIG. 15 and Table 11 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0280] Additional studies were conducted with antibody 01G06 to demonstrate the reversal of cachexia in this mouse model. HT-1080 cells were grown and inoculated subcutaneously into the flank of 8-week old female ICR SCID mice as described above. When body weight reached 93%, the mice were randomized into two groups of ten mice each. Each group received one of the following treatments: murine IgG control or 01G06 at 10 mg/kg. Treatment was administered once by intra-peritoneal injection. As shown in FIG. 16A, treatment with antibody 01G06 resulted in body weight increase to initial weight or 100% ($p < 0.001$) (FIG. 16A).

[0281] Food consumption was determined by weighing the food supply given to the mice daily (FIG. 16B). A significant increase in food consumption was observed in the 01G06 treated group for the first three days post treatment. After that time, no significant change was observed compared to the control group (mIgG).

[0282] Water consumption was determined by weighing the water supply given to the mice daily. No significant change in water consumption was observed between groups.

[0283] In this experiment, a group of ten mice were sacrificed at the time of the dose (baseline or 93% body weight, without treatment) and at the end of study (seven days post dose, either mIgG or 01G06). Gonadal fat and the gastrocnemius muscles were removed surgically and weighed as described above in Example 4 and tissues were snap frozen in liquid nitrogen. RNA was isolated from the gastrocnemius muscle samples to determine the levels of mMuRF1 and mAtrogin mRNA by RT-PCR, as described in Example 4.

[0284] As shown in FIG. 16C, a significant reduction in gonadal fat mass was observed seven days post dose with mIgG, but not in the group treated with antibody 01G06. In addition, mice treated with mIgG displayed significant gastrocnemius muscle loss compared to the baseline group, while the group of mice treated with antibody 01G06 did not (FIG. 16D). Further, the levels of muscular degradation markers, mMuRF1 and mAtrogin, were significantly higher in the mIgG group compared to the 01G06 group (FIG. 16E).

[0285] These results indicate that the disclosed anti-GDF15 antibodies can reverse cachexia measured by the loss of muscle mass, the loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

Example 11: Reversal of Cachexia in an HT-1080 Xenograft Tumor Model

[0286] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Eagle's Minimum Essential Medium (ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10^6 cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 80%, the mice were randomized into two groups of five mice each. Each group received one of the following treatments: murine IgG control, 01G06 dosed at 2 mg/kg on day 1 and day 7. Treatment was administered by intra-peritoneal injection. Treatment with antibody 01G06 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 17A and Table 12).

TABLE-US-00080 TABLE 12 Treatment % Body ANOVA Analysis Gr. Agent mg/kg Weight

(compared to mIgG) 1 mIgG 2 66.4 NA 2 01G06 2 97.16 p < 0.001

[0287] The data in FIGS. 17A-B and Table 12 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0288] In this experiment, a group of five mice were sacrificed at the time of dosing (baseline or 80% body weight loss, without treatment) and at the end of study (seven days post dose, either mIgG or 01G06). Liver, heart, spleen, kidney, gonadal fat and the gastrocnemius muscles were removed surgically and weighed. As shown in FIG. 17B, a significant loss in liver, heart, spleen, kidney, gonadal fat and gastrocnemius muscle mass was observed seven days post dose with mIgG, but not in the group treated with antibody 01G06. In addition, mice treated with antibody 01G06 displayed significant liver and gonadal muscle gain compared to the baseline group (FIG. 17B). [0289] These results indicate that the disclosed anti-GDF15 antibodies can reverse cachexia measured by the loss of key organ mass, loss of muscle mass, loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

Example 12: Reversal of Cachexia in a K-562 Xenograft Tumor Model

[0290] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 in a K-562 leukemia xenograft model. K-562 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Iscove's Modified Dulbecco's Medium (ATCC Catalog No. 30-2005) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female CB17SCRFMF mice with 2.5×10⁶ cells per mouse in 50% matrigel. Body weight was measured daily. When body weight reached 93%, the mice were randomly distributed into eight groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 at 10 mg/kg. Treatment was administered every three days by intra-peritoneal injection. Treatment with antibody 01G06, 03G05, 04F08, 06C11, 08G01, 14F11 or 17B11 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 18 and Table 13).

TABLE-US-00081 TABLE 13 Treatment % Body ANOVA Analysis Gr. Agent mg/kg Weight (compared to mIgG) 1 mIgG 10 90.4 NA 2 01G06 10 106.5 p < 0.001 3 03G05 10 109.8 p < 0.001 4 04F08 10 108.9 p < 0.001 5 06C11 10 109.5 p < 0.001 6 08G01 10 107.2 p < 0.001 7 14F11 10 107.0 p < 0.001 8 17B11 10 105.3 p < 0.001

[0291] The data in FIG. 18 and Table 13 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in a K-562 xenograft tumor model.

Example 13: Additional Xenograft Tumor Models

[0292] Antibody 01G06 was tested in additional tumor xenograft models including the TOV-21G ovarian xenograft model and the LS1034 colon xenograft model. In each model, antibody 01G06 reversed body weight loss compared to a PBS control (p<0.001 for the TOV-21G model and p<0.01 for the LS1034 model).

Example 14: Humanization of Anti-GDF15 Antibodies

[0293] This Example describes the humanization and chimerization of three murine antibodies, designated 01G06, 06C11, and 14F11, and the characterization of the resulting humanized antibodies. The humanized anti-GDF15 antibodies were designed, affinity matured by targeted CDR mutagenesis, and optimized using methods known in the art. The amino acid sequences were converted to codon-optimized DNA sequences and synthesized to include (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, humanized variable region, human IgG1 or Kappa constant region, stop codon, and a 3' EcoRI restriction site. [0294] Chimeric (murine variable region and human constant region) 01G06, 06C11, and 14F11 heavy (human IgG1) and light (human Kappa) chains were also constructed. To generate chimeric antibodies, the murine variable regions were fused to the human constant region, and codon-optimized DNA sequences were synthesized, including (in the following order): 5' HindIII restriction site, Kozak consensus sequence, amino terminal signal sequence, mouse variable region,

human IgG1 or Kappa constant region, stop codon, and 3' EcoRI restriction site.

[0295] The humanized and chimeric heavy chains were subcloned into pEE6.4 (Lonza, Basel, Switzerland) via HindIII and EcoRI sites using In-Fusion™ PCR cloning (Clontech, Mountain View, CA). The humanized and chimeric Kappa light chains were subcloned into pEE14.4 (Lonza) via HindII and EcoRI sites using In-Fusion™ PCR cloning.

[0296] Humanized antibody chains or chimeric antibody chains were transiently transfected into 293T cells to produce antibody. Antibody was either purified or used in cell culture media supernatant for subsequent in vitro analysis. Binding of the chimeric and humanized antibodies to human GDF15 was measured as described below. The results are summarized in Tables 24-27.

[0297] Each of the possible combinations of the chimeric or humanized K1G06 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 14.

TABLE-US-00082 TABLE 14 Antibody Light Chain Variable Heavy Chain Variable Name Region
Region Hu01G06-1 Ch01G06 Chimeric Kappa Ch01G06 Chimeric Heavy (SEQ ID NO: 76) (SEQ ID NO: 40) Hu01G06-14 Ch01G06 Chimeric Kappa Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 76) (SEQ ID NO: 54) Hu01G06-15 Ch01G06 Chimeric Kappa Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 76) (SEQ ID NO: 56) Hu01G06-147 Ch01G06 Chimeric Kappa Sh01G06 IGHV1-18 M69L (SEQ ID NO: 76) Heavy (SEQ ID NO: 58) Hu01G06-148 Ch01G06 Chimeric Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 76) G44S Heavy (SEQ ID NO: 60) Hu01G06-149 Ch01G06 Chimeric Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 76) Heavy (SEQ ID NO: 62) Hu01G06-150 Ch01G06 Chimeric Kappa Sh01G06 IGHV1-69 T30S I69L (SEQ ID NO: 76) Heavy (SEQ ID NO: 64) Hu01G06-151 Ch01G06 Chimeric Kappa Sh01G06 IGHV1-69 T30S K64Q (SEQ ID NO: 76) I69L Heavy (SEQ ID NO: 66) Hu01G06-4 Hu01G06 IGKV1-39 Kappa Ch01G06 Chimeric Heavy (SEQ ID NO: 90) (SEQ ID NO: 40) Hu01G06-46 Hu01G06 IGKV1-39 Kappa Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 90) (SEQ ID NO: 54) Hu01G06-52 Hu01G06 IGKV1-39 Kappa Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 90) (SEQ ID NO: 56) Hu01G06-100 Hu01G06 IGKV1-39 Kappa Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 90) (SEQ ID NO: 58) Hu01G06-102 Hu01G06 IGKV1-39 Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 90) G44S Heavy (SEQ ID NO: 60) Hu01G06-101 Hu01G06 IGKV1-39 Kappa Sh01G06 IGHV1-18 M69L (SEQ ID NO: 90) K64Q Heavy (SEQ ID NO: 62) Hu01G06-103 Hu01G06 IGKV1-39 Kappa Sh01G06 IGHV1-69 T30S (SEQ ID NO: 90) I69L Heavy (SEQ ID NO: 64) Hu01G06-104 Hu01G06 IGKV1-39 Kappa Sh01G06 IGHV1-69 T30S K64Q (SEQ ID NO: 90) I69L Heavy (SEQ ID NO: 66) Hu01G06-152 Hu01G06 IGKV1-39 S43A V48I Ch01G06 Chimeric Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 40) Hu01G06-71 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-18 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 54) Hu01G06-77 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-69 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 56) Hu01G06-110 Hu01G06 IGKV1-39 S43A V48I Sh01G06 IGHV1-18 M69L Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 58) Hu01G06-112 Hu01G06 IGKV1-39 S43A V48I Sh01G06 IGHV1-18 M69L K64Q Kappa (SEQ ID NO: 92) G44S Heavy (SEQ ID NO: 60) Hu01G06-111 Hu01G06 IGKV1-39 S43A V48I Sh01G06 IGHV1-18 M69L K64Q Kappa (SEQ ID NO: 92) Heavy (SEQ ID NO: 62) Hu01G06-113 Hu01G06 IGKV1-39 S43A V48I Sh01G06 IGHV1-69 T30S I69L Kappa (SEQ ID NO: 92) Heavy (SEQ ID NO: 64) Hu01G06-114 Hu01G06 IGKV1-39 S43A V48I Sh01G06 IGHV1-69 T30S K64Q Kappa (SEQ ID NO: 92) I69L Heavy (SEQ ID NO: 66) Hu01G06-122 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-18 F1 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 246) Hu01G06-119 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-18 F2 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 248) Hu01G06-135 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-69 F1 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 250) Hu01G06-138 Hu01G06 IGKV1-39 S43A V48I Hu01G06 IGHV1-69 F2 Heavy Kappa (SEQ ID NO: 92) (SEQ ID NO: 252) Hu01G06-153 Hu01G06 IGKV1-39 V48I Kappa Ch01G06 Chimeric Heavy (SEQ ID NO: 94) (SEQ ID NO: 40) Hu01G06-69 Hu01G06 IGKV1-39 V48I Kappa Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 94) (SEQ ID NO: 54) Hu01G06-75 Hu01G06 IGKV1-39 V48I Kappa

Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 94) (SEQ ID NO: 56) HU01G06-105 Hu01G06 IGKV1-39 V48I Kappa Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 94) (SEQ ID NO: 58) Hu01G06-107 Hu01G06 IGKV1-39 V48I Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 94) G44S Heavy (SEQ ID NO: 60) Hu01G06-106 Hu01G06 IGKV1-39 V48I Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 94) Heavy (SEQ ID NO: 62) Hu01G06-108 Hu01G06 IGKV1-39 V48I Kappa Sh01G06 IGHV1-69 T30S I69L (SEQ ID NO: 94) Heavy (SEQ ID NO: 64) Hu01G06-109 Hu01G06 IGKV1-39 V48I Kappa Sh01G06 IGHV1-69 T30S K64Q (SEQ ID NO: 94) I69L Heavy (SEQ ID NO: 66) Hu01G06-154 Hu01G06 IGKV1-39 F2 Kappa Ch01G06 Chimeric Heavy (SEQ ID NO: 254) (SEQ ID NO: 40) Hu01G06-155 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-18 Heavy (SEQ ID NO: 254) (SEQ ID NO: 54) Hu01G06-156 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-69 Heavy (SEQ ID NO: 254) (SEQ ID NO: 56) Hu01G06-157 Hu01G06 IGKV1-39 F2 Kappa Sh01G06 IGHV1-18 M69L Heavy (SEQ ID NO: 254) (SEQ ID NO: 58) Hu01G06-158 Hu01G06 IGKV1-39 F2 Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 254) G44S Heavy (SEQ ID NO: 60) Hu01G06-159 Hu01G06 IGKV1-39 F2 Kappa Sh01G06 IGHV1-18 M69L K64Q (SEQ ID NO: 254) Heavy (SEQ ID NO: 62) Hu01G06-160 Hu01G06 IGKV1-39 F2 Kappa Sh01G06 IGHV1-69 T30S I69L (SEQ ID NO: 254) Heavy (SEQ ID NO: 64) Hu01G06-161 Hu01G06 IGKV1-39 F2 Kappa Su01G06 IGHV1-69 T30S K64Q (SEQ ID NO: 254) I69L Heavy (SEQ ID NO: 66) Hu01G06-130 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-18 F1 Heavy (SEQ ID NO: 254) (SEQ ID NO: 246) Hu01G06-127 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-18 F2 Heavy (SEQ ID NO: 254) (SEQ ID NO: 248) Hu01G06-143 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-69 F1 Heavy (SEQ ID NO: 254) (SEQ ID NO: 250) Hu01G06-146 Hu01G06 IGKV1-39 F2 Kappa Hu01G06 IGHV1-69 F2 Heavy (SEQ ID NO: 254) (SEQ ID NO: 252)

[0298] Each of the possible combinations of the chimeric or humanized 06C11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 15.

TABLE-US-00083 TABLE 15 Antibody Light Chain Variable Heavy Chain Variable Name Region Region Hu06C11-1 Ch06C11 Chimeric Kappa Ch06C11 Chimeric Heavy (SEQ ID NO: 82) (SEQ ID NO: 46) Hu06C11-7 Ch06C11 Chimeric Kappa HE LM 06C11 IGHV2-70 (SEQ ID NO: 82) Heavy (SEQ ID NO: 68) Hu06C11-10 Ch06C11 Chimeric Kappa Hu06C11 IGHV2-5 Heavy (SEQ ID NO: 82) (SEQ ID NO: 70) Hu06C11-12 Sh06C11 IGKV1-16 Kappa Ch06C11 Chimeric Heavy (SEQ ID NO: 96) (SEQ ID NO: 46) Hu06C11-27 Sh06C11 IGKV1-16 Kappa HE LM 06C11 IGHV2-70 (SEQ ID NO: 96) Heavy (SEQ ID NO: 68) Hu06C11-30 Sh06C11 IGKV1-16 Kappa Hu06C11 IGHV2-5 Heavy (SEQ ID NO: 96) (SEQ ID NO: 70)

[0299] Each of the possible combinations of the chimeric or humanized 14F11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 16.

TABLE-US-00084 TABLE 16 Antibody Light Chain Variable Heavy Chain Variable Name Region Region Hu14F11-1 Ch14F11 Chimeric Kappa Ch14F11 Chimeric Heavy (SEQ ID NO: 86) (SEQ ID NO: 50) Hu14F11-14 Ch14F11 Chimeric Kappa Sh14F11 IGHV2-5 Heavy (SEQ ID NO: 86) (SEQ ID NO: 72) Hu14F11-15 Ch14F11 Chimeric Kappa Sh14F11 IGHV2-70 Heavy (SEQ ID NO: 86) (SEQ ID NO: 74) Hu14F11-11 Hu14F11 IGKV1-16 Kappa Ch14F11 Chimeric Heavy (SEQ ID NO: 98) (SEQ ID NO: 50) Hu14F11-39 Hu14F11 IGKV1-16 Kappa Sh14F11 IGHV2-5 Heavy (SEQ ID NO: 98) (SEQ ID NO: 72) Hu14F11-47 Hu14F11 IGKV1-16 Kappa Sh14F11 IGHV2-70 Heavy (SEQ ID NO: 98) (SEQ ID NO: 74)

[0300] Each of the possible combinations of the chimeric 04F08, 06C11, and 14F11 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 17.

TABLE-US-00085 TABLE 17 Light Chain Variable Region Heavy Chain Variable Region 04F08 Chimeric Kappa Ch06C11 Chimeric Heavy (SEQ ID NO: 80) (SEQ ID NO: 46) 04F08 Chimeric Kappa Ch14F11 Chimeric Heavy (SEQ ID NO: 80) (SEQ ID NO: 50) Ch06C11 Chimeric Kappa 04F08 Chimeric Heavy (SEQ ID NO: 82) (SEQ ID NO: 44) Ch06C11 Chimeric Kappa Ch14F11

Chimeric Heavy (SEQ ID NO: 82) (SEQ ID NO: 50) Ch14F11 Chimeric Kappa 04F08 Chimeric Heavy (SEQ ID NO: 86) (SEQ ID NO: 44) Ch14F11 Chimeric Kappa Ch06C11 Chimeric Heavy Light Chain Variable Region Heavy Chain Variable Region (SEQ ID NO: 86) (SEQ ID NO: 46) [0301] Each of the possible combinations of the chimeric 01G06 and chimeric 08G01 immunoglobulin heavy chain and immunoglobulin light chain variable regions is set forth below in Table 18.

TABLE-US-00086 TABLE 18 Light Chain Variable Region Heavy Chain Variable Region Ch01G06 Chimeric Kappa 08G01 Chimeric Heavy (SEQ ID NO: 76) (SEQ ID NO: 48) 08G01 Chimeric Kappa Ch01G06 Chimeric Heavy (SEQ ID NO: 84) (SEQ ID NO: 40)

[0302] The nucleic acid sequences and the encoded protein sequences defining variable regions of the chimeric and humanized 01G06, 06C11, and 14F11 antibodies are summarized below (amino terminal signal peptide sequences are not shown). CDR sequences (Kabat definition) are shown in bold and are underlined in the amino acid sequences.

Nucleic Acid Sequence Encoding the Ch01G06 Chimeric Heavy Chain Variable Region (SEQ ID NO:127)

TABLE-US-00087 1 gaagtgtgtg tgcagcagtc agggccggag ttggtaaaac cgggagcgtc
ggtgaaaatc 61 ccgtgcaaag cgtcggggta tacgtttacg gactataaca tggattgggt gaaacagtcg 121
catgggaaat cgcttgaatg gattggtcag atcaatccga ataattggagg aatcttctt 181 aatcagaagt
ttaaaggaaa agcgacgctt acagtcgata agtcgtcgaa cacggcgctt 241 atggaagtac ggctcgctac
gtcgggaagt acggcggtct attactgtgc gagggaggcg 301 attacgacgg tgggagcgat ggactattgg
ggacaaggga cgctcggtcac ggtatcgtcg

Protein Sequence Defining the Ch01G06 Chimeric Heavy Chain Variable Region (SEQ ID NO:40)

TABLE-US-00088 1 evllqqsgpe lvkpgasvki pckasgytft **dynmd**wvkqs hgkslewigq
inpnnggiff 61 **nqkfk**katl tvdkssntaf mevrsltsed tavyycarea **ittvgamdy**w gqgtsvtvss
Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18 Heavy Chain Variable Region (SEQ ID NO:53)

TABLE-US-00089 1 caagtgaac ttgtgcagtc ggggtgcggaa gtcaaaaagc cgggagcgtc
ggtgaaaagta 61 tcgtgtaaag cgtcggggata tacgtttacg gactataaca tggactgggt acgacaggca 121
ccggggaaat cgttggaatg gatcggacag attaattccga acaatggggg aattttctt 181 aatcagaaat
tcaaaggacg ggcgacgttg acggtcgata catcgacgaa tacggcgtat 241 atggaattga ggctcgcttcg
ctcggacgat acggcggtct attactgcgc cagggaggcg 301 atcacgacgg taggggcat ggattattgg
ggacagggga cgcttgtgac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-18 Heavy Chain Variable Region (SEQ ID NO:54)

TABLE-US-00090 1 qvqlvqsgae vkkpgasvk sckasgytft **dynmd**wvrqa pgkslewigq
inpnnggiff 61 **nqkfk**gratl tvdtstntay melrslrdd tavyycarea **ittvgamdy**w gqgtltvtss
Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69 Heavy Chain Variable Region (SEQ ID NO:55)

TABLE-US-00091 1 caagtccagc ttgtccagtc gggagcggaa gtgaagaaac cggggtcgtc
ggtcaaagta 61 tcgtgtaaag cgtcggggata tacgtttacg gactataaca tggattgggt acgacaggct 121
ccgggaaaat cattggaatg gattggacag attaattccga ataattggggg tatcttctt 181 aatcaaaagt
ttaaaggagg ggcgacgttg acggtggaaca aatcgacaaa tacggcgtat 241 atggaattgt cgctcgcttcg
gtcggaggac acggcggtgt attactgcgc gagggaggcg 301 atcacgacgg tcggggcgat ggattattgg
ggacagggga cgcttgtgac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-69 Heavy Chain Variable Region (SEQ ID NO:56)

TABLE-US-00092 1 qvqlvqsgae vkkpgssvk sckasgytft **dynmd**wvrqa pgkslewigq
inpnnggiff 61 **nqkfk**gratl tvdkstntay melsslrdd tavyycarea **ittvgamdy**w gqgtltvtss
Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region (SEQ ID NO:57)

TABLE-US-00093 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 181 aatcagaaat
tcaaaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggctgttgcg
atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg
Protein Sequence Defining the Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region (SEQ ID NO:58)

TABLE-US-00094 1 qvqlvqsgae vkkpgasvkv sckasgytft **dynmd**wvrqa pgqglewmgq
inpnnggiff 61 **nqkfkgr**vtl ttdtststay melrslrsdd tavyycarea **ittvgamdy**w gqgtlvtvss
Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18 M69L K640 G44S Heavy Chain
Variable Region (SEQ ID NO:59)

TABLE-US-00095 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacaga gccttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 181 aatcagaaat
tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggctgttgcg
atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg
Protein Sequence Defining the Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain Variable
Region (SEQ ID NO:60)

TABLE-US-00096 1 qvqlvqsgae vkkpgasvkv sckasgytft **dynmd**wvrqa pgqslewmgq
inpnnggiff 61 **nqkfkgr**vtl ttdtststay melrslrsdd tavyycarea **ittvgamdy**w gqgtlvtvss
Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable
Region (SEQ ID NO:61)

TABLE-US-00097 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttcttt 181 aatcagaaat
tccaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggctgttgcg
atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg
Protein Sequence Defining the Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable Region
(SEQ ID NO:62)

TABLE-US-00098 1 qvqlvqsgae vkkpgasvkv sckasgytft **dynmd**wvrqa pgqglewmgq
inpnnggiff 61 **nqkfkgr**vtl ttdtststay melrslrsdd tavyycarea **ittvgamdy**w gqgtlvtvss
Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable
Region (SEQ ID NO:63)

TABLE-US-00099 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc
gggtcaaagt 61 tcgtgtaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc 181 aatcagaagt
ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat 241 atggagttgt cgtcgttgcg
gtcggaggac acggcgggtct attactgcgc gaggggaagcg 301 attacgacgg tgggagcgat ggattattgg
gggcagggga cgttggtaac ggtgtcatcg
Protein Sequence Defining the Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable Region
(SEQ ID NO:64)

TABLE-US-00100 1 qvqlvqsgae vkkpgssvkv sckasgytfs **dynmd**wvrqa pgqglewmgq
inpnnggiff 61 **nqkfkgr**vtl tadkststay melsslrdd tavyycarea **ittvgamdy**w gqgtlvtvss
Nucleic Acid Sequence Encoding the Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable
Region (SEQ ID NO:65)

TABLE-US-00101 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc
gggtcaaagt 61 tcgtgtaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121

cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatggggg aatctttttc 181 aatcagaagt
ttcaggggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat 241 atggagttgt cgtcgttgcg
gtcggaggac acggcgggtct attactgcgc gaggggaagcg 301 attacgacgg tgggagcgat ggattattgg
gggcagggaa cgcttgtaac ggtgtcatcg

Protein Sequence Defining the Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable Region (SEQ ID NO:66)

TABLE-US-00102 1 qvqlvqsgae vkkpgssvk sckasgytfs **dynmd**wvrqa pgqglewmgq
inpnnggiff 61 **nqkfq**grvtl tadtststay melsslr sed tavyycarea **ittvgamdy**w gqgtltvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region (SEQ ID NO:245)

TABLE-US-00103 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacaga gccttgaatg gatggggcag attaaccgt acaatcacct gatcttcttt 181 aatcagaaat
tccagggaaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggtcgttgcg
atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgtttgtaac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region (SEQ ID NO:246)

TABLE-US-00104 1 qvqlvqsgae vkkpgasvk sckasgytft **dynmd**wvrqa pgqslewmqg
inpynhliff 61 **nqkfq**grvtl ttdtststay melrslrdd tavyycarea **ittvgamdy**w gqgtltvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region (SEQ ID NO:247)

TABLE-US-00105 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacaga gccttgaatg gatggggcag attaaccga ataatggact gatcttcttt 181 aatcagaaat
tccagggaaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggtcgttgcg
atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgtttgtaac ggtatcgtcg

Protein Sequence Defining the Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region (SEQ ID NO:248)

TABLE-US-00106 1 qvqlvqsgae vkkpgasvk sckasgytft **dynmd**wvrqa pgqslewmqg
inpnngliff 61 **nqkfq**grvtl ttdtststay melrslrdd tavyycarea **ittvgamdy**w gqgtltvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region (SEQ ID NO:249)

TABLE-US-00107 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc
ggtaaaagtg 61 tcgtgtaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121
cctgggcagg gacttgaatg gatgggtcag atcaatccga ataatgggt gatcttttc 181 aatcagaagt
ttaaagggag ggtaacgctg acggcggata aaagcacgtc aacggcgtat 241 atggagttgt cgtcgttgcg
gtcggaggac acggcgggtct attactgcgc gaggggaagcg 301 attacgacgg tgggagcgat ggattattgg
gggcagggaa cgcttgtaac ggtgtcatcg

Protein Sequence Defining the Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region (SEQ ID NO:250)

TABLE-US-00108 1 qvqlvqsgae vkkpgssvk sckasgytfs **dynmd**wvrqa
pgqglewmgq **inpnngliff** 61 **nqkfk**grvtl tadtststay melsslr sed tavyycarea **ittvgamdy**w
gqgtltvss

Nucleic Acid Sequence Encoding the Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region (SEQ ID NO:251)

TABLE-US-00109 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc
ggtaaaagtg 61 tcgtgtaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121
cctgggcagg gacttgaatg gatgggtcag atcaatccgt acaatcacct gatcttttc 181 aatcagaagt

ttaaagggag ggaacgctg acggcggata aaagcacgtc aacggcgtat 241 atggagtgt cgtcgttgcg
gtcggaggac acggcggctt attactgcgc gagggaaagcg 301 attacgacgg tgggagcgat ggattattgg
gggcagggaa cgcttgtaac ggtgtcatcg

Protein Sequence Defining the Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region (SEQ ID NO:252)

TABLE-US-00110 1 qvqlvqsgae vkkpgssvk sckasgytfs **dynmd**wvrqa pgqglewmgq
inpynhliff 61 **nqkfk**grvtl tadkststay melsslr sed tavyycare**a** **ittvgamdyw** gggtlvtvss

Nucleic Acid Sequence Encoding the Ch06C11 Chimeric Heavy Chain Variable Region (SEQ ID NO:129)

TABLE-US-00111 1 caggtgacac tcaaagaatc aggaccgga atccttcagc ccagccagac
cttgcgctg 61 acttggtcgt tctccggtt cagcctgaat acttatggga tgggtgtgtc atggatcagg 121
caaccgtccg ggaagagatt ggagtggctc gcgcacatct actgggacga tgacaaacgc 181 tacaatcctt
cgctgaagag ccgattgacg attccaagg atgcctcgaa caaccgggta 241 tttcttaaga tcacgtcgg
cgatacggca gacacggcga cctattactg cgcccaaaga 301 gggtacgatg actattgggg atattggggc
caggggacac tcgtcacaat ttcagct

Protein Sequence Defining the Ch06C11 Chimeric Heavy Chain Variable Region (SEQ ID NO:46)

TABLE-US-00112 1 qvtlkesgpg ilqpsqtlsl tcsfsgfsln **tygm**gvswir qpsgkglewl

ahiywdddkr 61 **ynpslks**rlt iskdasnrv flkitsvda dtatyycar**r** **gyddywgywg** qgtlvtisa

Nucleic Acid Sequence Encoding the HE LM 06C11 IGHV2-70 Heavy Chain Variable Region (SEQ ID NO:67)

TABLE-US-00113 1 caggtgactt tgaaagaatc cggccccga ttggtaaagc caaccagac
acttacgctc 61 acatgtacat ttccggatt cagcttgaac acttacggga tgggagtgtc gtggattcgg 121
caacctccgg ggaaggctct ggagtggctg gcgcacatct actgggatga tgacaaaagg 181 tataaccctt
cacttaaac gagactgacg atctcgaagg acacaagcaa gaatcaggtc 241 gtcctcacga ttacgaatgt
agaccgggtg gatactgccg tctattactg cgcgcaacgc 301 gggtatgatg actactgggg atattggggt
cagggcaccc tcgtgacat ctcgtca

Protein Sequence Defining the HE LM 06C11 IGHV2-70 Heavy Chain Variable Region (SEQ ID NO:68)

TABLE-US-00114 1 qvtlkesgpa lvkptqtltl tctfsgfsln **tygm**gvswir qppgkalewl

ahiywdddkr 61 **ynpslkt**rlt iskdtknqv vltitndpv dtavyycar**r** **gyddywgywg** qgtlvtiss

Nucleic Acid Sequence Encoding the Hu06C11 IGHV2-5 Heavy Chain Variable Region (SEQ ID NO:69)

TABLE-US-00115 1 caagtaacgc tcaaggagtc cggaccacc ttggtgaagc cgacgcagac cttgactctt
61 acgtgcactt tctcgggggtt ttcactgaat acgtacggga tgggtgtctc atggatcagg 121 caacctccgg
ggaagagatt ggaatggctg gcgcacatct actgggatga cgataagaga 181 tataacccaa gcctcaagtc
gcggctcacc attacaaaag atacatcgaa aaatcaggtc 241 gtacttacta tcacgaacat ggaccccggtg
gacacagcaa catattactg tgcccagcgc 301 ggctatgacg attattgggg ttactgggga caggaacac
tggtcacggt gtccagc

Protein Sequence Defining the Hu06C11 IGHV2-5 Heavy Chain Variable Region (SEQ ID NO:70)

TABLE-US-00116 1 qvtlkesgpt lvkptqtltl tctfsgfsln **tygm**gvswir qppgkglewl **ahiywdddkr**
61 **ynpslks**rlt itkdtknqv vltitnmdpv dtatyycar**r** **gyddywgywg** qgtlvtvss

Nucleic Acid Sequence Encoding the Ch14F11 Chimeric Heavy Chain Variable Region (SEQ ID NO:131)

TABLE-US-00117 1 caggtcacgc tgaaagagtc aggtcccga atccttcaac cttcgacagac attgtcactc
61 acatgttctt tctccgggtt ctcgtctcgc acttatggca tgggttagg atggatcgg 121 cagcccagcg
ggaaggggct tgagtgggtg gcggatatct ggtgggacga cgacaaatac 181 tacaatccga gcctgaagtc
ccgctcacc atttcgaaag atacgtcatc aaacgaagtc 241 ttttgaaga tcgccatcgt ggacacggcg
gatacagcga cgtattactg cgccagaagg 301 ggacactaca gcgcaatgga ttattgggga caggggacct
cggtgactgt gtcgtcc

Protein Sequence Defining the Ch14F11 Chimeric Heavy Chain Variable Region (SEQ ID NO:50)

TABLE-US-00118 1 qvtlkesgpg ilqpsqtll tcsfsgfsls **tygmvgvwir** qpsgkglewl
adiwwdddky 61 **ynpslks**rslt iskdtsnev flkiaivdta dtatyycarr **ghysamdy**wg qgtsvtvss
Nucleic Acid Sequence Encoding the Sh14F11 IGHV2-5 Heavy Chain Variable Region (SEQ ID
NO:71)

TABLE-US-00119 1 cagatcactt tgaagaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc
61 acgtgtacat ttcgggggtt ctcgcttca acttacggga tgggagtagg gtggattcgc 121 cagccgcctg
gtaaagcggt ggagtggctt gcagacatct ggtgggacga cgataagtac 181 tataatccct cgctcaagtc
cagactgacc atcacgaaag atacgagcaa gaaccaggtc 241 gtgctgacaa tgactaacat ggaccagtg
gatacggcta catattactg cgccaggcgg 301 ggtcactact cagcgatgga ttattggggc cagggaaacac
tggtaacggt gtcgtcc

Protein Sequence Defining the Sh14F11 IGHV2-5 Heavy Chain Variable Region (SEQ ID NO:72)

TABLE-US-00120 1 qitlkesgpt lvkptqtll tctfsgfsls **tygmvgvwir** qppgkalewl **adiwwddd**ky
61 **ynpslks**rslt itkdtsknqv vltmtnmdpv dtatyycarr **ghysamdy**wg qgtltvtvss
Nucleic Acid Sequence Encoding the Sh14F11 IGHV2-70 Heavy Chain Variable Region (SEQ ID
NO:73)

TABLE-US-00121 1 caagtgactc tcaaggagtc cggacccgcc ctggtcaaac caacgcagac actgacgctc
61 acatgcacct tcagcggatt ttcgtgtca acgtacggca tgggtgtggg gtggattcgc 121 cagcctccgg
ggaaagccct tgaatgggtg gcggacatct ggtgggatga tgacaagtac 181 tataatccct cacttaagtc
acggttgacg atctcgaaag acaccagcaa gaaccaggta 241 gtgctgacaa tgactaacat ggacccggtc
gatacagcgg tctactattg tgctagaagg 301 ggacactact ccgcaatgga ttattgggggt caggggacgc
tcgtaaccgt gtcgtcg

Protein Sequence Defining the Sh14F11 IGHV2-70 Heavy Chain Variable Region (SEQ ID
NO:74)

TABLE-US-00122 1 qvtlkesgpa lvkptqtll tctfsgfsls **tygmvgvwir** qppgkalewl
adiwwdddky 61 **ynpslks**rslt iskdtsknqv vltmtnmdpv dtavyycarr **ghysamdy**wg qgtltvtvss
Nucleic Acid Sequence Encoding the Ch01G06 Chimeric Kappa Chain Variable Region (SEQ ID
NO:133)

TABLE-US-00123 1 gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg
61 atcacgtgcc ggacatcaga gaatctccat aactacctg cgtggtatca acagaagcag 121 gggaagtcgc
cccagttgct tgtatacat gcgaaaacgt tggcggatgg ggtgccgtcc 181 agattctcgg gatcgggctc
ggggacgcag tactcgctca agatcaattc gctgcagccg 241 gaggactttg ggctgtacta ttgtcagcat
tttgggtcat caccgtatac atttgaggt 301 ggaacgaaac ttgagattaa g

Protein Sequence Defining the Ch01G06 Chimeric Kappa Chain Variable Region (SEQ ID NO:76)

TABLE-US-00124 1 diqmtqspas lsasvgetvt itc**rtsenlh** **nylawyqqkq** gkspqllvyd
aktladgvps 61 rfsgsgsgtq yslkinslqp edfsgsyyc**qh** **fwsspyt**fgg gtleik
Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 Kappa Chain Variable Region (SEQ ID
NO:89)

TABLE-US-00125 1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcatt aactacctg cgtggtatca gcagaagccc 121 gggaagtcac
cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtagcgac tatacgtta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttgggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa g

Protein Sequence Defining the Hu01G06 IGKV1-39 Kappa Chain Variable Region (SEQ ID
NO:90)

TABLE-US-00126 1 diqmtqspss lsasvgdrvt itc**rtsenlh** **nylawyqqkp** gkspkllvyd
aktladgvps 61 rfsgsgsgtd yltisslqp edfatyyc**qh** **fwsspyt**fgq gtleik
Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable
Region (Also Referred to Herein as Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region; SEQ
ID NO:91)

TABLE-US-00127 1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca

61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121 gggaaggccc
cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtacggac tatacgtta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa g
Protein Sequence Defining the Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable Region
(Also Referred to Herein as Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region; SEQ ID
NO:92)

TABLE-US-00128 1 diqmtqspss lsasvgdrvt itc**rtsenlh** **nylawyqqkp** gkapklli**d**
aktladgvps 61 rfsgsgsgtd yltisslqp edfatyyc**qh** **fwsspytfgq** gtkleik
Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 V48I Kappa Chain Variable Region
(SEQ ID NO:93)

TABLE-US-00129 1 gacatccaaa tgaccagtc gccgtcgctcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121 gggaagtcac
cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtacggac tatacgtta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa g
Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 V48I Kappa Chain Variable Region
(SEQ ID NO:94)

TABLE-US-00130 1 diqmtqspss lsasvgdrvt itc**rtsenlh** **nylawyqqkp** gkspklli**d**
aktladgvps 61 rfsgsgsgtd yltisslqp edfatyyc**qh** **fwsspytfgq** gtkleik
Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 F1 Kappa Chain Variable
Region (Also Referred to Herein as Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable
Region; SEQ ID NO:91)

TABLE-US-00131 1 gacatccaaa tgaccagtc gccgtcgctcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121 gggaaggccc
cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtacggac tatacgtta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa g
Protein Sequence Defining the Hu01G06 IGKV1-39 F1 Kappa Chain Variable Region (Also
Referred to Herein as Hu01G06 IGKV1-39 S43A V48I Kappa Chain Variable Region; SEQ ID
NO:92)

TABLE-US-00132 1 diqmtqspss lsasvgdrvt itc**rtsenlh** **nylawyqqkp** gkapklli**d**
aktladgvps 61 rfsgsgsgtd yltisslqp edfatyyc**qh** **fwsspytfgq** gtkleik
Nucleic Acid Sequence Encoding the Hu01G06 IGKV1-39 F2 Kappa Chain Variable Region (SEQ
ID NO:253)

TABLE-US-00133 1 gacatccaaa tgaccagtc gccgtcgctcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121 gggaagtcac
cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtacggac tatacgtta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttggtcgg acccctacac atttgggcag 301 gggaccaagt tggaaatcaa g
Protein Sequence Defining the Hu01G06 IGKV1-39 F2 Kappa Chain Variable Region (SEQ ID
NO:254)

TABLE-US-00134 1 diqmtqspss lsasvgdrvt itc**rtsenlh** **nylawyqqkp** gkspklli**d**
aktladgvps 61 rfsgsgsgtd yltisslqp edfatyyc**qh** **fwsdpytfgq** gtkleik
Nucleic Acid Sequence Encoding the Ch06C11 Chimeric Kappa Chain Variable Region (SEQ ID
NO:135)

TABLE-US-00135 1 gatatcgta tgaccagtc ccagaagttc atgtcaactt cagtgggaga cagagtgtcc
61 gtcacatgta aagcctcgca aaatgtggga accaacgtag cgtggtcca gcagaaacct 121 ggccaatcac
cgaaggcact gatctactcg gccagctata ggtactcggg agtaccagat 181 cggtttacgg ggtcggggag
cgggacggac ttatcctca ctatttcaa tgccagtcg 241 gaggaccttg cggaaactt ctgccagcag

tataacaact atccccctcac gtttggtgct 301 ggtacaaaat tggagttgaa g
Protein Sequence Defining the Ch06C11 Chimeric Kappa Chain Variable Region (SEQ ID NO:82)
TABLE-US-00136 1 divmtsqskf mstsvgdrvs vtc**kasqnv**g **tnvawf**qqkp gqspkaliys
asyrysgvpd 61 rftgsgsgtd ftiltsnvqs edlaeyfc**qq ynnyp**ltfga gtklelk
Nucleic Acid Sequence Encoding the Sh06C11 IGKV1-16 Kappa Chain Variable Region (SEQ ID NO:95)

TABLE-US-00137 1 gacatccaaa tgaccaatc gccctcctcc ctctccgcat cagtagggga ccgcgtcaca
61 attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc 121 ggaaaagctc
cgaagagctt gatctactcg gcctcatata ggtattcggg tgtgccgagc 181 cggtttagcg ggtcgggggc
aggtagctat ttcacgtcga caatttcacg gttgcagcca 241 gaagatttcg ccacatatta ctgtagcag
tacaacaatt accctctgac gttcggccag 301 ggaaccaaac ttgagatcaa g
Protein Sequence Defining the Sh06C11 IGKV1-16 Kappa Chain Variable Region (SEQ ID NO:96)

TABLE-US-00138 1 diqmtqspss lsasvgdrvt itc**kasqnv**g **tnvawf**qqkp gkapksliys
asyrysgvps 61 rfsgsgsgtd ftiltsllqp edfatyy**cqq ynnyp**ltfgq gtkleik
Nucleic Acid Sequence Encoding the Ch14F11 Chimeric Kappa Chain Variable Region (SEQ ID NO:137)

TABLE-US-00139 1 gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcgggtga tagagtatcc 61
gtcacgtgta aggcctcgca aaacgtagga actaatgtgg cgtggatatca acagaagcca 121 ggacagtcac
ccaaagcact catctacagc ccctcatatc ggtacagcgg ggtgccggac 181 aggttcacgg gatcggggag
cgggaccgat ttactactga ccatttcgaa tgtccagtcg 241 gaggacctg cggaatactt ctgccagcag
tataactcgt accctcacac gtttgagggt 301 ggcactaagt tggagatgaa a
Protein Sequence Defining the Ch14F11 Chimeric Kappa Chain Variable Region (SEQ ID NO:86)

TABLE-US-00140 1 divmtsqskf mstsvgdrvs vtc**kasqnv**g **tnvaw**yqqkp gqspkaliys
psyrysgvpd 61 rftgsgsgtd ftiltsnvqs edlaeyfc**qq ynsyp**htfgg gtklemk
Nucleic Acid Sequence Encoding the Hu14F11 IGKV1-16 Kappa Chain Variable Region (SEQ ID NO:97)

TABLE-US-00141 1 gatatccaga tgacacagtc accctcgtcg ctctcagctt ccgtaggcga cagggtcact
61 attacgtgta aagcatcaca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc 121 gggaagagcc
ccaaagcgct tatctactcc ccgtcgtatc ggtattccgg tgtgccaagc 181 agattttcgg ggtaggttc
gggaactgac ttaccctga ccatctcgtc cctccaaccg 241 gaagatttcg ccacgtactt ctgccagcag
tacaacagct atctcacac attcggacaa 301 gggacaaagt tggagattaa a
Protein Sequence Defining the Hu14F11 IGKV1-16 Kappa Chain Variable Region (SEQ ID NO:98)

TABLE-US-00142 1 diqmtqspss lsasvgdrvt itc**kasqnv**g **tnvawf**qqkp gkspkaliys
psyrysgvps 61 rfsgsgsgtd ftiltsllqp edfatyfc**qq ynsyp**htfgq gtkleik
[0303] The amino acid sequences defining the immunoglobulin heavy chain variable regions for the antibodies produced in Example 13 are aligned in FIG. 19. Amino terminal signal peptide sequences (for proper expression/secretion) are not shown. CDR.sub.1, CDR.sub.2, and CDR.sub.3 (Kabat definition) are identified by boxes. FIG. 20 show an alignment of the separate CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the variable region sequences shown in FIG. 19.
[0304] The amino acid sequences defining the immunoglobulin light chain variable regions for the antibodies in Example 13 are aligned in FIG. 21. Amino terminal signal peptide sequences (for proper expression/secretion) are not shown. CDR.sub.1, CDR.sub.2 and CDR.sub.3 are identified by boxes. FIG. 22 shows an alignment of the separate CDR.sub.1, CDR.sub.2, and CDR.sub.3 sequences for each of the variable region sequences shown in FIG. 21.

[0305] Table 19 is a concordance chart showing the SEQ ID NO. of each sequence discussed in this Example.

TABLE-US-00143 TABLE 19 SEQ. ID NO. Nucleic Acid or Protein 127 Ch01G06 Chimeric Heavy Chain Variable Region - nucleic acid 40 Ch01G06 Chimeric Heavy Chain Variable Region -

protein 1 Ch01G06 Chimeric Heavy Chain CDR.sub.1 7 Ch01G06 Chimeric Heavy Chain CDR.sub.2 15 Ch01G06 Chimeric Heavy Chain CDR.sub.3 53 Hu01G06 IGHV1-18 Heavy Chain Variable Region - nucleic acid 54 Hu01G06 IGHV1-18 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-18 Heavy Chain CDR.sub.1 7 Hu01G06 IGHV1-18 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-18 Heavy Chain CDR.sub.3 55 Hu01G06 IGHV1-69 Heavy Chain Variable Region - nucleic acid 56 Hu01G06 IGHV1-69 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-69 Heavy Chain CDR.sub.1 7 Hu01G06 IGHV1-69 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-69 Heavy Chain CDR.sub.3 57 Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region - nucleic acid 58 Sh01G06 IGHV1-18 M69L Heavy Chain Variable Region - protein 1 Sh01G06 IGHV1-18 M69L Heavy Chain CDR.sub.1 7 Sh01G06 IGHV1-18 M69L Heavy Chain CDR.sub.2 15 Sh01G06 IGHV1-18 M69L Heavy Chain CDR.sub.3 59 Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain Variable Region - nucleic acid 60 Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain Variable Region - protein 1 Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR.sub.1 13 Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR.sub.2 15 Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain CDR.sub.3 61 Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable Region - nucleic acid 62 Sh01G06 IGHV1-18 M69L K64Q Heavy Chain Variable Region - protein 1 Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR.sub.1 13 Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR.sub.2 15 Sh01G06 IGHV1-18 M69L K64Q Heavy Chain CDR.sub.3 63 Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable Region - nucleic acid 64 Sh01G06 IGHV1-69 T30S I69L Heavy Chain Variable Region - protein 1 Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR.sub.1 7 Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR.sub.2 15 Sh01G06 IGHV1-69 T30S I69L Heavy Chain CDR.sub.3 65 Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable Region - nucleic acid 66 Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain Variable Region - protein 1 Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR.sub.1 13 Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR.sub.2 15 Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain CDR.sub.3 245 Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region - nucleic acid 246 Hu01G06 IGHV1-18 F1 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-18 F1 Heavy Chain CDR.sub.1 236 Hu01G06 IGHV1-18 F1 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-18 F1 Heavy Chain CDR.sub.3 247 Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region - nucleic acid 248 Hu01G06 IGHV1-18 F2 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-18 F2 Heavy Chain CDR.sub.1 237 Hu01G06 IGHV1-18 F2 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-18 F2 Heavy Chain CDR.sub.3 259 Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region - nucleic acid 250 Hu01G06 IGHV1-69 F1 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-69 F1 Heavy Chain CDR.sub.1 238 Hu01G06 IGHV1-69 F1 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-69 F1 Heavy Chain CDR.sub.3 251 Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region - nucleic acid 252 Hu01G06 IGHV1-69 F2 Heavy Chain Variable Region - protein 1 Hu01G06 IGHV1-69 F2 Heavy Chain CDR.sub.1 239 Hu01G06 IGHV1-69 F2 Heavy Chain CDR.sub.2 15 Hu01G06 IGHV1-69 F2 Heavy Chain CDR.sub.3 129 Ch06C11 Chimeric Heavy Chain Variable Region - nucleic acid 46 Ch06C11 Chimeric Heavy Chain Variable Region - protein 4 Ch06C11 Chimeric Heavy Chain CDR.sub.1 9 Ch06C11 Chimeric Heavy Chain CDR.sub.2 18 Ch06C11 Chimeric Heavy Chain CDR.sub.3 67 HE LM 06C11 IGHV2-70 Heavy Chain Variable Region - nucleic acid 68 HE LM 06C11 IGHV2-70 Heavy Chain Variable Region - protein 4 HE LM 06C11 IGHV2-70 Heavy Chain CDR.sub.1 14 HE LM 06C11 IGHV2-70 Heavy Chain CDR.sub.2 18 HE LM 06C11 IGHV2-70 Heavy Chain CDR.sub.3 69 Hu06C11 IGHV2-5 Heavy Chain Variable Region - nucleic acid 70 Hu06C11 IGHV2-5 Heavy Chain Variable Region - protein 4 Hu06C11 IGHV2-5 Heavy Chain CDR.sub.1 9 Hu06C11 IGHV2-5 Heavy Chain CDR.sub.2 18 Hu06C11 IGHV2-5 Heavy Chain CDR.sub.3 131 Ch14F11 Chimeric Heavy Chain Variable Region - nucleic acid 50 Ch14F11 Chimeric Heavy Chain Variable Region - protein 5 Ch14F11 Chimeric Heavy Chain CDR.sub.1 11 Ch14F11 Chimeric Heavy Chain CDR.sub.2 19 Ch14F11 Chimeric Heavy Chain CDR.sub.3 71 Sh14F11

IGHV2-5 Heavy Chain Variable Region - nucleic acid 72 Sh14F11 IGHV2-5 Heavy Chain Variable Region - protein 5 Sh14F11 IGHV2-5 Heavy Chain CDR.sub.1 11 Sh14F11 IGHV2-5 Heavy Chain CDR.sub.2 19 Sh14F11 IGHV2-5 Heavy Chain CDR.sub.3 73 Sh14F11 IGHV2-70 Heavy Chain Variable Region - nucleic acid 74 Sh14F11 IGHV2-70 Heavy Chain Variable Region - protein 5 Sh14F11 IGHV2-70 Heavy Chain CDR.sub.1 11 Sh14F11 IGHV2-70 Heavy Chain CDR.sub.2 19 Sh14F11 IGHV2-70 Heavy Chain CDR.sub.3 133 Ch01G06 Chimeric Light (kappa) Chain Variable Region - nucleic acid 76 Ch01G06 Chimeric Light (kappa) Chain Variable Region - protein 21 Ch01G06 Chimeric Light (kappa) Chain CDR.sub.1 26 Ch01G06 Chimeric Light (kappa) Chain CDR.sub.2 32 Ch01G06 Chimeric Light (kappa) Chain CDR.sub.3 89 Hu01G06 IGKV1-39 Light (kappa) Chain Variable Region - nucleic acid 90 Hu01G06 IGKV1-39 Light (kappa) Chain Variable Region - protein 21 Hu01G06 IGKV1-39 Light (kappa) Chain CDR.sub.1 26 Hu01G06 IGKV1-39 Light (kappa) Chain CDR.sub.2 32 Hu01G06 IGKV1-39 Light (kappa) Chain CDR.sub.3 91 Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain Variable Region - nucleic acid 92 Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain Variable Region - protein 21 Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR.sub.1 26 Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR.sub.2 32 Hu01G06 IGKV1-39 S43A V48I Light (kappa) Chain CDR.sub.3 93 Hu01G06 IGKV1-39 V48I Light (kappa) Chain Variable Region - nucleic acid 94 Hu01G06 IGKV1-39 V48I Light (kappa) Chain Variable Region - protein 21 Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR.sub.1 26 Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR.sub.2 32 Hu01G06 IGKV1-39 V48I Light (kappa) Chain CDR.sub.3 91 Hu01G06 IGKV1-39 F1 Light (kappa) Chain Variable Region - nucleic acid 92 Hu01G06 IGKV1-39 F1 Light (kappa) Chain Variable Region - protein 21 Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR.sub.1 26 Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR.sub.2 32 Hu01G06 IGKV1-39 F1 Light (kappa) Chain CDR.sub.3 253 Hu01G06 IGKV1-39 F2 Light (kappa) Chain Variable Region - nucleic acid 254 Hu01G06 IGKV1-39 F2 Light (kappa) Chain Variable Region - protein 21 Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR.sub.1 26 Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR.sub.2 244 Hu01G06 IGKV1-39 F2 Light (kappa) Chain CDR.sub.3 135 Ch06C11 Chimeric Light (kappa) Chain Variable Region - nucleic acid 82 Ch06C11 Chimeric Light (kappa) Chain Variable Region - protein 23 Ch06C11 Chimeric Light (kappa) Chain CDR.sub.1 28 Ch06C11 Chimeric Light (kappa) Chain CDR.sub.2 35 Ch06C11 Chimeric Light (kappa) Chain CDR.sub.3 95 Sh06C11 IGKV1-16 Light (kappa) Chain Variable Region - nucleic acid 96 Sh06C11 IGKV1-16 Light (kappa) Chain Variable Region - protein 23 Sh06C11 IGKV1-16 Light (kappa) Chain CDR.sub.1 28 Sh06C11 IGKV1-16 Light (kappa) Chain CDR.sub.2 35 Sh06C11 IGKV1-16 Light (kappa) Chain CDR.sub.3 137 Ch14F11 Chimeric Light (kappa) Chain Variable Region - nucleic acid 86 Ch14F11 Chimeric Light (kappa) Chain Variable Region - protein 23 Ch14F11 Chimeric Light (kappa) Chain CDR.sub.1 30 Ch14F11 Chimeric Light (kappa) Chain CDR.sub.2 36 Ch14F11 Chimeric Light (kappa) Chain CDR.sub.3 97 Hu14F11 IGKV1-16 Light (kappa) Chain Variable Region - nucleic acid 98 Hu14F11 IGKV1-16 Light (kappa) Chain Variable Region - protein 23 Hu14F11 IGKV1-16 Light (kappa) Chain CDR.sub.1 30 Hu14F11 IGKV1-16 Light (kappa) Chain CDR.sub.2 36 Hu14F11 IGKV1-16 Light (kappa) Chain CDR.sub.3

[0306] Humanized monoclonal antibody heavy chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 20.

TABLE-US-00144	TABLE	20	Variable	Region	CDR1	CDR2	CDR3	SEQ	ID	NO:	Kabat
Ch01G06	Chimeric	DYNMD	QINPNNGG	IFFNQKFKG	EAITTVG	GAMDY	40	(SEQ	ID	NO:	1)
								(SEQ	ID	NO:	7)
								(SEQ	ID	NO:	15)
Hu01G06	IGHV1-	DYNMD	QINPNNGG	IFFNQKFKG	EAITTVG	GAMDY	54	18	(SEQ	ID	NO:
									(SEQ	ID	NO:
									(SEQ	ID	NO:
Hu01G06	IGHV1-	DYNMD	QINPNNGG	IFFNQKFKG	EAITTVG	GAMDY	56	69	(SEQ	ID	NO:
									(SEQ	ID	NO:
									(SEQ	ID	NO:
Sh01G06	IGHV1-	DYNMD	QINPNNGG	IFFNQKFKG	EAITTVG	GAMDY	58	18			M69L

(SEQ ID NO: 1)(SEQ ID NO: 7)(SEQ ID NO: 15)Sh01G06 IGHV1- DYNMD
QINPNNGGIFFNQKFQG EAITTVGAMDY 60 18 M69L K64Q(SEQ ID NO: 1)
(SEQ ID NO: 13)(SEQ ID NO: 15)G44S Sh01G06 IGHV1- DYNMD
QINPNNGGIFFNQKFQG EAITTVGAMDY 62 18 M69L K64Q(SEQ ID NO: 1)
(SEQ ID NO: 13)(SEQ ID NO: 15)Sh01G06 IGHV1- DYNMD
QINPNNGGIFFNQKFKG EAITTVGAMDY 64 69 T30S I69L(SEQ ID NO: 1)(SEQ
ID NO: 7)(SEQ ID NO: 15)Sh01G06 IGHV1- DYNMD QINPNNGGIFFNQKFQG
EAITTVGAMDY 66 69 T30S K64Q(SEQ ID NO: 1)(SEQ ID NO: 13)(SEQ
ID NO: 15)I69L Hu01G06 IGHV1- DYNMD QINPYNHLIFFNQKFQG EAITTVGAMDY
246 18F1(SEQ ID NO: 1)(SEQ ID NO: 236)(SEQ ID NO: 15)Hu01G06
IGHV1- DYNMD QINPNNGLIFFNQKFQG EAITTVGAMDY 248 18F2(SEQ ID NO: 1)
(SEQ ID NO: 237)(SEQ ID NO: 15)Hu01G06 IGHV1- DYNMD
QINPNNGLIFFNQKFKG EAITTVGAMDY 250 69F1(SEQ ID NO: 1)(SEQ ID NO:
238)(SEQ ID NO: 15)Hu01G06 IGHV1- DYNMD QINPYNHLIFFNQKFKG
EAITTVGAMDY 252 69F2(SEQ ID NO: 1)(SEQ ID NO: 239)(SEQ ID NO:
15)Ch06C11 Chimeric TYGMGVS HIYWDDDDKRYNPSLKS RGYDDYWGY 46(SEQ ID
NO: 4)(SEQ ID NO: 9)(SEQ ID NO: 18)HE LM 06C11 TYGMGVS
HIYWDDDDKRYNPSLKT RGYDDYWGY 68 IGHV2-70(SEQ ID NO: 4)(SEQ ID
NO: 14)(SEQ ID NO: 18)Hu06C11 IGHV2- TYGMGVS HIYWDDDDKRYNPSLKS
RGYDDYWGY 70 5(SEQ ID NO: 4)(SEQ ID NO: 9)(SEQ ID NO: 18)
Ch14F11 Chimeric TYGMGVG DIWWDDDDKYYNPSLKS RGHYSAMDY 50(SEQ ID
NO: 5)(SEQ ID NO: 11)(SEQ ID NO: 19)Sh14F11 IGHV2-5 TYGMGVG
DIWWDDDDKYYNPSLKS RGHYSAMDY 72 (SEQ ID NO: 5)(SEQ ID NO: 11)
(SEQ ID NO: 19)Sh14F11 IGHV2- TYGMGVG DIWWDDDDKYYNPSLKS
RGHYSAMDY 74 70(SEQ ID NO: 5)(SEQ ID NO: 11)(SEQ ID NO: 19)
Chothia Ch01G06 GYTFTDY NPNNGG EAITTVGAMDY 40 Chimeric(SEQ ID NO: 38)
(SEQ ID NO: 143)(SEQ ID NO: 15)Hu0106 IGHV1- GYTFTDY NPNNGG
EAITTVGAMDY 54 18(SEQ ID NO: 38)(SEQ ID NO: 143)(SEQ ID NO: 15)
Hu01G06 IGHV1- GYTFTDY NPNNGG EAITTVGAMDY 56 69(SEQ ID NO: 38)
(SEQ ID NO: 143)(SEQ ID NO: 15)Sh01G06 IGHV1- GYTFTDY NPNNGG
EAITTVGAMDY 58 18 M69L(SEQ ID NO: 38)(SEQ ID NO: 143)(SEQ ID
NO: 15)Sh01G06 IGHV1- GYTFTDY NPNNGG EAITTVGAMDY 60 18 M69L K64Q
(SEQ ID NO: 38)(SEQ ID NO: 143)(SEQ ID NO: 15)G44S Sh01G06 IGHV1-
GYTFTDY NPNNGG EAITTVGAMDY 62 18 M69L K64Q(SEQ ID NO: 38)(SEQ
ID NO: 143)(SEQ ID NO: 15)Sh01G06 IGHV1- GYTFSDY NPNNGG
EAITTVGAMDY 64 69 T30S I69L(SEQ ID NO: 234)(SEQ ID NO: 143)(SEQ
ID NO: 15)Sh01G06 IGHV1- GYTFSDY NPNNGG EAITTVGAMDY 66 69 T30S
K64Q(SEQ ID NO: 234)(SEQ ID NO: 143)(SEQ ID NO: 15)I69L Hu01G06
IGHV1- GYTFTDY NPYNHL EAITTVGAMDY 246 18 F1(SEQ ID NO: 38)(SEQ
ID NO: 240)(SEQ ID NO: 15)Hu01G06 IGHV1- GYTFTDY NPNNGL
EAITTVGAMDY 248 18 F2(SEQ ID NO: 38)(SEQ ID NO: 241)(SEQ ID
NO: 15)Hu01G06 IGHV1- GYTFSDY NPNNGL EAITTVGAMDY 250 69 F1(SEQ ID
NO: 234)(SEQ ID NO: 241)(SEQ ID NO: 15)Hu01G06 IGHV1- GYTFSDY
NPYNHL EAITTVGAMDY 252 69 F2(SEQ ID NO: 234)(SEQ ID NO: 240)
(SEQ ID NO: 15)Ch06C11 GFSLNTYGM YWDDD RGYDDYWGY 46 Chimeric(SEQ
ID NO: 132)(SEQ ID NO: 145)(SEQ ID NO: 18)HE LM 06C11
GFSLNTYGM YWDDD RGYDDYWGY 68 IGHV2-70(SEQ ID NO: 132)(SEQ ID
NO: 145)(SEQ ID NO: 18)Hu06C11 IGHV2- GFSLNTYGM YWDDD RGYDDYWGY
70 5(SEQ ID NO: 132)(SEQ ID NO: 145)(SEQ ID NO: 18)Ch14F11
Chimeric GFSLSTYGM WWDDD RGHYSAMDY 50(SEQ ID NO: 130)(SEQ ID

NO: 146) (SEQ ID NO: 19) GFSLSTYGM WWDDD RGHYSAMDY 72 Sh14F11
 IGHV2-5 (SEQ ID NO: 130) (SEQ ID NO: 146) (SEQ ID NO: 19) Sh14F11
 IGHV2- GFSLSTYGM WWDDD RGHYSAMDY 74 70 (SEQ ID NO: 130) (SEQ ID
 NO: 146) (SEQ ID NO: 19) IMGT Ch01G06 GYTFTDYN INPNNGGI
 AREAITTVGAMDY 40 Chimeric (SEQ ID NO: 136) (SEQ ID NO: 148) (SEQ ID
 NO: 154) Hu01G06 GYTFTDYN INPNNGGI AREAITTVGAMDY 54 IGHV1-18 (SEQ ID
 NO: 136) (SEQ ID NO: 148) (SEQ ID NO: 154) Hu01G06 GYTFTDYN INPNNGGI
 AREAITTVGAMDY 56 IGHV1-69 (SEQ ID NO: 136) (SEQ ID NO: 148) (SEQ
 ID NO: 154) Sh01G06 GYTFTDYN INPNNGGI AREAITTVGAMDY 58 IGHV1-18 M69L
 (SEQ ID NO: 136) (SEQ ID NO: 148) (SEQ ID NO: 154) Sh01G06 GYTFTDYN
 INPNNGGI AREAITTVGAMDY 60 IGHV1-18 M69L (SEQ ID NO: 136) (SEQ ID
 NO: 148) (SEQ ID NO: 154) K64Q G44S Sh01G06 GYTFTDYN INPNNGGI
 AREAITTVGAMDY 62 IGHV1-18 M69L (SEQ ID NO: 136) (SEQ ID NO: 148)
 (SEQ ID NO: 154) K64Q Sh01G06 GYTFTDYN INPNNGGI AREAITTVGAMDY 64
 IGHV1-69 T30S (SEQ ID NO: 235) (SEQ ID NO: 148) (SEQ ID NO: 154) I69L
 Sh01G06 GYTFTDYN INPNNGGI AREAITTVGAMDY 66 IGHV1-69 T30S (SEQ ID
 NO: 235) (SEQ ID NO: 148) (SEQ ID NO: 154) K64Q I69L Hu01G06
 GYTFTDYN INPNNGGI AREAITTVGAMDY 246 IGHV1-18 F1 (SEQ ID NO: 136)
 (SEQ ID NO: 242) (SEQ ID NO: 154) Hu01G06 GYTFTDYN INPNGLI
 AREAITTVGAMDY 248 IGHV1-18 F2 (SEQ ID NO: 136) (SEQ ID NO: 243)
 (SEQ ID NO: 154) Hu01G06 GYTFTDYN INPNGLI AREAITTVGAMDY 250 IGHV1-
 69 F1 (SEQ ID NO: 235) (SEQ ID NO: 243) (SEQ ID NO: 154) Hu01G06
 GYTFTDYN INPNNGGI AREAITTVGAMDY 252 IGHV1-69 F2 (SEQ ID NO: 235)
 (SEQ ID NO: 242) (SEQ ID NO: 154) Ch06C11 GFSLNTYGMG IYWDDDK
 AQRGYDDYWGY 46 Chimeric (SEQ ID NO: 141) (SEQ ID NO: 150) (SEQ ID
 NO: 157) HE LM 06C11 GFSLNTYGMG IYWDDDK AQRGYDDYWGY 68 IGHV2-70
 (SEQ ID NO: 141) (SEQ ID NO: 150) (SEQ ID NO: 157) Hu06C11
 GFSLNTYGMG IYWDDDK AQRGYDDYWGY 70 IGHV2-5 (SEQ ID NO: 141) (SEQ
 ID NO: 150) (SEQ ID NO: 157) Ch14F11 GFSLSTYGMG IYWDDDK
 ARRGHYSAMDY 50 Chimeric (SEQ ID NO: 140) (SEQ ID NO: 152) (SEQ ID
 NO: 158) Sh14F11 IGHV2- GFSLSTYGMG IYWDDDK ARRGHYSAMDY 72 5 (SEQ
 ID NO: 140) (SEQ ID NO: 152) (SEQ ID NO: 158) Sh14F11 IGHV2-
 GFSLSTYGMG IYWDDDK ARRGHYSAMDY 74 70 (SEQ ID NO: 140) (SEQ ID
 NO: 152) (SEQ ID NO: 158)

[0307] Humanized monoclonal antibody Kappa light chain CDR sequences (Kabat, Chothia, and IMGT definitions) are shown in Table 21.

TABLE-US-00145 TABLE 21 Variable Region CDR1 CDR2 CDR3 SEQ ID NO:
 Kabat/Chothia Ch01G06 Chimeric RTSENHNYLA DAKTLAD QHFWSSPYT 76 (SEQ
 ID NO: 21) (SEQ ID NO: 26) (SEQ ID NO: 32) Hu01G06 IGKV1-39
 RTSENHNYLA DAKTLAD QHFWSSPYT 90 (SEQ ID NO: 21) (SEQ ID NO: 26)
 (SEQ ID NO: 32) Hu01G06 IGKV1-39 RTSENHNYLA DAKTLAD QHFWSSPYT 92
 S43A V48I (also (SEQ ID NO: 21) (SEQ ID NO: 26) (SEQ ID NO: 32)
 known as Hu0106 IGKV1-39 F1) Hu01G06 IGKV1-39 RTSENHNYLA DAKTLAD
 QHFWSSPYT 94 V48I (SEQ ID NO: 21) (SEQ ID NO: 26) (SEQ ID NO: 32)
 Hu01G06 IGKV1-39 RTSENHNYLA DAKTLAD QHFWSSPYT 92 F1 (also known as
 (SEQ ID NO: 21) (SEQ ID NO: 26) (SEQ ID NO: 32) Hu01G06 IGKV1-39
 S43A V48I) Hu01G06 IGKV1-39 RTSENHNYLA DAKTLAD QHFWSDPYT 254 F2
 (SEQ ID NO: 21) (SEQ ID NO: 26) (SEQ ID NO: 244) Ch06C11 Chimeric
 KASQNVGTNVA SASYRYS QQYNNYPLT 82 (SEQ ID NO: 23) (SEQ ID NO: 28)
 (SEQ ID NO: 35) Sh06C11 IGKV1-16 KASQNVGTNVA SASYRYS QQYNNYPLT 96

(SEQ ID NO: 23)(SEQ ID NO: 28)(SEQ ID NO: 35) Ch14F11 Chimeric KASQNVGTNVA SPSYRYS QQYNSYPHT 86 (SEQ ID NO: 23)(SEQ ID NO: 30) (SEQ ID NO: 36) Hu14F11 IGKV1-16 KASQNVGTNVA SPSYRYS QQYNSYPHT 98 (SEQ ID NO: 23)(SEQ ID NO: 30)(SEQ ID NO: 36) IMGT Ch01G06 Chimeric ENLHNY DAK QHFWSSPYT 76 (SEQ ID NO: 160)(SEQ ID NO: 32) Hu01G06 IGKV1-39 ENLHNY DAK QHFWSSPYT 90 (SEQ ID NO: 160)(SEQ ID NO: 32) Hu01G06 IGKV1-39 ENLHNY DAK QHFWSSPYT 92 S43A V48I (also (SEQ ID NO: 160)(SEQ ID NO: 32) known as Hu01G06 IGKV1-39 F1) Hu01G06 IGKV1-39 ENLHNY DAK QHFWSSPYT 94 V48I (SEQ ID NO: 160) DAK (SEQ ID NO: 32) Hu01G06 IGKV1-39 ENLHNY QHFWSSPYT 92 F1 (also known as (SEQ ID NO: 160)(SEQ ID NO: 32) Hu01G06 IGKV1-39 S43A V48I) Hu01G06 IGKV1-39 ENLHNY DAK QHFWSDPYT 254 F2 (SEQ ID NO: 160)(SEQ ID NO: 244) Ch06C11 Chimeric QNVGTN SAS QQYNNYPLT 82 (SEQ ID NO: 162)(SEQ ID NO: 35) Sh06C11 IGKV1-16 QNVGTN SAS QQYNNYPLT 96 (SEQ ID NO: 162) (SEQ ID NO: 35) Ch14F11 Chimeric QNVGTN SPS QQYNSYPHT 86 (SEQ ID NO: 162)(SEQ ID NO: 36) Hu14F11 IGKV1-16 QNVGTN SPS QQYNSYPHT 98 (SEQ ID NO: 162)(SEQ ID NO: 36)

[0308] To create the complete chimeric and humanized heavy or kappa chain antibody sequences, each variable sequence above is combined with its respective human constant region. For example, a complete heavy chain comprises a heavy variable sequence followed by a human IgG1 heavy chain constant sequence. A complete kappa chain comprises a kappa variable sequence followed by the human kappa light chain constant sequence.

Nucleic Acid Sequence Encoding the Human IgG1 Heavy Chain Constant Region (SEQ ID NO:171)

TABLE-US-00146 1 gcctcaacaa aaggaccaag tgtgttccca ctgccccta gcagcaagag tacatccggg
61 ggcaactgcag cactcggctg cctcgtaag gattatttc cagagccagt aaccgtgagc 121 tggaacagtg
gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 181 ggctgtact cactcagctc
cgctgtgacc gtgccatctt catctctggg cactcagacc 241 tacatctgta atgtaaacca caagcctagc
aataactaagg tcgataagcg ggtggaaccc 301 aagagctgcg acaagactca cactgtccc ccatgccctg
cccctgaact tctgggcggt 361 cccagcgtct tttgttccc accaaagcct aaagatactc tgatgataag
tagaacaccc 421 gaggtgacat gtgtgtgtgt agacgtttcc cacgaggacc cagaggtaa gttcaactgg 481
tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 541 agtacatacc
gtgtagtcatg tgttctcaca gtgctgcacc aagactggct caacggcaaa 601 gaatacaaat gcaaagtgtc
caacaaagca ctcccagccc ctatcgagaa gactattagt 661 aaggcaaagg ggcagcctcg tgaaccacag
gtgtacactc tgccaccag tagagaggaa 721 atgacaaaga accaagtctc attgacctgc ctggtgaaag
gcttctaccc cagcgacatc 781 gccgttgagt gggagagtaa cggctcgcct gagaacaatt acaagacaac
ccccccagtg 841 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa gtcccgtgg 901
cagcagggta acgtcttcag ctgtccgtg atgcacgagg cattgcacaa ccactacacc 961 cagaagtcac
tgagcctgag cccaggggaag

Protein Sequence Defining the Human IgG1 Heavy Chain Constant Region (SEQ ID NO:172)

TABLE-US-00147 1 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsalgtsgv htfpavlqss
61 glylssvvt vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 121 psvflfppkp
kdtlmisrtp evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 181 styrvvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 241 mtknqvsitc lvkgfypsdi avewesngqp
ennyktpppv ldsdgsffly skltvdksw 301 qqgnvfscsv mhealhnhyt qkslspsgk

[0309] Nucleic Acid Sequence Encoding the Human Kappa Light Chain Constant Region (SEQ ID NO:173)

TABLE-US-00148 1 cgcacagttg ctgccccag cgtgttcatt ttccaccta gcgatgagca gctgaaaagc
61 ggtactgcct ctgtcgtatg cttgctcaac aacttttacc cacgtgaggc taaggtgcag 121 tggaagtg
ataatgcact tcaatctgga aacagtcaag agtccgtgac agaacaggac 181 agcaaagact caacttattc

actctcttcc accctgactc tgtccaaggc agactatgaa 241 aaacacaagg tatacgctg cgaggttaca

caccaggggt tgtctagtcc tgtcaccaag 301 tccttcaata ggggcgaatg t

Protein Sequence Defining the Human Kappa Light Chain Constant Region (SEQ ID NO:174)

TABLE-US-00149 1 rtvaapsvfi fppsdeqlks gtasvclln nfypreakvq wkvdnalqsg nsqesvteqd

61 skdstyslss tltlskadye khkvyacevt hqglsspvtk sfnrgec

[0310] The following sequences represent the actual or contemplated full length heavy and light chain sequence (i.e., containing both the variable and constant regions sequences) for each antibody described in this Example. Signal sequences for proper secretion of the antibodies (e.g., signal sequences at the 5' end of the DNA sequences or the amino terminal end of the protein sequences) are not shown in the full length heavy and light chain sequences disclosed herein and are not included in the final secreted protein. Also not shown are stop codons for termination of translation required at the 3' end of the DNA sequences. It is within ordinary skill in the art to select a signal sequence and/or a stop codon for expression of the disclosed full length immunoglobulin heavy chain and light chain sequences. It is also contemplated that the variable region sequences can be ligated to other constant region sequences to produce active full length immunoglobulin heavy and light chains.

Nucleic Acid Sequence Encoding the Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:175)

TABLE-US-00150 1 gaagtgtgtg tgcagcagtc agggccggag ttggtaaaac cgggagcgtc ggtgaaaatc

61 ccgtgcaaag cgtcggggta tacgtttacg gactataaca tggattgggt gaaacagtcg 121 catgggaaat

cgcttgaatg gattgtcag atcaatccga ataatggagg aatcttctt 181 aatcagaagt ttaaagggaa

agcgacgctt acagtcgata agtcgtcgaa cacggcgctc 241 atggaagtac ggctcgcttac gtcggaagat

acggcggtct attactgtgc gagggaggcg 301 attacgacgg tgggagcgat ggactattgg ggacaaggga

cgtcgggtcac ggtatcgctg 361 gcctcaacaa aaggaccaag tgtgttccca ctcgccccta gcagcaagag

tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481

tggaacagtg gagcactcac ttctggtgtc catactttc ctgctgtcct gcaaagctct 541 ggcctgtact

cactcagctc cgctcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca

caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgtccc

ccatgccctg ccctgaact tctgggcggt 721 cccagcgtct tttgttccc accaaagcct aaagatactc

tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtg agacgtttcc cacgaggacc cagagggtta

gttcaactgg 841 tacgttgatg gtagtgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901

agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat

gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaaagg ggcagcctcg

tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc

ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt

acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa

gtcccgtgg 1261 cagcagggtg acgtctcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321

cagaagtcac tgagcctgag cccagggaag

Protein Sequence Defining the Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:176)

TABLE-US-00151 1 evllqqsgpe lvkpgasvki pckasgytft dynmdwvkqs hgkslewigq

inpnnggiff 61 nqkfkkgatl tvdkssntaf mevrsltsed tavyycarea ittvgamdyw gqgtsvtvss 121

astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsgv htfpavlqss 181 glylssvvt

vpssslgtqt yicvnkhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrt

evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk

eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp

ennykttppv ldsdgsffly skltvdksw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:177)

TABLE-US-00152 1 caagtgaac ttgtgcagtc ggggtcggaa gtcaaaaagc cgggagcgtc ggtgaaagta

61 tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggactgggt acgacaggca 121 ccggggaaat
 cgttggaatg gatcggacag attaatccga acaatggggg aattttcttt 181 aatcagaaat tcaaaggacg
 ggcgacgttg acggtcgata catcgacgaa tacggcgat 241 atggaattga ggctcgcttcg ctcggacgat
 acggcgggtct attactgcgc cagggaggcg 301 atcacgacgg taggggcat ggattattgg ggacagggga
 cgcttgtgac ggtatcgctg 361 gcctcaaca aaggaccaag tgtgttcca ctcgccccta gcagcaagag
 tacatccggg 421 ggactgcag cactcggtg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481
 tggaacagtg gagcactcac ttctggtgtc catactttc ctgctgtcct gcaaagctct 541 ggctgtact
 cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca
 caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgtccc
 ccatgccctg ccctgaact tctgggcggt 721 ccagcgtct tttgttccc accaaagcct aaagatactc
 tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtg agacgtttcc cacgaggacc cagagggtta
 gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901
 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
 gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
 tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
 ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
 acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa
 gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321
 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:178)

TABLE-US-00153 1 qvqlvqgae vkkpgasvkv sckasgyft dynmdwvrqa pgkslewigq
 inpnnggiff 61 nqkfkgratl tvdtstntay melrslrdd tavyycarea ittvgamdyw gqgtlvvtss 121
 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsгv htфpavlqss 181 glyslssvvt
 vpssslgtqt yicvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfpkp kdtlmisrtp
 evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlnk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:179)

TABLE-US-00154 1 caagtccagc ttgtccagtc gggagcggaa gtgaagaaac cggggtcgtc ggtcaaagta
 61 tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggattgggt acgacaggct 121 ccgggaaaat
 cattggaatg gattggacag attaatccga ataatggggg tatcttcttt 181 aatcaaaagt ttaaaggag
 ggcgacgttg acggtggaca aatcgacaaa tacggcgat 241 atggaattgt cgtcgcttcg gtcggaggac
 acggcgggtgt attactgcgc gagggaggcg 301 atcacgacgg tcggggcgat ggattattgg ggacagggaa
 cgcttgtgac ggtatcgctg 361 gcctcaaca aaggaccaag tgtgttcca ctcgccccta gcagcaagag
 tacatccggg 421 ggactgcag cactcggtg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481
 tggaacagtg gagcactcac ttctggtgtc catactttc ctgctgtcct gcaaagctct 541 ggctgtact
 cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca
 caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgtccc
 ccatgccctg ccctgaact tctgggcggt 721 ccagcgtct tttgttccc accaaagcct aaagatactc
 tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtg agacgtttcc cacgaggacc cagagggtta
 gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901
 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
 gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
 tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
 ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
 acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa
 gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321
 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:180)

TABLE-US-00155 1 qvqlvqsgae vkkpgssvk sckasgytft dynmdwvrqa pgkslewigq
inpnnggiff 61 nqkfkgratl tvdkstntay melsslrsed tavyycarea ittvgamdyw gqgtlvtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsqv htfpavlqss 181 glyslssvvt
vpssslgtqt yicvnvhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Sh01G06 IGHV1-18 M69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:181)

TABLE-US-00156 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacagg gtcttgaatg gatggggcag attaataccga ataattggagg gatcttcttt 181 aatcagaaat
tcaaaggaag ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggtcgttgcg
atcagatgat acggcggctt actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg 361 gcctcaacaa aaggaccaag tgtgttccca ctcgccccta
gcagcaagag tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt
aaccgtgagc 481 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta
atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaacc 661 aagagctgcg acaagactca
cacttgctcc ccatgccctg cccctgaact tctgggcggg 721 ccagcgtct tttgttccc accaaagcct
aaagatactc tgatgataag tagaacacc 781 gaggtgacat gtgtgtgtgt agacgtttcc cagaggacc
cagaggtaa gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga
gcagtataat 901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961
gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg
ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc
attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct
gagaacaatt acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagtga
ctgtggacaa gtcccgtgg 1261 cagcaggga acgtcttcag ctgtccgtg atgcacgagg cattgcacaa
ccactacacc 1321 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Sh01G06 IGHV1-18 M69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:182)

TABLE-US-00157 1 qvqlvqsgae vkkpgasvk sckasgytft dynmdwvrqa pgqglewmgq
inpnnggiff 61 nqkfkgrvlt tdtststay melrslrdd tavyycarea ittvgamdyw gqgtlvtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsqv htfpavlqss 181 glyslssvvt
vpssslgtqt yicvnvhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Sh01G06 IGHV1-18 M69L K640 G44S Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:183)

TABLE-US-00158 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacaga gccttgaatg gatggggcag attaataccga ataattggagg gatcttcttt 181 aatcagaaat
tccagggaa ggtaacgctg acgacagaca cgtcaacatc gacggcctat 241 atggaattgc ggtcgttgcg
atcagatgat acggcggctt actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg 361 gcctcaacaa aaggaccaag tgtgttccca ctcgccccta
gcagcaagag tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt

aaccgtgagc 481 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta
atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca
cacttgctcc ccatgccctg cccctgaact tctgggcggg 721 cccagcgtct tttgttccc accaaagcct
aaagatactc tgatgataag tagaacaccc 781 gaggtgacat gtgttgtgt agacgtttcc cagcaggacc
cagagggttaa gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga
gcagtataat 901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961
gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg
ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc
attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct
gagaacaatt acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga
ctgtggacaa gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa
ccactacacc 1321 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain
(Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:184)

TABLE-US-00159 1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqslwmgq
inpnnggiff 61 nqkfqgrvtl tdtststay melrslrdd tavyycarea ittvgamdyw gqgtlvtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsqv htfpavlqss 181 glyslssvvt
vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
ennyktpppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslspsgk

Nucleic Acid Sequence Encoding the Full Length Sh01G06 IGHV1-18 M69L K64Q Heavy Chain
(Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:185)

TABLE-US-00160 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
cctggacagg gtcttgaatg gatggggcag attaatccga ataatggagg gatcttctt 181 aatcagaaat
tccagggaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 241 atggaattgc ggtcgttgcg
atcagatgat acggcggctc actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
ggacagggga cgttggtaac ggtatcgtcg 361 gcctcaacaa aaggaccaag tgtgttccca ctcgccccta
gcagcaagag tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt
aaccgtgagc 481 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541
ggcctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta
atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca
cacttgctcc ccatgccctg cccctgaact tctgggcggg 721 cccagcgtct tttgttccc accaaagcct
aaagatactc tgatgataag tagaacaccc 781 gaggtgacat gtgttgtgt agacgtttcc cagcaggacc
cagagggttaa gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga
gcagtataat 901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961
gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg
ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc
attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct
gagaacaatt acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga
ctgtggacaa gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa
ccactacacc 1321 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Sh01G06 IGHV1-18 M69L K64Q Heavy Chain
(Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO: 186)

TABLE-US-00161 1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqglewmgq
inpnnggiff 61 nqkfqgrvtl tdtststay melrslrdd tavyycarea ittvgamdyw gqgtivtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsqv htfpavlqss 181 glyslssvvt
vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp

evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksw 421 qqgnvfscsv mhealhnhyt qkslslspgk
 Nucleic Acid Sequence Encoding the Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain
 (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:187)
 TABLE-US-00162 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaac cgggatcgtc ggtcaaagtg
 61 tcgtgtaaag cgctcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121 cctgggcagg
 gacttgaatg gatgggtcag atcaatccga ataatggggg aatcttttc 181 aatcagaagt ttaaagggag
 ggtaacgctg acggcgggata aaagcacgtc aacggcgat 241 atggagtgt cgctgttcg gtcggaggac
 acggcggctt attactgcgc gagggaaagcg 301 attacgacgg tgggagcgat ggattattgg gggcagggaa
 cgcttgaac ggtgtcatcg 361 gcctcaaaa aaggaccaag tgtgttcca ctcgccccta gcagcaagag
 tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481
 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541 ggctgtact
 cactcagctc cgctcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca
 caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgctcc
 ccatgccctg cccctgaact tctgggcggt 721 ccagcgtct tttgttccc accaaagcct aaagatactc
 tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtgt agacgtttcc cacgaggacc cagagggtta
 gttcaactgg 841 tacgttgatg gtagcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901
 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
 gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
 tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
 ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
 acaagacaac cccccagtg 1201 ctggatagtg acgggtctt ctttctgtac agtaagctga ctgtggacaa
 gtcccgtgg 1261 cagcagggta acgtctcag ctgtccgtg atgcacgagg cattgcacaa ccactacacc 1321
 cagaagtcac tgagcctgag cccagggaag

Protein Sequence Defining the Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain
 (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:188)

TABLE-US-00163 1 qvqlvqgae vkkpgssvk sckasgytfs dynmdwvrqa pgqglewmgq
 inpnnggiff 61 nqkfkgrvtl tadkststay melsslrsed tavyycarea ittvgamdyw gqgtivtvss 121
 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsgv htfpavlqss 181 glyslssvvt
 vpssslgtqt yicvnkhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
 evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksw 421 qqgnvfscsv mhealhnhyt qkslslspgk
 Nucleic Acid Sequence Encoding the Full Length Sh01G06 IGHV1-69 T30S K64Q I69L Heavy
 Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID
 NO:189)

TABLE-US-00164 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaac cgggatcgtc ggtcaaagtg
 61 tcgtgtaaag cgctcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121 cctgggcagg
 gacttgaatg gatgggtcag atcaatccga ataatggggg aatcttttc 181 aatcagaagt ttcaggggag
 ggtaacgctg acggcgggata aaagcacgtc aacggcgat 241 atggagtgt cgctgttcg gtcggaggac
 acggcggctt attactgcgc gagggaaagcg 301 attacgacgg tgggagcgat ggattattgg gggcagggaa
 cgcttgaac ggtgtcatcg 361 gcctcaaaa aaggaccaag tgtgttcca ctcgccccta gcagcaagag
 tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481
 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541 ggctgtact
 cactcagctc cgctcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca
 caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgctcc
 ccatgccctg cccctgaact tctgggcggt 721 ccagcgtct tttgttccc accaaagcct aaagatactc
 tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtgt agacgtttcc cacgaggacc cagagggtta
 gttcaactgg 841 tacgttgatg gtagcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901

agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
 gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
 tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
 ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
 acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa
 gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321
 cagaagtcac tgagcctgag cccagggaag

Protein Sequence Defining the Full Length Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Chain
 (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:190)

TABLE-US-00165 1 qvqlvqsgae vkkpgssvk sckasgytfs dynmdwvrqa pgqglewmgq
 inpnnggiff 61 nqkfqgrvtl tadkststay melsslr sed tavyycarea ittvgamdyw gqgtivtvss 121
 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsgv htfpavlqss 181 glyslssvvt
 vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
 evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-18 F1 Heavy Chain
 (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:255)

TABLE-US-00166 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc
 ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt gcgccaagcg 121
 cctggacaga gccttgaatg gatggggcag attaaccgt acaatcacct gatctcttt 181 aatcagaaat
 tccagggaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 241 atggaattgc ggctgttgcg
 atcagatgat acggcggctc actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
 ggacagggga cgttggtaac ggtatcgtcg 361 gcctcaacaa aaggaccaag tgtgttccca ctgccccta
 gcagcaagag tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt
 aaccgtgagc 481 tggaacagtg gagcactcac ttctgtgtc cactctttc ctgctgtcct gcaaagctct 541
 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta
 atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca
 cacttgtccc ccatgccctg cccctgaact tctgggcggg 721 ccagcgtct tttgttccc accaaagcct
 aaagatactc tgatgataag tagaacaccc 781 gaggtgacat gtgttgtgt agacgtttcc cacgaggacc
 cagaggttaa gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga
 gcagtataat 901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961
 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg
 ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc
 attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct
 gagaacaatt acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga
 ctgtggacaa gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa
 ccactacacc 1321 cagaagtcac tgagcctgag cccagggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-18 F1 Heavy Chain (Humanized
 Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:256)

TABLE-US-00167 1 qvqlvqsgae vkkpgasvk sckasgytft dynmdwvrqa pgqslwmgq
 inpynhliff 61 nqkfqgrvtl tdtststay melslrsdd tavyycarea ittvgamdyw gqgtivtvss 121
 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsaltsgv htfpavlqss 181 glyslssvvt
 vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
 evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvsvlt vlhqdwlngk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-18 F2 Heavy Chain
 (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:257)

TABLE-US-00168 1 caggtccagc ttgtgcaatc gggagcggaa gtgaagaaac cgggagcgtc

ggtaaaagtc 61 tcgtgcaaag cgtcggggta tacgtttacg gactataaca tggactgggt ggcgcaagcg 121
 cctggacaga gccttgaatg gatggggcag attaatccga ataatggact gatcttcttt 181 aatcagaaat
 tccagggaaag ggtaacgctg acgacagaca cgtaacatc gacggcctat 241 atggaattgc ggctgtgctg
 atcagatgat acggcgggtct actattgtgc gagggaggcg 301 attacgacgg tgggagcgat ggattattgg
 ggacaggggga cgttggtaac ggtatcgtcg 361 gcctcaacaa aaggaccaag tgtgttccca ctgccccta
 gcagcaagag tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt
 aaccgtgagc 481 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541
 ggctgtact cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta
 atgtaaacca caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca
 cacttgtccc ccatgccctg cccctgaact tctgggcggg 721 ccagcgtct tttgttccc accaaagcct
 aaagatactc tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtgt agacgtttcc cacgaggacc
 cagagggttaa gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga
 gcagtataat 901 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961
 gaatacaaat gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg
 ggcagcctcg tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc
 attgacctgc ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct
 gagaacaatt acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga
 ctgtggacaa gtcccgtgg 1261 cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa
 ccactacacc 1321 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-18 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:258)

TABLE-US-00169 1 qvqlvqsgae vkkpgasvkv sckasgytft dynmdwvrqa pgqslwmgq
 inpnngliff 61 nqkfqgrvtl tdtststay melrslrdd tavyycarea ittvgamdyw gqgtivtvss 121
 astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsгv htфpavlqss 181 glyslssvvt
 vpssslgtqt yicvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
 evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvvsvlt vlhqdwlnk
 eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
 ennykttppv ldsdgsffly skltvdksw 421 qqgnvfscsv mhealthnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:259)

TABLE-US-00170 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
 61 tcgtgtaaag cgtcgggata tacgtttacg gactataaca tggattgggt gcgacaagcg 121 cctgggcagg
 gacttgaatg gatgggtcag atcaatccga ataatgggt gatctttttc 181 aatcagaagt ttaaagggag
 ggtaacgctg acggcggata aaagcacgtc aacggcgat 241 atggagtgt cgtcgttgcg gtcggaggac
 acggcgggtct attactgcgc gaggggaagcg 301 attacgacgg tgggagcgat ggattattgg gggcagggaa
 cgcttgtaac ggtgtcatcg 361 gcctcaacaa aaggaccaag tgtgttccca ctgccccta gcagcaagag
 tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattattttc cagagccagt aaccgtgagc 481
 tggaacagtg gagcactcac ttctggtgtc catacttttc ctgctgtcct gcaaagctct 541 ggctgtact
 cactcagctc cgtcgtgacc gtgccatctt catctctggg cactcagacc 601 tacatctgta atgtaaacca
 caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgtccc
 ccatgccctg cccctgaact tctgggcggg 721 ccagcgtct tttgttccc accaaagcct aaagatactc
 tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtgt agacgtttcc cacgaggacc cagagggttaa
 gttcaactgg 841 tacgttgatg gagtcgaagt acataatgct aagaccaagc ctagagagga gcagtataat 901
 agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
 gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
 tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
 ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
 acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa
 gtcccgtgg 1261 cagcagggta acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321
 cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:260)

TABLE-US-00171 1 qvqlvqsgae vkkpgssvkv sckasgytfs dynmdwvrqa pgqglewmgq
inpnngliff 61 nqkfkgrvtl tadkststay melsslr sed tavyycarea ittvgamdyw gqgtivtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss 181 glyslssvvt
vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:261)

TABLE-US-00172 1 caagtacagc ttgtacagtc gggagcggaa gtcaagaaac cgggatcgtc ggtcaaagtg
61 tcgtgtaaaag cgtcgggata tacgtttagc gactataaca tggattgggt gcgacaagcg 121 cctgggcagg
gacttgaatg gatgggtcag atcaatccgt acaatcacct gatcttttc 181 aatcagaagt ttaaagggag
ggtaacgctg acggcggata aaagcacgtc aacggcgat 241 atggagtgt cgtcgttcg gtcggaggac
acggcggctc attactgcgc gagggaaagcg 301 attacgacgg tgggagcgat ggattattgg gggcagggaa
cgcttgtaac ggtgtcatcg 361 gcctcaacaa aaggaccaag tgtgttcca ctcgcccta gcagcaagag
tacatccggg 421 ggactgcag cactcggctg cctcgtcaag gattatttc cagagccagt aaccgtgagc 481
tggaacagtg gagcactcac ttctgggtgc catactttc ctgctgtcct gcaaagctct 541 ggctgtact
cactcagctc cgtcgtgacc gtgccatct catctctggg cactcagacc 601 tacatctgta atgtaaacca
caagcctagc aatactaagg tcgataagcg ggtggaaccc 661 aagagctgcg acaagactca cacttgtccc
ccatgccctg cccctgaact tctgggcggt 721 cccagcgtct tttgttccc accaaagcct aaagatactc
tgatgataag tagaacaccc 781 gaggtgacat gtgtgtgtgt agacgtttcc cagaggacc cagagggtta
gttcaactgg 841 tacgttgatg gagtcaagt acataatgct aagaccaagc ctagagagga gcagtataat 901
agtacatacc gtgtagtcag tgttctcaca gtgctgcacc aagactggct caacggcaaa 961 gaatacaaat
gcaaagtgtc caacaaagca ctcccagccc ctatcgagaa gactattagt 1021 aaggcaaagg ggcagcctcg
tgaaccacag gtgtacactc tgccaccag tagagaggaa 1081 atgacaaaga accaagtctc attgacctgc
ctggtgaaag gcttctaccc cagcgacatc 1141 gccgttgagt gggagagtaa cggtcagcct gagaacaatt
acaagacaac cccccagtg 1201 ctggatagtg acgggtcttt ctttctgtac agtaagctga ctgtggacaa
gtcccgtgg 1261 cagcagggtg acgtcttcag ctgttccgtg atgcacgagg cattgcacaa ccactacacc 1321
cagaagtcac tgagcctgag cccaggggaag

Protein Sequence Defining the Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:262)

TABLE-US-00173 1 qvqlvqsgae vkkpgssvkv sckasgytfs dynmdwvrqa pgqglewmgq
inpnynhliff 61 nqkfkgrvtl tadkststay melsslr sed tavyycarea ittvgamdyw gqgtivtvss 121
astkgpsvfp lapsskstsg gtaalgclvk dyfpepvtvs wnsгалtsgv htfpavlqss 181 glyslssvvt
vpssslgtqt yicnvnhkps ntkvdkrvep kscdkthtcp pcpapellgg 241 psvflfppkp kdtlmisrtp
evtcvvvdvs hedpevkfnw yvdgvevhna ktkpreeqyn 301 styrvvsvlt vlhqdwlngk
eykckvsnka lpapiektis kakgqprepq vytlppsree 361 mtknqvsltc lvkgfypsdi avewesngqp
ennykttppv ldsdgsffly skltvdksrw 421 qqgnvfscsv mhealhnhyt qkslslspgk

Nucleic Acid Sequence Encoding the Full Length Ch06C11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:191)

TABLE-US-00174 1 caggtgacac tcaaagaatc aggaccgga atccttcagc ccagccagac cttgtcgtg
61 acttgttcgt tctccggtt cagcctgaat acttatggga tgggtgtgtc atggatcagg 121 caaccgtccg
ggaaaggatt ggagtggctc gcgcacatct actgggacga tgacaaacgc 181 tacaatcctt cgctgaagag
ccgattgacg attccaagg atgcctcgaa caaccgggta 241 tttcttaaga tcacgtcggc cgatacggca
gacacggcga cctattactg cgcccaaaga 301 gggtagatg actattgggg atattggggc caggggacac
tcgtcacaat ttcagctgcc 361 tcaacaaaag gaccaagtgt gtcccactc gccctagca gcaagagtac
atccgggggc 421 actgcagcac tcggctgcct cgtaaggat tattttccag agccagtaac cgtgagctgg 481
aacagtggag cactcactc tggtgtccat acttttctg ctgtcctgca aagctctggc 541 ctgtactcac

tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
 gcctagcaat actaaggtcg ataagcgggt ggaacccaag 661 agctgcgaca agactcacac ttgtcccca
 tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgttcccacc aaagcctaaa gatactctga
 tgataagtag aacacccgag 781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt
 caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa 961 taaaaatgca
 aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga
 accacaggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gacctgcctg
 gtgaaaggct tctaccccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
 agacaacccc cccagtgtg 1201 gatagtacg ggtctttctt tctgtacagt aagctgactg tggacaagtc
 ccgctggcag 1261 cagggtaacg tcttcagctg ttccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
 aagtcactga gcctgagccc aggggaag

Protein Sequence Defining the Full Length Ch06C11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:192)

TABLE-US-00175 1 qvtlkespgg ilqpsqtls tcsfsgfsln tygmgswwir qpsgkglewl ahiywdddkr
 61 ynpslksrlt iskdasnry flkitsvda dtatycaqr gyddywgywg qgtivtisaa 121 stkgpsvfpl
 apsskstsgg taalgclvkd yfpepvtvsw nsgaltsgvh tfpavlqssg 181 lyslssvvtv pssslgtqy
 icvnvhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfppkpk dtlmisrtpc vtcvvvdvsh
 edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvvsvltv lhqdwlngke ykckvsnkal papiektisk
 akqpprepqv ytlppsreem 361 tknqvsltel vkgfypsdiawewesngqpe nnykttppvl dsdgsfflys
 kltdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length HE LM 06C11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:193)

TABLE-US-00176 1 caggtgactt tgaaagaatc cggctccgca ttggtaaagc caaccagac acttacgctc
 61 acatgtacat ttccggatt cagcttgaac acttacggga tgggagtgtc gtggattcgg 121 caacctccgg
 ggaaggctct ggagtggctg ggcacatct actgggatga tgacaaaagg 181 tataaccct cacttaaac
 gagactgacg atctgaagg acacaagcaa gaatcaggtc 241 gtcctcacga ttacgaatgt agaccgggtg
 gatactgccg tctattactg cgcgcaacgc 301 gggtatgatg actactgggg atattgggggt cagggcacc
 tcgtgacat ctcgtcagcc 361 tcaacaaaag gaccaagtgt gttccactc gccctagca gcaagagtac
 atccgggggc 421 actgcagcac tcggctgcct cgtcaaggat tttttccag agccagtaac cgtgagctgg 481
 aacagtggag cactcacttc tgggtgcat acttttctg ctgtctgca aagctctggc 541 ctgtactcac
 tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
 gcctagcaat actaaggtcg ataagcgggt ggaacccaag 661 agctgcgaca agactcacac ttgtcccca
 tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgttcccacc aaagcctaaa gatactctga
 tgataagtag aacacccgag 781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagtt
 caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
 acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa 961 taaaaatgca
 aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga
 accacaggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gacctgcctg
 gtgaaaggct tctaccccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
 agacaacccc cccagtgtg 1201 gatagtacg ggtctttctt tctgtacagt aagctgactg tggacaagtc
 ccgctggcag 1261 cagggtaacg tcttcagctg ttccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
 aagtcactga gcctgagccc aggggaag

Protein Sequence Defining the Full Length HE LM 06C11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:194)

TABLE-US-00177 1 qvtlkesgpa lvkptqltl tctfsgfsln tygmgswwir qppgkalewl ahiywdddkr
 61 ynpslktrlt iskdtsknqv vltitnvdv dtavycaqr gyddywgywg qgtivtissa 121 stkgpsvfpl
 apsskstsgg taalgclvkd yfpepvtvsw nsgaltsgvh tfpavlqssg 181 lyslssvvtv pssslgtqy
 icvnvhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfppkpk dtlmisrtpc vtcvvvdvsh
 edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvvsvltv lhqdwlngke ykckvsnkal papiektisk

akgqpprepqv ytlppsreem 361 tknqvsltlc vkgfypsdiawewesngqpe nnykttppvl dsdgsfflys
kltdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length Hu06C11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:195)

TABLE-US-00178 1 caagtaacgc tcaaggagtc cggaccacc ttggtgaagc cgacgcagac cttgactctt
61 acgtgcactt tctcgggggtt tctactgaat acgtacggga tgggtgtctc atggatcagg 121 caacctccgg
ggaaaggatt ggaatggctg gcgcacatct actgggatga cgataagaga 181 tataacccaa gcctcaagtc
gcggctcacc attacaaaag atacatcgaa aaatcaggtc 241 gtacttacta tcacgaacat ggaccccgtg
gacacagcaa catattactg tgcccagcgc 301 ggctatgacg attattgggg ttactgggga cagggaacac
tggtcacggt gtccagcgc 361 tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac
atccgggggc 421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg 481
aacagtggag cactcacttc tgggtgccat acttttctg ctgtcctgca aagctctggc 541 ctgtactcac
tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
gcctagcaat actaaggctg ataagcgggt ggaaccaag 661 agctgcgaca agactcacac ttgtcccca
tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgttcccacc aaagcctaaa gatactctga
tgataagtag aacacccgag 781 gtgacatgtg ttgtttaga cgtttcccac gaggaccag aggttaagtt
caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa 961 tacaatgca
aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga
accacagggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gacctgcctg
gtgaaaggct tctaccccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
agacaacccc ccagtgctg 1201 gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc
ccgctggcag 1261 cagggtaacg tcttcagctg ttccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
aagtcactga gcctgagccc agggag

Protein Sequence Defining the Full Length Hu06C11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:196)

TABLE-US-00179 1 qvtlkesgpt lvkptqtl tctfsgfsln tygmvgvswir qppgkglewl ahiywdddkr
61 ynpslksrlt itkdtsknqv vltitnmdpv dtatycaqr gyddywgywg qgtlvtvssa 121 stkgpsvfpl
apsskstsgg taalgclvkd yfpepvtvsw nsgaltgsvh tfpavlqssg 181 lyslssvvtv pssslgtqty
icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfpkpk dtlmisrtp vtcvvvdvsh
edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvsvltv lhqdwlngke ykckvsnkal papiektisk
akgqpprepqv ytlppsreem 361 tknqvsltlc vkgfypsdiawewesngqpe nnykttppvl dsdgsfflys
kltdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length Ch14F11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:197)

TABLE-US-00180 1 caggtcacgc tgaagagtc aggtcccga atcctcaac ctcgcagac attgtcactc
61 acatgttctt tctccgggtt ctgcctctg acttatggca tgggtgtagg atggattcgg 121 cagcccagcg
ggaaggggct tgagtgggtg gcggatatct ggtgggacga cgacaaatac 181 tacaatccga gcctgaagtc
ccgctcacc atttcgaaag atacgtcatc aaacgaagtc 241 ttttgaaga tcgccatcgt ggacacggcg
gatacagcga cgtattactg cgccagaagg 301 ggacactaca gcgcaatgga ttattgggga caggggacct
cggtgactgt gtcgtccgcc 361 tcaacaaaag gaccaagtgt gttcccactc gccctagca gcaagagtac
atccgggggc 421 actgcagcac tcggctgcct cgtcaaggat tattttccag agccagtaac cgtgagctgg 481
aacagtggag cactcacttc tgggtgccat acttttctg ctgtcctgca aagctctggc 541 ctgtactcac
tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
gcctagcaat actaaggctg ataagcgggt ggaaccaag 661 agctgcgaca agactcacac ttgtcccca
tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgttcccacc aaagcctaaa gatactctga
tgataagtag aacacccgag 781 gtgacatgtg ttgtttaga cgtttcccac gaggaccag aggttaagtt
caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa 961 tacaatgca
aagtgtccaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga

accacaggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gactgcctg
gtgaaaggct tctacccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
agacaacccc cccagtgtg 1201 gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc
ccgctggcag 1261 cagggtaacg tcttcagctg ttccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
aagtcactga gcctgagccc agggaag

Protein Sequence Defining the Full Length Ch14F11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:198)

TABLE-US-00181 1 qvtlkespgg ilqpsqtls tcsfsgfsls tygmvgvwir qpsgkglewl
adiwwdddky 61 ynpslksrlt iskdtssnev flkiaivda dtatyycarr ghysamdywg qgtsvtvssa 121
stkgpsvfpl apsskstsgg taalglvkd yfpepvtvsw nsgaltsgvh tfpavlqssg 181 lyslssvvtv
pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfppkpk dtlmisrtp
vtcvvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvsvltv lhqdwlngke
ykckvsnkakl papiektisk akgqprepqv ytlppsreem 361 tknqvsltl vkgfypsdiawewesngqpe
nnykttppvl dsdgsfflys kltvdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length Sh14F11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:199)

TABLE-US-00182 1 cagatcactt tgaagaaag cggaccgacc ttggtcaagc ccacacaaac cctcacgctc
61 acgtgtacat ttcgggggtt ctgctttca acttacggga tgggagtagg gtggattcgc 121 cagccgcctg
gtaaagcggt ggagtggctt gcagacatct ggtgggacga cgataagtac 181 tataatccct cgctcaagtc
cagactgacc atcacgaaag atacgagcaa gaaccaggtc 241 gtgctgacaa tgactaacat ggaccagtg
gatacggcta catattactg cgccaggcgg 301 ggtcactact cagcgatgga ttattggggc caggaacac
tggtaacggt gtcgtccgcc 361 tcaacaaaag gaccaagtgt gtccactc gccctagca gcaagagtac
atccgggggc 421 actgcagcac tggctgcct cgtcaaggat tatttccag agccagtaac cgtgagctgg 481
aacagtggag cactcacttc tgggtccat actttctg ctgtctgca aagctctggc 541 ctgtactcac
tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
gcctagcaat actaaggtcg ataagcgggt ggaaccaag 661 agctgcgaca agactcacac ttgtcccca
tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgtcccacc aaagcctaaa gatactctga
tgataagtag aacaccgag 781 gtgacatgtg ttgtttaga cgttccac gaggaccag aggttaagt
caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
acataccgtg tagtcagtgt tctcacagt ctgcaccaag actggctcaa cggcaaagaa 961 tacaatgca
aagtgtcaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga
accacaggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gactgcctg
gtgaaaggct tctacccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
agacaacccc cccagtgtg 1201 gatagtgcg ggtctttctt tctgtacagt aagctgactg tggacaagtc
ccgctggcag 1261 cagggtaacg tcttcagctg ttccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
aagtcactga gcctgagccc agggaag

Protein Sequence Defining the Full Length Sh14F11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:200)

TABLE-US-00183 1 qitlkesgpt lvkptqtl tctfsgfsls tygmvgvwir qppgkalewl adiwwdddky
61 ynpslksrlt itkdtsknqv vltmtnmdpv dtatyycarr ghysamdywg qgtltvssa 121 stkgpsvfpl
apsskstsgg taalglvkd yfpepvtvsw nsgaltsgvh tfpavlqssg 181 lyslssvvtv pssslgtqty
icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfppkpk dtlmisrtp vtcvvdvsh
edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvsvltv lhqdwlngke ykckvsnkakl papiektisk
akgqprepqv ytlppsreem 361 tknqvsltl vkgfypsdiawewesngqpe nnykttppvl dsdgsfflys
kltvdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length Sh14F11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:201)

TABLE-US-00184 1 caagtgactc tcaaggagtc cggacccgcc ctggtcaaac caacgcagac actgacgctc
61 acatgcacct tcagcggatt ttcgtgtca acgtacggca tgggtgtggg gtggattcgc 121 cagcctccgg
ggaaagccct tgaatggtg gcggacatct ggtgggatga tgacaagtac 181 tataatccct cacttaagtc

acgggtgacg atctcgaaag acaccagcaa gaaccaggta 241 gtgctgacaa tgactaacat ggaccgggtc
gatacagcgg tctactattg tgctagaagg 301 ggacactact ccgcaatgga ttattggggt caggggacgc
tcgtaaccgt gtcgtcggcc 361 tcaacaaaag gaccaagtgt gttccactc gcccttagca gcaagagtac
atccggggggc 421 actgcagcac tcggctgcct cgtcaaggat tttttccag agccagtaac cgtgagctgg 481
aacagtggag cactcacttc tgggttccat acttttctg ctgtcctgca aagctctggc 541 ctgtactcac
tcagctccgt cgtgaccgtg ccatcttcat ctctgggcac tcagacctac 601 atctgtaatg taaaccacaa
gcctagcaat actaaggctg ataagcgggt ggaaccaag 661 agctgcgaca agactcacac ttgtccccc
tgccctgccc ctgaacttct gggcgggtccc 721 agcgtctttt tgttcccacc aaagcctaaa gatactctga
tgataagtag aacacccgag 781 gtgacatgtg ttgtttaga cgtttccac gaggaccag aggttaagt
caactggtac 841 gttgatggag tcgaagtaca taatgctaag accaagccta gagaggagca gtataatagt 901
acataccgtg tagtcagtgt tctcacagtg ctgcaccaag actggctcaa cggcaaagaa 961 taciaaatga
aagtgtcaa caaagcactc ccagccccta tcgagaagac tattagtaag 1021 gcaaaggggc agcctcgtga
accacaggtg tacactctgc caccagtag agaggaaatg 1081 acaaagaacc aagtctcatt gacctgcctg
gtgaaaggct tctacccag cgacatcgcc 1141 gttgagtggg agagtaacgg tcagcctgag aacaattaca
agacaacccc ccagtgctg 1201 gatagtgcg ggtctttct tctgtacagt aagctgactg tggacaagtc
ccgctggcag 1261 cagggtaacg tcttcagctg tccgtgatg cacgaggcat tgcacaacca ctacaccag 1321
aagtcactga gcctgagccc aggggaag

Protein Sequence Defining the Full Length Sh14F11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:202)

TABLE-US-00185 1 qvtlkesgpa lvkptqltl tctfsgfsls tygmvgwir qppgkalewl adiwwdddky
61 ynpslksrlt iskdtsknqv vltmtndpv dtavyycarr ghysamdywg qgtlvtvssa 121
stkgpsvfpl apsskstsgg taalglvkd yfpepvtvsw nsgaltsgvh tfpavlqssg 181 lyslssvvtv
pssslgtqty icnvnhkpsn tkvdkrvepk scdkthtcpp cpapellggp 241 svflfppkpk dtlmisrtp
vtcvvvdvsh edpevkfnwy vdgvevhnak tkpreeqyns 301 tyrvsvltv lhqdwlngke
ykckvsnkak papiektisk akgqprepqv ytlppsreem 361 tknqvsltl vkgfypsdiawewesngqpe
nnykttppvl dsdgsfflys kltvdksrwq 421 qgnvfscsvm healhnhytq kslslspgk

Nucleic Acid Sequence Encoding the Full Length Ch01G06 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:203)

TABLE-US-00186 1 gacatccaaa tgaccagtc acccgcgagc ctttcggcgt cggtcggaga aacggtcacg
61 atcacgtgcc ggacatcaga gaatctccat aactacctg cgtggatatca acagaagcag 121 gggaagtcgc
cccagttgct tgtatacgat gcgaaaacgt tggcggatgg ggtgccgtcc 181 agattctcgg gatcgggctc
ggggacgcag tactcgctca agatcaattc gctgcagccg 241 gaggactttg ggctgtacta ttgtcagcat
ttttggtcat caccgtatac atttgagggt 301 ggaacgaaac ttgagattaa gcgcacagtt gctgccccca
gcgtgttcat ttccacact 361 agcgatgagc agctgaaaag cggtagtgc tctgtcgtat gcttgcctaa
caacttttac 421 ccacgtgagg ctaaggtgca gtggaaagtg gataatgcac ttcaatctgg aaacagtcaa 481
gagtcctgta cagaacagga cagcaaagac tcaacttatt cactctctc caccctgact 541 ctgtccaagg
cagactatga aaaacacaag gtatacgct gcgagggtac acaccagggt 601 ttgtctagtc ctgtcaccaa
gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Ch01G06 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:204)

TABLE-US-00187 1 diqmtqspas lsasvgetvt itcrtsenlh nylawyqqkq gkspqllvyd aktladgvps
61 rfsgsgsgtq yslkinslqp edfsgsyqcw fwsspytfgg gtleikrtv aapsvfifpp 121 sdeqlkshta
svvcllnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt 181 lskadyekhk vyacevthqg
lsspvtksfn rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:205)

TABLE-US-00188 1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaattgcat aactacctg cgtggatatca gcagaagccc 121 gggaagtcac
cgaaactcct tgtctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtagcgac tatacgctta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta ttgtcagcat

tttgggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa gcgcacagtt gctgccccca
gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtgcc tctgtcgat gcttgctcaa
caacttttac 421 ccacgtgagg ctaaggtgca gtggaaaagt gataatgcac ttcaatctgg aaacagtcaa 481
gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541 ctgtccaagg
cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 601 ttgtctagtc ctgtcaccaa
gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:206)

TABLE-US-00189 1 diqmtqspss lsasvgrvt itcrtsenlh nylawyqqkp gkspkllvyd aktladgvps
61 rfsrgsgsgtd ylttisslqp edfatyyqch fwsspytfgq gtleikrtv aapsvfifpp 121 sdeqlksgta
svvcllnfy preakvqwkv dnalqsgnsq esvteqdsd styslsstlt 181 lskadyekhk vyacevthqg
lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (Also Referred to Herein as the Full Length Hu01G06 IGFV1-39 F1 Light Chain; SEQ ID NO:207)

TABLE-US-00190 1 gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga tcgggtcaca
61 attacgtgcc gaacgtcaga gaatttgcatt aactacctcg cgtggtatca gcagaagccc 121 gggaaggccc
cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg gaagcggatc
cggtagcgac tatacgctta cgatctcatc gtcacagccc 241 gaggactttg cgacgtacta ttgtcagcat
tttgggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa gcgcacagtt gctgccccca
gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtgcc tctgtcgat gcttgctcaa
caacttttac 421 ccacgtgagg ctaaggtgca gtggaaaagt gataatgcac ttcaatctgg aaacagtcaa 481
gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541 ctgtccaagg
cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 601 ttgtctagtc ctgtcaccaa
gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (also Referred to Herein as the Full Length Hu01G06 IGFV1-39 F1 Light Chain; SEQ ID NO:208)

TABLE-US-00191 1 diqmtqspss lsasvgrvt itcrtsenlh nylawyqqkp gkapklliyd
aktladgvps 61 rfsrgsgsgtd ylttisslqp edfatyyqch fwsspytfgq gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwkv dnalqsgnsq esvteqdsd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:209)

TABLE-US-00192 1 gacatccaaa tgaccagtc gccgtcgctg ctttcagcgt cggtagggga
tcgggtcaca 61 attacgtgcc gaacgtcaga gaatttgcatt aactacctcg cgtggtatca gcagaagccc 121
gggaagtcac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg
gaagcggatc cggtagcgac tatacgctta cgatctcatc gtcacagccc 241 gaggactttg cgacgtacta
ttgtcagcat tttgggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa gcgcacagtt
gctgccccca gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtgcc tctgtcgat
gcttgctcaa caacttttac 421 ccacgtgagg ctaaggtgca gtggaaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 gagtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatacgctt gcgagggtac acaccagggt 601 ttgtctagtc
ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:210)

TABLE-US-00193 1 diqmtqspss lsasvgrvt itcrtsenlh nylawyqqkp gkspklliyd
aktladgvps 61 rfsrgsgsgtd ylttisslqp edfatyyqch fwsspytfgq gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwkv dnalqsgnsq esvteqdsd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (also referred to herein as the Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain; SEQ ID NO:207)

TABLE-US-00194 1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga
tcgggtcaca 61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121
gggaaggccc cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg
gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta
ttgtcagcat ttttggtcgt cgccctacac atttgggcag 301 gggaccaagt tggaaatcaa gcgcacagtt
gctgccccca gcgtgttcat tttccacac 361 agcgtatgagc agctgaaaag cggtagtgc tctgtcgtat
gcttgctcaa caacttttac 421 ccacgtgagg ctaagggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgaggttac acaccagggg 601 ttgtctagtc
ctgtcacaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (also referred to herein as the Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain; SEQ ID NO:208)

TABLE-US-00195 1 diqmtqspss lsavsgdrvt itcrtsenlh nylawyqqkp gkapklli
aktladgvps 61 rfsqsgsgtd ylttisslqp edfatyyqh fwsspytfgq gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwk dnalqsgnsq esvteqdsd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:263)

TABLE-US-00196 1 gacatccaaa tgaccagtc gccgtcgtcg ctttcagcgt cggtagggga
tcgggtcaca 61 attacgtgcc gaacgtcaga gaatttgcac aactacctcg cgtggtatca gcagaagccc 121
gggaagtac cgaaactcct tatctacgat gcgaaaacgc tggcggatgg agtgccgtcg 181 agattctcgg
gaagcggatc cggtagcgac tatacgctta cgatctcatc gctccagccc 241 gaggactttg cgacgtacta
ttgtcagcat ttttggtcgg acccctacac atttgggcag 301 gggaccaagt tggaaatcaa gcgcacagtt
gctgccccca gcgtgttcat tttccacac 361 agcgtatgagc agctgaaaag cggtagtgc tctgtcgtat
gcttgctcaa caacttttac 421 ccacgtgagg ctaagggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgaggttac acaccagggg 601 ttgtctagtc
ctgtcacaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:264)

TABLE-US-00197 1 diqmtqspss lsavsgdrvt itcrtsenlh nylawyqqkp gksplli
aktladgvps 61 rfsqsgsgtd ylttisslqp edfatyyqh fwsdpytfgq gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwk dnalqsgnsq esvteqdsd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Ch06C11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:211)

TABLE-US-00198 1 gatatcgta tgaccagtc ccagaagttc atgtcaactt cagtgggaga
cagagtgtcc 61 gtcacatgta aagcctcgca aatgtggga accaacgtag cgtggttcca gcagaaacct 121
ggccaatcac cgaaggcact gatctactcg gccagctata ggtactcggg agtaccagat 181 cggtttacgg
ggtcggggag cgggacggac ttatcctca ctatttcaa tgcagtcg 241 gaggaccttg cggaatactt
ctgccagcag tataacaact atcccctcac gtttggtgct 301 ggtacaaaat tggagttgaa gcgcacagtt
gctgccccca gcgtgttcat tttccacac 361 agcgtatgagc agctgaaaag cggtagtgc tctgtcgtat
gcttgctcaa caacttttac 421 ccacgtgagg ctaagggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgaggttac acaccagggg 601 ttgtctagtc
ctgtcacaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Ch06C11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:212)

TABLE-US-00199 1 divmtsqskf mstsvgdrvs vtckasqnvq tnvwfqqkp gqspkaliys
asyrysgvpd 61 rftgsgsgtd ftiltlnvqs edlaeyfcqq ynnyppltfga gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:213)

TABLE-US-00200 1 gacatccaaa tgaccaatc gccctcctcc ctctccgcat cagtagggga
ccgcgtcaca 61 attacttgca aagcgtcgca gaacgtcgga acgaatgtgg cgtggtttca gcagaagccc 121
ggaaaagctc cgaagagctt gatctactcg gcctcatata ggtattcggg tgtgccgagc 181 cggtttagcg
ggtcgggggtc aggtactgat ttcacgtcga caatttcac gttgcagcca 241 gaagatttcg ccacatatta
ctgtcagcag tacaacaatt accctctgac gttcggccag 301 ggaaccaaac ttgagatcaa gcgcacagtt
gctgccccca gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtccc tctgtcgtat
gcttgctcaa caacttttac 421 ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaactatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgaggttac acaccagggg 601 ttgtctagtc
ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:214)

TABLE-US-00201 1 diqmtqspss lsasvgdrvt itckasqnvq tnvwfqqkp gkapksliys
asyrysgvps 61 rftgsgsgtd ftiltlnvqs edfatyycqq ynnyppltfgg gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Ch14F11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:215)

TABLE-US-00202 1 gacatcgtga tgacacagtc acagaaattc atgtccacat ccgtcgggtga
tagagtatcc 61 gtcacgtgta aggcctcgca aaacgtagga actaatgtgg cgtgggtatca acagaagcca 121
ggacagtcac ccaaagcact catctacagc ccctcatatc ggtacagcgg ggtgccggac 181 aggttcacgg
gatcgggggag cgggaccgat tttactactga ccatttcgaa tgtccagtcg 241 gaggaccttg cggaaactt
ctgccagcag tataactcgt accctcacac gtttgagggt 301 ggactaagt tggagatgaa acgcacagtt
gctgccccca gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtccc tctgtcgtat
gcttgctcaa caacttttac 421 ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaactatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgaggttac acaccagggg 601 ttgtctagtc
ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Ch14F11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:216)

TABLE-US-00203 1 divmtsqskf mstsvgdrvs vtckasqnvq tnvwyyqqkp gqspkaliys
psyrysgvpd 61 rftgsgsgtd ftiltlnvqs edlaeyfcqq ynsyphtfgg gtleikrtv aapsvfifpp 121
sdeqlksgta svvcllnfy preakvqwkv dnalqsgnsq esvteqdskd styslsstlt 181 lskadyekhk
vyacevthqg lsspvtksf n rgec

Nucleic Acid Sequence Encoding the Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:217)

TABLE-US-00204 1 gatatccaga tgacacagtc accctcgtcg ctctcagctt ccgtaggcga
cagggtcact 61 attacgtgta aagcatcaca gaacgtcgga acgaatgtgg cgtgggttca gcagaagccc 121
gggaagagcc ccaaagcgct tatctactcc ccgtcgtatc ggtattccgg tgtccaagc 181 agatttcgg
ggtcagggttc gggaactgac tttaccctga ccctcgtc cctccaaccg 241 gaagatttcg ccacgtactt
ctgccagcag tacaacagct atcctcacac attcggacaa 301 gggacaaagt tggagattaa acgcacagtt
gctgccccca gcgtgttcat tttccacct 361 agcgatgagc agctgaaaag cggtagtccc tctgtcgtat

gcttgctcaa caacttttac 421 ccacgtgagg ctaaggtgca gtggaaagt gataatgcac ttcaatctgg
aaacagtcaa 481 ggtccgtga cagaacagga cagcaaagac tcaacttatt cactctcttc caccctgact 541
ctgtccaagg cagactatga aaaacacaag gtatagcct gcgagggtac acaccagggt 601 ttgtctagtc
ctgtcaccaa gtccttcaat aggggcgaat gt

Protein Sequence Defining the Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:218)

TABLE-US-00205 1 diqmtqspss lsasvdrvt itckasqnvgt tnvwfqqkp gkspkaliys
psyrsgvps 61 rfsgsgsgtd fltisslqp edfatyfcqq ynsyphtfgg gtleikrtv aapsvfifpp 121
sdeqlkshta svvcllnfy preakvqwkvd dnalqsgnsq esvteqdskd styslsstlt 181 lskadyekhh
vyacevthqg lsspvtksfv rgec

[0311] Table 22 is a concordance chart showing the SEQ ID NO. of each sequence discussed in this Example.

TABLE-US-00206 TABLE 22 SEQ ID NO. Nucleic Acid or Protein 171 Human IgG1 constant - nucleic acid 172 Human IgG1 constant - protein 173 Human Kappa constant - nucleic acid 174 Human Kappa constant - protein 175 Humanized Ch01G06 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid 176 Humanized Ch01G06 Chimeric Heavy Human Variable + Human IgG1 constant - protein 177 Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human IgG1 constant - nucleic acid 178 Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human IgG1 constant - protein 179 Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human IgG1 constant - nucleic acid 180 Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human IgG1 constant - protein 181 Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant - nucleic acid 182 Humanized Sh01G06 IGHV1-18 M69L Heavy Human Variable + Human IgG1 constant - protein 183 Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant - nucleic acid 184 Humanized Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Human Variable + Human IgG1 constant - protein 185 Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant - nucleic acid 186 Humanized Sh01G06 IGHV1-18 M69L K64Q Heavy Human Variable + Human IgG1 constant - protein 187 Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant - nucleic acid 188 Humanized Sh01G06 IGHV1-69 T30S I69L Heavy Human Variable + Human IgG1 constant - protein 189 Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant - nucleic acid 190 Humanized Sh01G06 IGHV1-69 T30S K64Q I69L Heavy Human Variable + Human IgG1 constant - protein 255 Humanized Hu01G06 IGHV1-18 F1 Heavy Human Variable + Human IgG1 constant - nucleic acid 256 Humanized Hu01G06 IGHV1-18 F1 Heavy Human Variable + Human IgG1 constant - protein 257 Humanized Hu01G06 IGHV1-18 F2 Heavy Human Variable + Human IgG1 constant - nucleic acid 258 Humanized Hu01G06 IGHV1-18 F2 Heavy Human Variable + Human IgG1 constant - protein 259 Humanized Hu01G06 IGHV1-69 F1 Heavy Human Variable + Human IgG1 constant - nucleic acid 260 Humanized Hu01G06 IGHV1-69 F1 Heavy Human Variable + Human IgG1 constant - protein 261 Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant - nucleic acid 262 Humanized Hu01G06 IGHV1-69 F2 Heavy Human Variable + Human IgG1 constant - protein 191 Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid 192 Humanized Ch06C11 Chimeric Heavy Human Variable + Human IgG1 constant - protein 193 Humanized HE LM 06C11 IGHV2-70 Heavy Human Variable + Human IgG1 constant - nucleic acid 194 Humanized HE LM 06C11 IGHV2-70 Heavy Human Variable + Human IgG1 constant - protein 195 Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - nucleic acid 196 Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - protein 197 Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant - nucleic acid 198 Humanized Ch14F11 Chimeric Heavy Human Variable + Human IgG1 constant - protein 199 Humanized Sh14F11 IGHV2-5 Heavy Human Variable + Human IgG1 constant - nucleic acid 200 Humanized Sh14F11 IGHV2-5 Heavy Human

Variable + Human IgG1 constant - protein 201 Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human IgG1 constant - nucleic acid 202 Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human IgG1 constant - protein 203 Humanized Ch01G06 Chimeric Human Variable + Human Kappa constant - nucleic acid 204 Humanized Ch01G06 Chimeric Human Variable + Human Kappa constant - protein 205 Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant - nucleic acid 206 Humanized Hu01G06 IGKV1-39 Human Variable + Human Kappa constant - protein 207 Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant - nucleic acid 208 Humanized Hu01G06 IGKV1-39 S43A V48I Human Variable + Human Kappa constant - protein 209 Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant - nucleic acid 210 Humanized Hu01G06 IGKV1-39 V48I Human Variable + Human Kappa constant - protein 207 Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant - nucleic acid 208 Humanized Hu01G06 IGKV1-39 F1 Human Variable + Human Kappa constant - protein 263 Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant - nucleic acid 264 Humanized Hu01G06 IGKV1-39 F2 Human Variable + Human Kappa constant - protein 211 Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant - nucleic acid 212 Humanized Ch06C11 Chimeric Human Variable + Human Kappa constant - protein 213 Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant - nucleic acid 214 Humanized Sh06C11 IGKV1-16 Human Variable + Human Kappa constant - protein 215 Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant - nucleic acid 216 Humanized Ch14F11 Chimeric Human Variable + Human Kappa constant - protein 217 Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant - nucleic acid 218 Humanized Hu14F11 IGKV1-16 Human Variable + Human Kappa constant - protein

[0312] Table 23 below shows antibodies containing chimeric immunoglobulin heavy and light chains and exemplary combinations of the full-length chimeric or humanized immunoglobulin heavy and light chains.

TABLE-US-00207 TABLE 23 Antibody Name Light Chain Heavy Chain Hu01G06-1 Humanized Ch01G06 Chimeric Humanized Ch01G06 Chimeric Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 204) constant (SEQ ID NO: 176) Hu01G06-46 Humanized Hu01G06 IGKV1-39 Humanized Hu01G06 IGHV1-18 Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 206) constant (SEQ ID NO: 178) Hu01G06-52 Humanized Hu01G06 IGKV1-39 Humanized Hu01G06 IGHV1-69 Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 206) constant (SEQ ID NO: 180) Hu01G06-100 Humanized Hu01G06 IGKV1-39 Humanized Sh01G06 IGHV1-18 M69L Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 206) constant (SEQ ID NO: 182) Hu01G06-101 Humanized Hu01G06 IGKV1-39 Humanized Sh01G06 IGHV1-18 M69L Human Variable + Human Kappa K64Q Heavy Human Variable + Human constant (SEQ ID NO: 206) IgG1 constant (SEQ ID NO: 186) Hu01G06-102 Humanized Hu01G06 IGKV1-39 Humanized Sh01G06 IGHV1-18 M69L K64Q Human Variable + Human Kappa G44S Heavy Human Variable + Human constant (SEQ ID NO: 206) IgG1 constant (SEQ ID NO: 184) Hu01G06-103 Humanized Hu01G06 IGKV1-39 Humanized Sh01G06 IGHV1-69 T30S Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 206) IgG1 constant (SEQ ID NO: 188) Hu01G06-104 Humanized Hu01G06 IGKV1-39 Humanized Sh01G06 IGHV1-69 T30S K64Q Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 206) IgG1 constant (SEQ ID NO: 190) Hu01G06-105 Humanized Hu01G06 IGKV1-39 V48I Humanized Sh01G06 IGHV1-18 M69L Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 210) constant (SEQ ID NO: 182) Hu01G06-106 Humanized Hu01G06 IGKV1-39 V48I Humanized Sh01G06 IGHV1-18 M69L Human Variable + Human Kappa K64Q Heavy Human Variable + Human constant (SEQ ID NO: 210) IgG1 constant (SEQ ID NO: 186)

Hu01G06-107 Humanized Sh01G06 IGKV1-39 V48I Humanized Sh01G06 IGHV1-18 M69L K64Q Human Variable + Human Kappa G44S Heavy Human Variable + Human constant (SEQ ID NO: 210) IgG1 constant (SEQ ID NO: 184) Hu01G06-108 Humanized Hu01G06 IGKV1-39 V48I Humanized Sh01G06 IGHV1-69 T30S Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 210) IgG1 constant (SEQ ID NO: 188) Hu01G06-109 Humanized Hu01G06 IGKV1-39 V48I Humanized Sh01G06 IGHV1-69 T30S K64Q Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 210) IgG1 constant (SEQ ID NO: 190) Hu01G06-110 Humanized Hu01G06 IGKV1-39 S43A Humanized Sh01G06 IGHV1-18 M69L V48I Human Variable + Human Kappa Heavy Human Variable + Human constant (SEQ ID NO: 208) IgG1 constant (SEQ ID NO: 182) Hu01G06-111 Humanized Hu01G06 IGKV1-39 S43A Humanized Sh01G06 IGHV1-18 M69L V48I Human Variable + Human Kappa K64Q Heavy Human Variable + Human constant (SEQ ID NO: 208) IgG1 constant (SEQ ID NO: 186) Hu01G06-112 Humanized Hu01G06 IGKV1-39 S43A Humanized Sh01G06 IGHV1-18 M69L V48I Human Variable + Human Kappa K64Q G44S Heavy Human Variable + constant (SEQ ID NO: 208) Human IgG1 constant (SEQ ID NO: 184) Hu01G06-113 Humanized Hu01G06 IGKV1-39 S43A Humanized Sh01G06 IGHV1-69 T30S V48I Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 208) IgG1 constant (SEQ ID NO: 188) Hu01G06-114 Humanized Hu01G06 IGKV1-39 S43A Humanized Sh01G06 IGHV1-69 T30S K64Q V48I Human Variable + Human Kappa I69L Heavy Human Variable + Human constant (SEQ ID NO: 208) IgG1 constant (SEQ ID NO: 190) Hu01G06-122 Humanized Hu01G06 IGKV1-39 F1 Humanized Hu01G06 IGHV1-18 F1 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 208) constant (SEQ ID NO: 256) Hu01G06-127 Humanized Hu01G06 IGKV1-39 F2 Humanized Hu01G06 IGHV1-18 F2 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 264) constant (SEQ ID NO: 258) Hu01G06-135 Humanized Hu01G06 IGKV1-39 F1 Humanized Hu01G06 IGHV1-69 F1 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 208) constant (SEQ ID NO: 260) Hu01G06-138 Humanized Hu01G06 IGKV1-39 F1 Humanized Hu01G06 IGHV1-69 F2 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 208) constant (SEQ ID NO: 262) Hu01G06-146 Humanized Hu01G06 IGKV1-39 F2 Humanized Hu01G06 IGHV1-69 F2 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 264) constant (SEQ ID NO: 262) Hu06C11-1 Humanized Ch06C11 Chimeric Humanized Ch06C11 Chimeric Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 212) constant (SEQ ID NO: 176) Hu06C11-27 Humanized Sh06C11 IGKV1-16 Humanized HE LM 06C11 IGHV2-70 Human Variable + Human Kappa Heavy Human Variable + Human IgG1 constant (SEQ ID NO: 214) constant (SEQ ID NO: 194) Hu06C11-30 Humanized Sh06C11 IGKV1-16 Humanized Hu06C11 IGHV2-5 Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 214) constant (SEQ ID NO: 196) Hu14F11-1 Humanized Ch14F11 Chimeric Humanized Ch14F11 Chimeric Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 216) constant (SEQ ID NO: 198) Hu14F11-23 Humanized Ch14F11 Chimeric Humanized Ch06C11 Chimeric Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 216) constant (SEQ ID NO: 192) Hu14F11-24 Humanized Ch06C11 Chimeric Humanized Ch14F11 Chimeric Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 212) constant (SEQ ID NO: 198) Hu14F11-39 Humanized Hu14F11 IGKV1-16 Humanized Sh14F11 IGHV2-5 Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 218) constant (SEQ ID NO: 200) Hu14F11-47 Humanized Hu14F11 IGKV1-16 Humanized Sh14F11-IGHV2-70 Heavy Human Variable + Human Kappa Human Variable + Human IgG1 constant (SEQ ID NO: 218) constant (SEQ ID NO: 202)

[0313] The antibody constructs containing the full length chimeric heavy and light chains are

designated below: [0314] Chimeric 01G06 (Hu01G06-1)=Full Length Ch01G06 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:176) plus Full Length Ch01G06 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:204) [0315] Chimeric 06C11 (Hu06C11-1)=Full Length Ch06C11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:192) plus Full Length Ch06C11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:212) [0316] Chimeric 14F11 (Hu14F11-1)=Full Length Ch14F11 Chimeric Heavy Chain (Mouse Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:198) plus Full Length Ch14F11 Chimeric Light Chain (Mouse Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:216)

[0317] Fifteen of the possible antibody constructs containing the full length immunoglobulin heavy and light chains containing humanized variable regions are designated below: [0318] Hu01G06-46=Full Length Hu01G06 IGHV1-18 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:178) plus Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:206) [0319] Hu01G06-52=Full Length Hu01G06 IGHV1-69 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:180) plus Full Length Hu01G06 IGKV1-39 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:206) [0320] Hu01G06-107=Full Length Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:184) plus Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:210) [0321] Hu01G06-108=Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:188) plus Full Length Hu01G06 IGKV1-39 V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:210) [0322] Hu01G06-112=Full Length Sh01G06 IGHV1-18 M69L K64Q G44S Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:184) plus Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208) [0323] Hu01G06-113=Full Length Sh01G06 IGHV1-69 T30S I69L Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:188) plus Full Length Hu01G06 IGKV1-39 S43A V48I Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208) [0324] Hu01G06-122=Full Length Hu01G06 IGHV1-18 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:256) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208) [0325] Hu01G06-127=Full Length Hu01G06 IGHV1-18 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:258) plus Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:264) [0326] Hu01G06-135=Full Length Hu01G06 IGHV1-69 F1 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:260) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208) [0327] Hu01G06-138=Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:262) plus Full Length Hu01G06 IGKV1-39 F1 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:208) [0328] Hu01G06-146=Full Length Hu01G06 IGHV1-69 F2 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:262) plus Full Length Hu01G06 IGKV1-39 F2 Light Chain (Humanized Kappa Chain Variable

Region and Human Kappa Constant Region) (SEQ ID NO:264) [0329] Hu06C11-27=Full Length HE LM 06C11 IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:194) plus Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:214) [0330] Hu06C11-30=Full Length Hu06C11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:196) plus Full Length Sh06C11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:214) [0331] Hu14F11-39=Full Length Sh14F11 IGHV2-5 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:200) plus Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:218) [0332] Hu14F11-47=Full Length Sh14F11-IGHV2-70 Heavy Chain (Humanized Heavy Chain Variable Region and Human IgG1 Constant Region) (SEQ ID NO:202) plus Full Length Hu14F11 IGKV1-16 Light Chain (Humanized Kappa Chain Variable Region and Human Kappa Constant Region) (SEQ ID NO:218)

Example 15: Binding Affinities of Humanized and Chimeric Anti-GDF15 Monoclonal Antibodies [0333] The binding affinities and kinetics of binding of chimeric and humanized antibodies to mFc-rhGDF15 were measured by surface plasmon resonance, using a BIAcore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0334] Goat anti-human IgGs (Fc fragment specific, Jackson ImmunoResearch, West Grove, PA) were immobilized on carboxymethylated dextran CM4 sensor chips by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 L/minute. Injection time was varied for each antibody to yield an R_{max} between 30 and 60 RU. Buffer or mFc-rhGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60 L/minute. The dissociation phase was monitored for up to 1200 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 2.25, at a flow rate of 30 L/minute. The GDF15 concentration range tested was 20 nM to 0.625 nM.

[0335] Kinetic parameters were determined using the kinetic function of the BIAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters for each antibody, k_{sub.a} (association rate constant), k_{sub.d} (dissociation rate constant), and K_{sub.D} (equilibrium dissociation constant) were determined. Kinetic values of purified monoclonal antibodies on mFc-rhGDF15 are summarized in Table 24.

TABLE-US-00208 TABLE 24 Antibody k_{sub.a} (1/Ms) k_{sub.d} (1/s) K_{sub.D} (M) n
Hu01G06-1 4.2E+06 6.4E-04 1.6E-10 8
Hu01G06-46 3.6E+06 3.8E-04 1.1E-10 11
Hu01G06-52 3.6E+06 3.6E-04 9.9E-11 10
Hu06C11-1 5.3E+06 8.4E-04 1.8E-10 2
Hu06C11-27 4.7E+06 8.2E-04 1.8E-10 2
Hu06C11-30 4.8E+06 8.7E-04 1.8E-10 2
Hu14F11-1 3.0E+06 4.6E-04 1.6E-10 2
Hu14F11-39 3.0E+06 1.9E-04 6.6E-11 2
Hu14F11-47 3.3E+06 1.8E-04 6.5E-11 2

[0336] The results in Table 24 demonstrate that the chimeric and each of the humanized antibodies, have fast association rates (k_{sub.a}), very slow disassociation rates (k_{sub.d}) and very high affinities (K_{sub.D}). In particular, the antibodies have affinities ranging from about 65 pM to about 200 pM.

[0337] Kinetic values of chimeric 01G06 (Hu01G06-1), two initial lead humanized 01G06 monoclonal antibodies (Hu01G06-46 and -52), and sequence optimized humanized 01G06 monoclonal antibody variants Hu01G06-100 through -114 (in supernatant) on mFc-rhGDF15 are summarized in Table 25.

TABLE-US-00209 TABLE 25 Antibody k_{sub.a} (1/Ms) k_{sub.d} (1/s) K_{sub.D} (M) n
Hu01G06-1 4.9E+06 7.1E-04 1.4E-10 3
Hu01G06-46 4.1E+06 4.3E-04 1.0E-10 3
Hu01G06-52 5.0E+06 4.4E-04 8.9E-11 3
Hu01G06-100 4.1E+06 6.2E-04 1.5E-10 3
Hu01G06-101 4.4E+06 6.3E-04 1.4E-10 3
Hu01G06-102 4.4E+06 4.6E-04 1.1E-10 3
Hu01G06-103 4.4E+06 4.7E-04 1.1E-10 3
Hu01G06-104 4.5E+06 5.2E-04 1.2E-10 3
Hu01G06-105 4.3E+06 5.6E-04 1.3E-10 3
Hu01G06-

106 4.3E+06 7.0E-04 1.6E-10 3 Hu01G06-107 4.1E+06 4.7E-04 1.2E-10 3 Hu01G06-108 4.2E+06 4.6E-04 1.2E-10 3 Hu01G06-109 4.6E+06 5.6E-04 1.3E-10 4 Hu01G06-110 4.3E+06 5.8E-04 1.4E-10 4 Hu01G06-111 4.3E+06 6.6E-04 1.6E-10 3 Hu01G06-112 4.7E+06 5.3E-04 1.2E-10 3 Hu01G06-113 4.5E+06 4.8E-04 1.1E-10 3 Hu01G06-114 4.5E+06 5.4E-04 1.3E-10 3 [0338] The results in Table 25 demonstrate that the sequence optimized antibodies, Hu01G06-100 through -114, have binding affinities ranging from about 89 pM to about 160 pM.

[0339] Binding affinities and binding kinetics of mFc-rhGDF15 with chimeric 14F11 (Hu14F11-1), chimeric light 14F11 with chimeric heavy 06C11 (Hu14F11-23), and chimeric light 06C11 with chimeric heavy 14F11 (Hu14F11-24) heavy monoclonal antibody variants (in supernatant) were measured using biolayer interferometry (BLI) on an Octet™ QK instrument (ForteBio, Inc., Menlo Park, CA). The Octet analysis was performed at 30° C. using 1× Kinetics Buffer (ForteBio, Inc.) as assay buffer. Anti-human IgG Fc Capture (AHC) biosensors (ForteBio, Inc.) were used to capture human antibodies onto the sensors. Sensors were saturated in assay buffer for at 300 seconds before the assay. Antibodies were loaded onto sensors by dipping the sensors into antibody supernatant solution for 220 seconds, which typically resulted in capture levels of 1.5-2 nm. Baseline was established by dipping the sensors into 1× assay buffer for 200 seconds. Next, association was monitored for 220 seconds in 400 nM mFc-rhGDF15 protein, and dissociation was followed for 600 seconds in buffer alone.

[0340] Kinetic parameters for Hu14F11-1, Hu14F11-23, and Hu14F11-24 were determined using the kinetic function of the ForteBio Analysis Software Version 7.0. Kinetic parameters of the antibody, k.sub.a, k.sub.d, and K.sub.D were determined.

[0341] Kinetic values of Hu14F11-1, Hu14F11-23, and Hu14F11-24 heavy monoclonal antibody variants (in supernatant) on mFc-rhGDF15 are summarized in Table 26.

TABLE-US-00210 TABLE 26 Antibody k.sub.a (1/Ms) k.sub.d (1/s) K.sub.D (M) n Hu14F11-1 6.3E+05 1.9E-05 3.2E-11 3 Hu14F11-23 3.4E+05 6.2E-05 1.8E-10 1 Hu14F11-24 7.1E+05 2.2E-04 3.1E-10 1

[0342] The results in Table 26 demonstrate that Hu14F11-23 and Hu14F11-24, (i.e., antibodies that consist of one chimeric 06C11 chain (heavy or light) mixed with one chimeric 14F11 chain (heavy or light)), retain binding to GDF15. In particular, these antibodies have high affinities ranging from about 180 pM to about 310 pM.

Example 16: Binding Affinities of Affinity Matured Humanized Anti-GDF15 Monoclonal Antibodies

[0343] The binding affinities and kinetics of binding of chimeric and humanized antibodies to mFc-rhGDF15, cleaved-rhGDF15, rabbit Fc mature recombinant mouse GDF15 (rFc-rmGDF15), and mouse Fc mature recombinant cynomolgus monkey GDF15 (mFc-rcGDF15) were measured by surface plasmon resonance, using a BIAcore® T100 instrument (GE Healthcare, Piscataway, NJ).

[0344] Goat anti-human IgGs (Fc fragment specific, Jackson ImmunoResearch, West Grove, PA) were immobilized on carboxymethylated dextran CM4 sensor chips by amine coupling, according to a standard protocol. Analyses were performed at 37° C. using PBS containing 0.05% surfactant P20 as running buffer. The antibodies were captured in individual flow cells at a flow rate of 10 µL/minute. Injection time was varied for each antibody to yield an Rmax between 30 and 60 RU. Buffer, mFc-rhGDF15, cleaved-rhGDF15, rFc-rmGDF15, or mFc-rcGDF15 diluted in running buffer was injected sequentially over a reference surface (no antibody captured) and the active surface (antibody to be tested) for 240 seconds at 60 µL/minute. The dissociation phase was monitored for up to 1500 seconds. The surface was then regenerated with two 60-second injections of 10 mM Glycine-HCl, pH 2.25, at a flow rate of 30 µL/minute. The GDF15 concentration range tested for each GDF15 protein was 5 nM to 0.3125 nM (two-fold dilutions).

[0345] Kinetic parameters were determined using the kinetic function of the BIAevaluation software (GE Healthcare) with double reference subtraction. Kinetic parameters for each antibody, k.sub.a (association rate constant), k.sub.d (dissociation rate constant), and K.sub.D (equilibrium

dissociation constant) were determined. Kinetic values of purified monoclonal antibodies on mFc-rhGDF15, mature human GDF15, rFc-rmGDF15, and mFc-rcGDF15 are summarized in Table 27.

TABLE-US-00211

TABLE 27	Protein	Antibody	k.sub.a (1/Ms)	k.sub.d (1/s)	K.sub.D (M)	n				
mFc-Hu01G06-122	5.9E+06	2.1E-05	6.5E-12	5	rhGDF15	Hu01G06-127	4.6E+06	4.2E-05	1.8E-11	4
Hu01G06-135	5.3E+06	4.4E-05	1.4E-11	5	Hu01G06-138	5.9E+06	4.1E-05	1.1E-11	5	
Hu01G06-146	5.3E+06	2.6E-05	9.3E-12	5	Cleaved- Hu01G06-122	7.9E+06	3.4E-05	7.9E-12	4	
rhGDF15	Hu01G06-127	6.1E+06	3.6E-05	1.0E-11	4	Hu01G06-135	7.3E+06	6.2E-05	1.0E-11	4
Hu01G06-138	7.9E+06	2.5E-05	4.5E-12	4	Hu01G06-146	6.5E+06	5.2E-05	1.1E-11	4	
mFc- Hu01G06-122	2.3E+06	2.4E-05	1.0E-11	4	rcGDF15	Hu01G06-127	1.8E+06	1.6E-05	9.5E-12	4
Hu01G06-135	2.2E+06	7.9E-05	3.8E-11	4	Hu01G06-138	2.3E+06	5.3E-05	2.5E-11	4	
Hu01G06-146	2.0E+06	1.5E-05	8.0E-12	4	rFc- Hu01G06-122	2.2E+07	1.4E-03	6.3E-11	2	
rmGDF15	Hu01G06-127	3.9E+07	2.1E-03	5.1E-11	2	Hu01G06-135	3.7E+07	1.9E-03	5.5E-11	2
Hu01G06-138	1.9E+07	8.0E-04	4.4E-11	2	Hu01G06-146	1.1E+07	7.2E-04	6.3E-11	2	

[0346] The results in Table 27 demonstrate that the chimeric and each of the humanized antibodies, have fast association rates (k.sub.a), very slow disassociation rates (k.sub.d) and very high affinities (K.sub.D). In particular, the antibodies have affinities ranging from less than 5 pM (e.g., about 4.5 pM) to about 65 pM.

Example 17: Reversal of Cachexia in an HT-1080 Fibrosarcoma Xenograft Model

[0347] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by humanized 01G06, 06C11, 14F11 antibodies in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. and inoculated subcutaneously into the flank of 8-week old female ICR-SCID mice as described above in Example 10. Body weight was measured daily. When body weight reached 93%, the mice were randomly divided into groups of ten mice each. Each group received one of the following treatments: murine IgG control, 01G06, 06C11, 14F11, and their respective humanized versions at 2 mg/kg. Treatment was administered once a day by intra-peritoneal injection. Antibody treatment with 01G06, Hu01G06-46 and Hu01G06-52 resulted in body weight increase to initial weight or 100% (p<0.001) (FIG. 23). Statistical analysis was performed using ANOVA. Results for reversal of body weight on day in the HT-1080 model are shown in FIG. 23 and Table 28, respectively.

TABLE-US-00212

TABLE 28	Treatment	% Body	ANOVA Analysis	Gr.	Agent	mg/kg	weight			
(compared to mIgG)	1	mIgG	2	79.5	NA	2	01G06	2	87.6	p < 0.001
	3	Hu01G06-46	2	95.4	p < 0.001					
	4	Hu01G06-52	2	87.8	p < 0.001					

[0348] The data in FIG. 23 and Table 28 indicated that antibodies 01G06, Hu01G06-46 and Hu01G06-52 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0349] Antibody treatment with 06C11, Hu06C11-27, and Hu06C11-30 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 24). Statistical analysis was performed using ANOVA. Results for reversal of body weight in the HT-1080 model are shown in FIG. 24 and Table 29.

TABLE-US-00213

TABLE 29	Treatment	% Body	ANOVA Analysis	Gr.	Agent	mg/kg	weight			
(compared to mIgG)	1	mIgG	2	87.7	NA	2	01G06	2	93.6	p < 0.001
	3	Hu06C11-27	2	93.2	p < 0.001					
	4	Hu06C11-30	2	89.8	p < 0.001					

[0350] The data in FIG. 24 and Table 29 indicate that antibodies 06C11, Hu06C11-27, and Hu06C11-30 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0351] Antibody treatment with 14F11, Hu14F11-39, and Hu14F11-47 resulted in body weight increase relative to initial weight or about 100% (p<0.001) (FIG. 25). Statistical analysis was performed using ANOVA. Results for reversal of body weight in the HT-1080 model are shown in FIG. 25 and Table 30.

TABLE-US-00214

TABLE 30	Treatment	% Body	ANOVA Analysis	Gr.	Agent	mg/kg	weight			
(compared to mIgG)	1	mIgG	2	87.7	NA	2	14F11	2	96.6	p < 0.001
	3	Hu14F11-39	2	90.5	p < 0.001					
	4	Hu14F11-47	2	90.7	p < 0.001					

[0352] The data in FIG. 25 and Table 30 indicated that antibodies 14F11, Hu14F11-39, and Hu14F11-47 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0353] Antibody treatment with humanized 01G06 antibodies (i.e., antibodies Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 and Hu01G06-146) resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 26). Statistical analysis was performed using ANOVA. Treatment with human IgG (hIgG) was used as a control. Results for reversal of body weights in the HT-1080 model are shown in FIG. 26 and Table 31.

TABLE-US-00215 TABLE 31 Treatment % Body ANOVA Analysis Gr. Agent mg/kg weight (compared to mIgG)

1	mIgG	2	84.2	NA	2	Hu01G06-122	2	96.3	$p < 0.001$	3	Hu01G06-127	2	96.1	$p < 0.001$	4	Hu01G06-135	2	93.5	$p < 0.001$	5	Hu01G06-138	2	91.9	$p < 0.001$	6	Hu01G06-146	2	92.7	$p < 0.001$
---	------	---	------	----	---	-------------	---	------	-------------	---	-------------	---	------	-------------	---	-------------	---	------	-------------	---	-------------	---	------	-------------	---	-------------	---	------	-------------

[0354] The data in FIG. 26 and Table 31 indicated that humanized anti-GDF15 antibodies Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 and Hu01G06-146 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

Example 18: Reversal of Cachexia in an mFc-rhGDF15-Induced Model

[0355] This Example demonstrates the reversal of cachexia (as indicated by body weight loss) by humanized 01G06 antibodies (i.e., antibody Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138, or Hu01G06-146) in an mFc-rhGDF15-induced cachexia model. mFc-rhGDF15 (1 $\mu\text{g/g}$) was administered subcutaneously into the flank of 8-week old female ICR-SCID mice. Body weight was measured daily. When body weight reached 93%, the mice were randomly divided into six groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG), Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 or Hu01G06-146 at 2 mg/kg. Treatment was administered once by intra-peritoneal injection. Treatment with antibody Hu01G06-122, Hu01G06-127, Hu01G06-135, Hu01G06-138 or Hu01G06-146 resulted in body weight increase relative to initial weight or about 100% ($p < 0.001$) (FIG. 27 and Table 32).

TABLE-US-00216 TABLE 32 Treatment % Body ANOVA Analysis Gr. Agent mg/kg weight (compared to mIgG)

1	mIgG	2	70.6	NA	2	Hu01G06-122	2	101.7	$p < 0.001$	3	Hu01G06-127	2	103.2	$p < 0.001$	4	Hu01G06-135	2	102.5	$p < 0.001$	5	Hu01G06-138	2	101.8	$p < 0.001$	6	Hu01G06-146	2	102.5	$p < 0.001$
---	------	---	------	----	---	-------------	---	-------	-------------	---	-------------	---	-------	-------------	---	-------------	---	-------	-------------	---	-------------	---	-------	-------------	---	-------------	---	-------	-------------

[0356] The data in FIG. 27 and Table 32 indicate that the disclosed anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced mouse model (i.e., a non-tumor bearing mouse model).

[0357] These results indicate that humanized anti-GDF15 antibodies can reverse cachexia in an mFc-rhGDF15-induced cachexia model.

Example 19: Dose Response Reversal of Cachexia in an HT-1080 Fibrosarcoma Xenograft Model

[0358] This Example demonstrates the dose response reversal of cachexia (as indicated by body weight loss) by humanized Hu01G06-127 and Hu01G06-135 antibodies in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. and inoculated subcutaneously into the flank of 8-week old female ICR-SCID mice as described above in Example 10. Body weight was measured daily. When body weight reached 93%, the mice were randomly divided into groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG; 20 mg/kg), Hu01G06-127 (20 mg/kg, 2 mg/kg, or 0.2 mg/kg) and Hu01G06-135 (20 mg/kg, 2 mg/kg, or 0.2 mg/kg). Treatment was administered once a day by intravenous injection. Antibody treatment with Hu01G06-127 and Hu01G06-135 at 20 mg/kg resulted in body weight increase above the initial weight or 108% ($p < 0.001$) (FIG. 28). Antibody treatment with Hu01G06-127 and Hu01G06-135 at 2 mg/kg resulted in limited body weight decrease compare to control (hIgG) from the initial weight or 88-85% ($p < 0.001$) (FIG. 28). Statistical analysis was performed using ANOVA. Results for changes of body weight at the end of the study in the HT-1080 model are shown in FIG. 28 and Table 33.

TABLE-US-00217 TABLE 33 Treatment % Body ANOVA Analysis Gr. Agent mg/kg weight

(compared to mIgG) 1 mIgG 20 75.2 NA 2 Hu01G06-127 20 108.9 $p < 0.001$ 3 Hu01G06-127 2.0 88.1 $p < 0.001$ 4 Hu01G06-127 0.2 80.0 NS 5 Hu01G06-135 20 108.6 $p < 0.001$ 6 Hu01G06-135 2.0 85.2 $p < 0.001$ 7 Hu01G06-135 0.2 77.3 NS

[0359] The data in FIG. 28 and Table 33 indicated that antibodies Hu01G06-127 and Hu01G06-135 can reverse cachexia in an HT-1080 fibrosarcoma xenograft model in a dose-dependent manner.

Example 20: Reversal of Muscle and Fat Loss in an HT-1080 Xenograft Tumor Model

[0360] This Example demonstrates the reversal of cachexia (as indicated by body weight loss, muscle mass loss and fat mass loss) by antibody 01G06 in an HT-1080 fibrosarcoma xenograft model. HT-1080 cells were grown in culture at 37° C. in an atmosphere containing 5% CO₂, using Eagle's Minimum Essential Medium (ATCC, Catalog No. 30-2003) containing 10% FBS. Cells were inoculated subcutaneously into the flank of 8-week old female ICR SCID mice with 5×10^6 cells per mouse in 50% matrigel. A cohort of ten 8-week old female ICR SCID mice with the same body weight was selected for subcutaneous inoculation into the flank with matrigel, as a non tumor (SHAM) control arm. Body weight was measured daily. When body weight reached 91% in the tumor bearing mice, the mice were randomly divided into two groups of ten mice each. Each group received one of the following treatments: human IgG control (hIgG) or Hu01G06-127 10 mg/kg on day 1, day 3 and day 6. Treatment was administered by intra-peritoneal injection. Treatment with antibody Hu01G06-127 resulted in body weight increase to 105% of initial weight compared to non-tumor bearing control mice (SHAM; $p < 0.001$) (FIG. 29A and Table 34).

TABLE-US-00218 TABLE 34 Treatment % Body ANOVA Analysis Gr. Agent mg/kg weight (compared to hIgG) 1 hIgG 10 84.3 NA 2 Hu01G06-127 10 105.4 $p < 0.001$ 2 SHAM no tumor control none 101.9 $p < 0.001$

[0361] The data in FIG. 29A and Table 34 indicate that the disclosed anti-GDF15 antibody can completely reverse cachexia in an HT-1080 fibrosarcoma xenograft model.

[0362] In this experiment, a group of ten mice were sacrificed at the time of dosing (baseline or 91% body weight loss, without treatment) and at the end of the study (eight days post dose, either hIgG or Hu01G06-127 as well SHAM non tumor control mice). Gonadal fat and the gastrocnemius muscles were removed surgically and weighed. As shown in FIG. 29B, significant gonadal fat mass loss was observed seven days post dose with hIgG, but not in the group treated with antibody Hu01G06-127 compared to baseline control (91% body weight loss). Moreover, treatment with Hu01G06-127 not only prevented further fat loss (compared to baseline group), but also, was able to restore the normal levels of gonadal fat (compared to SHAM non-tumor control) (FIG. 29B). In addition, significant gastrocnemius muscle mass loss was observed seven days post dose with hIgG, but not in the group treated with antibody Hu01G06-127 compared to baseline control (91% body weight loss) (FIG. 29C). Treatment with Hu01G06-127 not only prevented further muscle loss (compare to baseline group), but also was able to restore the normal levels of gastrocnemius muscle (compared to SHAM non tumor control) (FIG. 29C).

[0363] These results indicate that the disclosed anti-GDF15 antibodies can completely reverse cachexia measured by the loss of muscle mass, loss of fat and involuntary weight loss in an HT-1080 xenograft tumor model.

INCORPORATION BY REFERENCE

[0364] The entire disclosure of each of the patent documents and scientific articles referred to herein is incorporated by reference for all purposes.

EQUIVALENTS

[0365] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting on the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and the range of equivalency of the claims are intended to be embraced therein.

Claims

1. A method of treating cachexia in a mammal comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
2. The method of claim 1, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.
3. The method of claim 1, further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.
4. The method of claim 1, wherein the mammal is a human.
5. A method of inhibiting loss of muscle mass associated with cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
6. The method of claim 5, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.
7. The method of claim 5, wherein the loss of muscle mass is accompanied by a loss of fat mass.
8. The method of claim 5, further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.
9. A method of inhibiting loss of organ mass associated with cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
10. The method of claim 9, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.
11. The method of claim 9, wherein the organ is kidney, liver, heart or spleen.
12. The method of claim 9, wherein the loss of organ mass is accompanied by a loss of muscle mass, a loss of fat mass or involuntary weight loss.
13. The method of claim 9 further comprising administering a second agent to the mammal in need thereof, wherein the second agent is selected from the group consisting of an inhibitor of Activin-A, an inhibitor of ActRIIB, an inhibitor of IL-6, an inhibitor of IL-6R, a melanocortin peptide inhibitor, a melanocortin receptor inhibitor, a ghrelin, a ghrelin mimetic, a GHS-R1a agonist, a SARM, a TNF α inhibitor, an IL-1 α inhibitor, a myostatin inhibitor, a beta-blocker and an anti-cancer agent.
14. A method of treating sarcopenia associated with cachexia in a mammal comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
15. A method of decreasing the incidence and/or severity of cachexia in a mammal, thereby increasing the maximum tolerated dose of an anti-cancer agent capable of causing cachexia in the mammal, the method comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
16. The method of claim 14, wherein the antibody or antigen-binding fragment thereof is administered in combination with the anti-cancer agent.
17. The method of claim 15, wherein the mammal is a human.

- 18.** A method of increasing appetite in a mammal suffering from cachexia comprising administering an effective amount of an anti-GDF15 antibody or antigen-binding fragment thereof.
- 19.** The method of claim 18, wherein the cachexia is associated with an underlying disease selected from the group consisting of cancer, chronic heart failure, chronic kidney disease, COPD, AIDS, multiple sclerosis, rheumatoid arthritis, sepsis, and tuberculosis.
- 20.** The method of claim 16, wherein the mammal is a human.
-